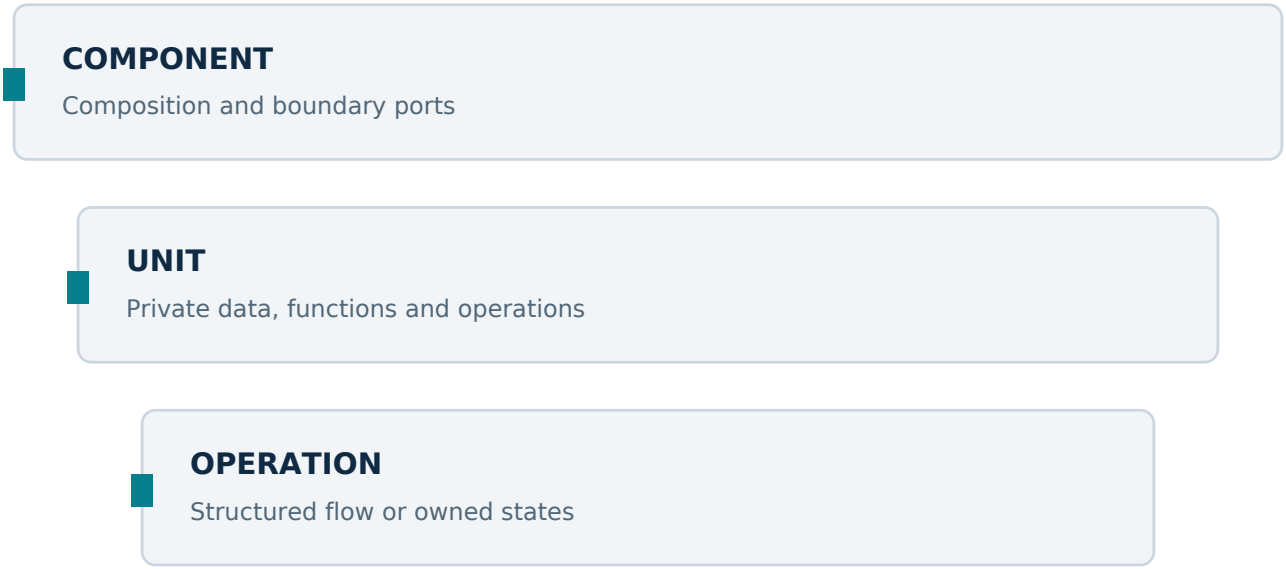
SOFTWARE ARCHITECTURE MODELING LANGUAGE

SWAML

Complete language specification

Specification 1.1.0 / Language 1.1 / Reference simulation 1.1



Typed interfaces at every software boundary.

Embedded realtime systems, services and event-driven architectures.

14 October 2026 / Catalin Ilie and SWAML contributors

Independent language design on a proposed KerML semantic foundation.

github.com/voidaliot/swaml

About this edition

VERSION 1.1.0 / COMPLETE LANGUAGE REFERENCE

This book contains the complete SWAML 1.1 language specification, a concern guide and every keyword in alphabetical order, formal execution rules, rendered notation examples, a reference simulation binding, the complete grammar and runnable model examples. The specification and package revision is 1.1.0. Source files begin with packages; hosts select and pin the 1.1 grammar. Explicit type definitions retain def; component usages form execution roots and operations own their states directly.

Normative chapters define syntax, validity, interface isolation, operation/state behavior and mapping obligations. Tutorials and figures illustrate these rules. This independently authored language is not an OMG UML release or an AUTOSAR implementation. KerML export remains a proposed semantic mapping contract.

Editorial revision 2026-10-16: platform-neutral tooling guidance and view descriptions. The historical language syntax and semantics are retained.

A working Langium parser/validator and deterministic logical-time simulator provide executable evidence. Physical timing proof, a kernel exporter, production runtime and graphical editor require separate implementations.

Navigation	How to use this PDF
Contents and bookmarks	Select a contents entry or use the PDF outline to open a chapter or section.
Cross-references	Links to included documents jump inside this book. Other repository links target version v1.1.0.
Code and tables	Code is selectable, numbered and wrapped without omitting source text. Tables repeat their headers across pages.
Source record	The build manifest lists every included source and its SHA-256 digest. Original source files are embedded as PDF attachments. The release includes a PDF checksum.

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USER REFERENCE / CONCERN AND KEYWORD NAVIGATION

Language reference by concern

Use this alphabetical map to navigate SWAML 1.1 by architectural question. The alphabetical keyword reference provides the other reading order; the complete Markdown edition contains both maps, all normative chapters, user guidance, grammar and model examples in one document. Rules are normative where their source chapter says so; a tutorial or illustration never overrides those rules.

Architecture selection and platform correspondence

Start with the architecture comparison. SWAML borrows strict interface boundaries from component/service architectures and familiar structural/behavioral concepts from UML/SysML. Its authoring model is deliberately software-specific. A Classic-like control loop uses sampled senderReceiver members, periodic operations and explicit timing. A service architecture uses operation/event contracts, correlated requests, timeout handling and static subscriber connections. A gateway explicitly translates between these through its own unit ports.

Choose a root component profile (`realtime`, `service` or `mixed`) according to the contracts you intend to supply. The choice does not turn the model into AUTOSAR configuration or prove a scheduling property. Use an adapter unit when hardware or middleware behavior matters to the architecture. External devices cannot enter a case through an untyped actor shortcut.

Behavior and state ownership

See operations, flow and states and the operational contract. Units own executable operations directly. A function is a pure returned calculation; an untriggered Void operation is a local imperative helper; a triggered operation is a root activation. Components compose these units and do not add a second behavior allocation layer.

Choose a statement body for sequential calculation and requests. Choose a state body when the operation remains in named states between inputs. States, guards and transitions remain inside that operation. There are no independent actions or performers. The operation's flow/state body is the subject of FV. Entry/exit cannot read a transition payload; save a needed value in private unit data in the effect.

Cases, requirements and evidence

See root components and cases. Requirements are named natural-language obligations. A case describes permitted interface exchanges in a chosen root component and may claim requirement coverage. Case steps reference actual top-level connectors and member paths. `after` gives causality; independent steps remain unordered.

A reply needs one unanswered matching request in its causal ancestry. Route validity is checked statically. A simulation or analysis witness is separate evidence that the operation bodies can realize the described case. Neither a `satisfies` keyword nor a green line in a diagram establishes that evidence. CV shows coverage claims; SV shows the causal exchanges.

Communication and interface boundaries

See interfaces and communication. Choose senderReceiver for data, clientServer for RPC, publishSubscribe for events, or service for RPC plus provider events. Contracts are nominal and versioned. Unit ports bind those contracts to operations; component boundary ports delegate them through the hierarchy. Connect only immediate child ports at each composition level.

A sender/publisher can fan out. A subscriber can receive multiple publisher routes, each with its own queue. A receiver/client/requires port has at most one route. Required ports must be connected in a selected closed

root component; optional unbound RPC reports Unavailable. Reading another unit's fields, invoking its operation directly, sending a free event, or adding a topic string outside a contract is invalid.

Queue delivery, order, capacity and overflow are mandatory. Data/event admission can partially succeed across fanout, so atomic broadcast needs an explicit protocol. Requests wait for one correlated result and require a failure branch. Timeout does not cancel server effects. A field-like service property is modeled as getter, setter and changed event, with private storage; see the setting example.

Conformance and formal accuracy

Read the obligation classification, formal rules and implementation matrix. S means grammar support, V static validity, K kernel mapping, E execution and G graphical editing. Their obligations are distinct. The reference implementation supplies parsing, broad static checks and a defined logical simulation binding; independent complete certification is not asserted.

A parsed but invalid model is not executable. A valid model may deliberately fail an assertion, overflow a queue, exceed a loop bound, time out, or violate a timing contract at runtime. Report those observations; do not weaken the language to make every accepted model look successful. Unknown syntax and unsupported requested editions must be rejected. The host pins the parser edition; historical file headers are not 1.1 syntax.

Graphical views and navigation

See graphical notation. The Model Browser exposes ownership and definition/occurrence identity. GV shows general structure, IV connections, CV cases/coverage, SV exchanges, GRV tabular properties, and FV operation flow or states. A selected occurrence retains its instance path when its shared declaration is displayed.

A diagram is a projection of the same semantic model as the text. Layout belongs in a sidecar. Port/member paths and connector identities must remain discoverable; a drawing cannot create a hidden channel. Source edits require parse/link/validation and undo.

Kernel foundation and interchange

Read KerML mapping and the normalized model. They specify identities, ownership, value types, routes, behavior lowering, metadata and reverse traces. The mapping inventory covers every grammar rule and type alias, including syntax that contributes to an owner's semantics rather than creating its own kernel node.

SWAML is defined over a proposed KerML-based semantic contract. This release does not export a kernel-validated artifact or publish a formal SWAML KerML library. The Langium AST is an authoring representation. A backend must preserve SWAML-specific queues, copying, snapshots, correlation and reaction order through a versioned library; kernel metadata alone is not execution support.

Names, structure and reusable types

See structure, names and types. A package owns top-level declarations; `A::T` selects a package member, while dots follow structural or local-value paths. Imports are direct, not transitive. Short-name ambiguity is an error across declaration kinds. Local operations/functions never resolve to another unit by fallback.

`component def` and `unit def` explicitly declare reusable types. Instances create distinct boundaries, stores and activations. A root component is a named occurrence, either inline composition or an instance of a component definition. Both root and nested components may contain units/components. Only a root carries an execution profile and node/deployment binding. Containment is acyclic. A unit variable belongs to one occurrence and is initialized before any startup work runs. Reusing a definition never creates shared singleton state.

Simulation and runtime failures

Use the reference execution profile. Every example has a checked run configuration. Supply the entire validated workspace, selected root component and finite horizon. The output records exact sources, options, chronological observations and final stores/states. Exercise adverse conditions with the offer-indexed fault schedule and inspect whether the intended offer occurred.

Simulation fixes one deterministic ordering among core-permitted schedules, uses bounded resources and zero local CPU cost. It can expose behavioral failures, queue effects and elapsed-time violations; it cannot prove all schedules safe or establish hardware schedulability. Keep evidence linked to exact source versions and target assumptions.

Timing, deployment and concurrency

See timing and deployment and dispatch rules. WCET is active execution, deadline is elapsed time from release, maxLatency ends at destination readiness, and maxAge measures original message age until dispatch. Periodic deadlines cannot exceed their periods. A state machine's contract applies per reaction, not to its whole lifetime.

Serial activation roots retain their unit lock across RPC waits; a callback to that same serial unit may have to wait. Reentrant activation roots use read-only dispatch-time snapshots and may not mutate private unit state through helpers. Allocation inherits from the nearest explicitly deployed ancestor. MCU/process/container labels do not supply a scheduler or bypass interface isolation.

Values and calculation

See value and expression rules. Values are Bool, bounded Int, finite binary64 Real, String, nominal records/enums and fixed arrays. Parameters, locals and messages contain copies. There are no pointers, dynamic collections, implicit remote references or global mutable data.

Use whole-value assignment. Record field and final array-index reads are local value operations; field/index mutation is absent. Pure functions cannot read unit variables. Int-to-Real widening is scalar only. Division returns Real; integer remainder follows the dividend's sign. Arithmetic range, general array bounds and finite loop/microstep behavior are runtime obligations.

Versioning and language evolution

The source is headerless; its parser/host is explicitly pinned to SWAML 1.1. This specification/package revision is 1.1.0. Interface versions remain independent nominal contract labels. Version 1.1 is an intentional syntax revision; it is not backward-compatible with 1.0 source despite the minor version number. Follow the migration guide and preserve historical release artifacts.

Inheritance, generics, dynamic instance/subscription/discovery lifecycle, unbounded arrays, reference payloads, hierarchical/orthogonal states, unrestricted fork/join and host-language code are outside this closed version. A proposal adding them must define syntax, ownership, typing, operational meaning, graphical projection, mapping, migration and tests before it is accepted. These concerns must not appear as undocumented backend shortcuts.

USER REFERENCE / CONCERN AND KEYWORD NAVIGATION

Alphabetical keyword reference

This user-facing reference covers every reserved word in the SWAML 1.1 grammar. Entries are sorted alphabetically, ignoring case for sorting only; source keywords remain case-sensitive. Example fragments illustrate a use and are not standalone models. Read the linked normative concern for full constraints. Punctuation and operator precedence follow the final section.

after · and · as · assert · atLeastOnce · atMostOnce · bestEffort · Bool · bound · call · capacity · case · client · clientServer · completion · component · concurrency · connect · container · CV · data · datatype · deadline · def · delegate · delivery · deploy · dropNewest · dropOldest · else · entry · enum · event · every · exit · fail · false · fifo · final · for · function · FV · GRV · GV · idempotent · if · import · initial · Int · interface · invoke · IV · label · let · manual · maxAge · maxLatency · mcu · mixed · ms · node · not · ns · of · on · operation · optional · or · ordering · otherwise · overflow · package · port · priority · process · provides · publish · publisher · publishSubscribe · qos · queued · Real · realtime · receive · receiver · record · reentrant · reject · reply · request · requirement · requires · return · s · sampled · satisfies · send · sender · senderReceiver · serial · server · service · start · state · String · subscriber · SV · timing · transition · true · unit · unordered · us · var · version · via · view · Void · wcet · when · while · within

Keyword: after

Concern: Behavior and states. **Grammar:** `Transition`, `ScenarioStep`.

In a transition, creates a timer measured from source-state entry completion; leaving the state cancels its generation. In a case step, names one earlier causal predecessor, not a delay or source-order convention.

Example fragment: `on after 10 ms / via link after sent label received;`

Keyword: and

Concern: Values and calculation. **Grammar:** `AndExpression`.

Short-circuit Boolean conjunction. Both operands must be Bool; the right operand is evaluated only when the left is true.

Example fragment: `ready and enabled`

Keyword: as

Concern: Communication. **Grammar:** `Request`.

Names the successful awaited RPC result. A Void result reserves the name without creating a readable value. The name must be fresh in the live lexical environment.

Example fragment: `request api.read() as value within 1 s otherwise { return; };`

Keyword: assert

Concern: Behavior and states. **Grammar:** `Assert`.

Evaluates a Bool; false fails the activation without rollback. A failed RPC handler returns RemoteFailure through its correlation. Assertions are executable checks, not static proofs.

Example fragment: `assert measured <= 100;`

Keyword: atLeastOnce

Concern: Communication. **Grammar:** `Qos`.

Permits retries and duplicate delivery subject to liveness, overflow and freshness. RPC members on such a connector must declare idempotent behavior. It never implies exactly-once effects.

Example fragment: `delivery atLeastOnce;`

Keyword: atMostOnce

Concern: Communication. **Grammar:** `Qos`.

Allows loss but no duplicate delivery for a logical message. A lost request or response can still lead to timeout. This policy does not guarantee admission or successful execution.

Example fragment: `delivery atMostOnce;`

Keyword: bestEffort

Concern: Communication. **Grammar:** `Qos`.

Permits loss without an eventual-delivery guarantee and without duplicate delivery. Queue capacity, overflow and declared timing bounds still apply.

Example fragment: `delivery bestEffort;`

Keyword: Bool

Concern: Values and calculation. **Grammar:** `TypeRef`.

Primitive two-valued Boolean type. Conditions and logical operators require Bool; numbers and strings are not truthy substitutes.

Example fragment: `let ready: Bool = true;`

Keyword: bound

Concern: Behavior and states. **Grammar:** `While`.

Sets the positive safe-integer maximum number of loop-body executions. After that many iterations the condition is reevaluated; a still-true condition fails execution.

Example fragment: `while (i < 3) bound 3 { i = i + 1; }`

Keyword: call

Concern: Behavior and states. **Grammar:** `Trigger`.

Activates a unit operation for an incoming RPC interface member. Parameters and result must match the contract, and the member has exactly one handler owner.

Example fragment: `operation read(): Int on call api.get { return current; }`

Keyword: capacity

Concern: Communication. **Grammar:** `Qos`.

Positive safe-integer retained-message limit per connector occurrence, direction and member. Transport-pending messages count; running handlers do not. Sampled data requires capacity 1.

Example fragment: `capacity 8;`

Keyword: case

Concern: Cases and evidence. **Grammar:** `Scenario`.

Defines a descriptive scenario for a root component with labeled interface exchanges and optional requirement traces. It does not execute or prove itself and cannot introduce hidden traffic.

Example fragment: `case Checkout for Shop satisfies Accepted { ... }`

Keyword: client

Concern: Communication. **Grammar:** `Port`.

Port role for the requesting side of a clientServer interface. A closed client port has at most one route; replies return through the same route and correlation.

Example fragment: `port api: Health client;`

Keyword: clientServer

Concern: Communication. **Grammar:** `Interface`.

Interface kind containing operation signatures only. A client issues request; a server implements on call handlers. Events belong in service or publishSubscribe contracts.

Example fragment: `interface clientServer Health version "1.0.0" { operation check(): Bool; }`

Keyword: completion

Concern: Behavior and states. **Grammar:** `Transition`.

Local state stimulus reconsidered after entry and each reaction until quiescent. It is not an interface event. Unconditional cycles are invalid; guarded cycles have a 1,024-edge runtime limit.

Example fragment: `transition next: A -> B on completion {}`

Keyword: component

Concern: Structure and names. **Grammar:** `Component`, `Instance`, `RootComponent`.

With def, declares a reusable composition type. Without def, a typed child occurrence uses name: Type; at package level a root occurrence either supplies inline unit/component children and connectors or instantiates a component definition. A typed root body may bind nodes/deployment only. The root profile defaults to service.

Example fragment: `component def Assembly { unit controller: Controller; } / component Demo: Assembly;`

Keyword: concurrency

Concern: Timing and deployment. **Grammar:** `Timing`.

Selects serial or reentrant operation activation policy in a timing contract. Helpers inherit the activation mode. Stateful operations never overlap their own reactions.

Example fragment: `concurrency serial;`

Keyword: connect

Concern: Communication. **Grammar:** `Connect`.

Creates an explicit connector between immediate child ports of a component definition or root component. Contracts must be identical and roles complementary. Every connector supplies QoS.

Example fragment: `connect samples: sensor.output -> controller.input qos { ... };`

Keyword: container

Concern: Timing and deployment. **Grammar:** `Node`.

Execution-node kind representing a container deployment location. It does not create a component, communication channel or container runtime configuration.

Example fragment: `node application container;`

Keyword: CV

Concern: Graphical views. **Grammar:** `View`.

Case View kind: displays cases, requirements and claimed coverage for a case or root component. Verified coverage requires separate linked evidence.

Example fragment: `view coverage CV of Shop;`

Keyword: data

Concern: Communication. **Grammar:** `DataMember`.

Declares one senderReceiver member with a value type and sampled or queued mode. Outgoing use is send; incoming handling is on receive with one typed argument.

Example fragment: `data sample: Real sampled;`

Keyword: datatype

Concern: Values and calculation. **Grammar:** `DataType`.

Declares a nominal finite record. Field names are unique, every literal supplies all fields once, and recursive value containment is invalid. Empty records have one value.

Example fragment: `datatype Point { x: Real; y: Real; }`

Keyword: deadline

Concern: Timing and deployment. **Grammar:** `Timing`.

Positive elapsed-time bound from activation release to completion, including queueing and request suspension. For stateful operations it applies per reaction. WCET must not exceed it.

Example fragment: `deadline 5 ms;`

Keyword: def

Concern: Structure and names. **Grammar:** `Component`, `Unit`.

Required marker for a reusable component or unit type. Definitions create no running singleton. An occurrence without def must either refer to the matching definition kind or, for a package-level component root, supply inline composition.

Example fragment: `unit def Worker { ... }`

Keyword: delegate

Concern: Communication. **Grammar:** `Delegate`.

Aliases a component boundary port to one immediate child port of the same role and interface. Every component port delegates once. Delegation adds neither conversion nor a queue.

Example fragment: `delegate api -> worker.api;`

Keyword: delivery

Concern: Communication. **Grammar:** `Qos`.

Chooses bestEffort, atMostOnce or atLeastOnce for all member queues of a connector. Delivery is separate from FIFO ordering, admission and application success.

Example fragment: `delivery atMostOnce;`

Keyword: deploy

Concern: Timing and deployment. **Grammar:** `Deployment`.

Assigns an existing component or descendant occurrence to a root component node. Descendants inherit the nearest explicit allocation. Duplicate exact-path allocations are invalid.

Example fragment: `deploy backend.worker on application;`

Keyword: dropNewest

Concern: Communication. **Grammar:** `Qos`.

On a full queue, discards the offered message observably. Admission does not fail the producer; an RPC can later time out. Existing retained messages stay.

Example fragment: `overflow dropNewest;`

Keyword: dropOldest

Concern: Communication. **Grammar:** `Qos`.

On a full queue, discards the oldest retained message and admits the new one. Required with capacity 1 for sampled data. It does not undo effects of already dispatched handlers.

Example fragment: `overflow dropOldest;`

Keyword: else

Concern: Behavior and states. **Grammar:** `If`.

Optional alternative block of an if statement. Only the chosen branch executes; branch-local declarations do not escape that branch.

Example fragment: `if (ready) { return; } else { fail "not ready"; }`

Keyword: entry

Concern: Behavior and states. **Grammar:** `State`.

State-entry block executed after that state becomes current. It has no incoming payload scope. Residence timers begin after entry finishes; return/fail terminates the owning operation.

Example fragment: `state Idle { entry { count = 0; } }`

Keyword: enum

Concern: Values and calculation. **Grammar:** `EnumType`.

Declares a nominal set of one or more unique literals. Values use Type.Literal or Package::Type.Literal. A live value first segment in the current expression environment takes precedence over an unqualified enum name, with no fallback after a failed field lookup; qualify to select the enum explicitly.

Example fragment: `enum Status { Ready, Failed } / Status.Ready / Contracts::Status.Ready`

Keyword: event

Concern: Communication. **Grammar:** `EventMember`.

Declares a typed asynchronous member in publishSubscribe or service. It must be emitted by an operation through a port and owned by one receiving operation; free global events are absent.

Example fragment: `event changed(value: Real);`

Keyword: every

Concern: Behavior and states. **Grammar:** `Trigger`.

Periodic parameterless Void activation at epoch plus k periods, k starting at 1. Releases do not merge when earlier activations are busy; pending work is bounded by the binding.

Example fragment: `operation sample(): Void on every 10 ms { ... }`

Keyword: exit

Concern: Behavior and states. **Grammar:** `State`.

State-exit block runs after timer cancellation and before the transition effect. It lacks payload scope. Return/failure skips remaining transition phases.

Example fragment: `state Running { exit { active = false; } }`

Keyword: fail

Concern: Behavior and states. **Grammar:** `Fail`.

Raises a local failure with a String literal reason, unwinds synchronous callers, and preserves committed effects. An RPC caller receives the correlated RemoteFailure outcome.

Example fragment: `fail "invalid operating mode";`

Keyword: false

Concern: Values and calculation. **Grammar:** `Literal`.

The false Bool literal. Distinct from numeric zero, empty strings or absent values; no implicit coercion exists.

Example fragment: `let ready: Bool = false;`

Keyword: fifo

Concern: Communication. **Grammar:** `Qos`.

Dispatches retained messages in queue order for one connector/direction/member. Independent queues and handler completion order are not globally ordered.

Example fragment: `ordering fifo;`

Keyword: final

Concern: Behavior and states. **Grammar:** `State`.

Marks a state that completes its owning operation after entry. It has no outgoing transition. Future input to that operation is undeliverable, not rerouted.

Example fragment: `state Stopped final {}`

Keyword: for

Concern: Cases and evidence. **Grammar:** `Scenario`.

Selects the root component described by a case. The steps use root-level connectors and immediate child unit/component port-member paths. Typed roots resolve topology through their component definition.

Example fragment: `case Startup for Product { ... }`

Keyword: function

Concern: Values and calculation. **Grammar:** `FunctionDef`.

Defines a pure expression with a non-Void result. It sees only copied parameters and other local pure functions, never unit variables. The call graph is acyclic.

Example fragment: `function twice(value: Int): Int = value * 2; / twice(3)`

Keyword: FV

Concern: Graphical views. **Grammar:** `View`.

Flow View of an operation. Projects either its statement body or its owned states/transitions in the same view family.

Example fragment: `view control FV of Controller.regulate;`

Keyword: GRV

Concern: Graphical views. **Grammar:** `View`.

Grid View of a package or named semantic declaration other than a view. Presents inventory, contracts, timing, allocation or trace tables; sorting does not change model meaning.

Example fragment: `view inventory GRV of Architecture;`

Keyword: GV

Concern: Graphical views. **Grammar:** `View`.

General View of a package or resolvable named semantic declaration other than a view. Shows ownership, types, attributes and relations, retaining definition/occurrence distinction.

Example fragment: `view overview GV of Product;`

Keyword: idempotent

Concern: Communication. **Grammar:** `Signature`.

Contract assertion required for RPC under atLeastOnce delivery. Repeated requests must have idempotent effects including published side effects. The keyword alone is not a proof.

Example fragment: `operation read(): Int idempotent;`

Keyword: if

Concern: Behavior and states. **Grammar:** `If`.

Evaluates one Bool and executes the selected lexical block. It supplies no concurrency or implicit variable merging. Value-returning operations must cover both paths or fail.

Example fragment: `if (value > limit) { return false; } else { return true; }`

Keyword: import

Concern: Structure and names. **Grammar:** `Import`.

Imports top-level names from one package using namespace::*; imports are not transitive. A short name must be unique across local and directly imported declarations.

Example fragment: `import Common::*;`

Keyword: initial

Concern: Behavior and states. **Grammar:** `StateBody`.

The first item in a stateful operation body. References exactly one state owned by that operation and distinguishes state behavior from a statement block without a wrapper keyword. Initialization enters it once per run epoch before input consumption.

Example fragment: `initial Idle;`

Keyword: Int

Concern: Values and calculation. **Grammar:** `TypeRef`.

Exact signed integer type from -(2^53-1) through 2^53-1. Overflow, invalid indexing and division/remainder by zero fail execution. Scalar Int may widen to Real.

Example fragment: `var count: Int = 0;`

Keyword: interface

Concern: Communication. **Grammar:** `Interface`.

Declares a nominal, versioned contract of one of four kinds. Port endpoints must reference the same declaration; matching names or version strings alone are insufficient.

Example fragment: `interface service Orders version "1.0.0" { ... }`

Keyword: invoke

Concern: Behavior and states. **Grammar:** `Invoke`.

Synchronously calls an untriggered Void operation in the same unit. It shares the activation lock and snapshot; copied parameters are local. Remote invocation requires request through a port.

Example fragment: `invoke prepare();`

Keyword: IV

Concern: Graphical views. **Grammar:** `View`.

Interconnect View of a root component/component definition or occurrence. Shows immediate children, boundary ports, connectors and delegation with exact interface paths.

Example fragment: `view wiring IV of Product;`

Keyword: label

Concern: Cases and evidence. **Grammar:** `ScenarioStep`.

Assigns a unique name to a scenario step. Later after clauses use that label as a causal predecessor; source position alone imposes no ordering.

Example fragment: `via orderLink label submitted;`

Keyword: let

Concern: Behavior and states. **Grammar:** `Let`.

Creates a typed mutable local value from a copied initializer. It cannot shadow a live local, parameter or unit variable, and disappears at block exit.

Example fragment: `let count: Int = 0;`

Keyword: manual

Concern: Behavior and states. **Grammar:** `Trigger`.

Parameterless Void operation trigger released by a harness. It is control of execution, not a route for injecting external model payloads or bypassing interfaces.

Example fragment: `operation refresh(): Void on manual { ... }`

Keyword: maxAge

Concern: Communication. **Grammar:** `Qos`.

Maximum message age measured from original emission through retries. Eligibility is inclusive; greater age discards before dispatch. Required for sampled data in realtime/mixed roots.

Example fragment: `maxAge 20 ms;`

Keyword: maxLatency

Concern: Communication. **Grammar:** `Qos`.

Positive bound from admitted emission to destination readiness, excluding dispatch wait. Exceeding it is an observable contract violation. Required on realtime/mixed connectors.

Example fragment: `maxLatency 1 ms;`

Keyword: mcu

Concern: Timing and deployment. **Grammar:** `Node`.

Execution-node kind for an embedded microcontroller location. The core does not infer interrupts, task priorities, processor speed or memory limits.

Example fragment: `node controller mcu;`

Keyword: mixed

Concern: Timing and deployment. **Grammar:** `RootComponent`.

Root component profile covering embedded/service combinations. All used operations carry timing and all connectors latency; sampled data also carries age. It does not silently exempt cloud-facing paths.

Example fragment: `component Product mixed { ... }`

Keyword: ms

Concern: Timing and deployment. **Grammar:** `Duration`.

Millisecond duration unit; one ms equals 1,000,000 ns. Duration magnitude must be a positive integer and normalize exactly within the safe range.

Example fragment: `within 25 ms`

Keyword: node

Concern: Timing and deployment. **Grammar:** `Node`.

Declares a named execution location in a root component with mcu, process or container kind. Nodes affect allocation only; traffic still follows interfaces.

Example fragment: `node edge process;`

Keyword: not

Concern: Values and calculation. **Grammar:** `UnaryExpression`.

Unary Bool negation with precedence above multiplication. No numeric or string coercion is allowed.

Example fragment: `not ready`

Keyword: ns

Concern: Timing and deployment. **Grammar:** `Duration`.

Nanosecond duration unit and the reference simulation clock's integral base. All specified durations are positive, with safe exact conversion.

Example fragment: `after 1 ns`

Keyword: of

Concern: Graphical views. **Grammar:** `View`.

Introduces a view subject path rooted in the owning package or its top-level declaration. Allowed subject kinds depend on the view kind.

Example fragment: `view flow FV of Product.controller.worker.run;`

Keyword: on

Concern: Behavior and states. **Grammar:** `Deployment`, `Trigger`, `Transition`.

Introduces an operation or transition trigger. Operations use start/manual/every/receive/call; transitions use completion/after/receive. Triggers do not introduce free signals.

Example fragment: `on receive input.sample`

Keyword: operation

Concern: Behavior and states. **Grammar:** `Signature`, `Operation`.

In an interface, declares an RPC signature. In a unit, declares executable statements or directly owned states. An omitted result annotation means Void. An incoming signature is implemented by exactly one on call operation.

Example fragment: `operation reset(); / operation reset() on call api.reset { return; }`

Keyword: optional

Concern: Communication. **Grammar:** `Port`.

Allows a port to remain unconnected in a closed root component. It does not change contract, role or handler rules. Unbound data/event sends succeed; unbound RPC reports Unavailable.

Example fragment: `port diagnostics: Notices publisher optional;`

Keyword: or

Concern: Values and calculation. **Grammar:** `OrExpression`.

Short-circuit Boolean disjunction. The right Bool operand is evaluated only when the left is false.

Example fragment: `ready or retryAllowed`

Keyword: ordering

Concern: Communication. **Grammar:** `Qos`.

Selects fifo or unordered independently of delivery reliability. Ordering applies per connector occurrence, direction and member, not across an entire interface or root.

Example fragment: `ordering fifo;`

Keyword: otherwise

Concern: Communication. **Grammar:** `Request`.

Mandatory request failure block with implicit copied failure:String. It must return or fail on every path and cannot fall through to successful result use.

Example fragment: `within 1 s otherwise { lastError = failure; return; };`

Keyword: overflow

Concern: Communication. **Grammar:** `Qos`.

Required admission policy when a member queue is full: dropOldest, dropNewest or reject. Policies apply independently to every route in fanout.

Example fragment: `overflow reject;`

Keyword: package

Concern: Structure and names. **Grammar:** `Package`.

Top-level, non-nested namespace for declarations and direct imports. A package is declared once in the loaded workspace; multiple packages may share a file.

Example fragment: `package Control { ... }`

Keyword: port

Concern: Communication. **Grammar:** `Port`.

Owned communication boundary feature typed by one interface, with a compatible role and optionality. Component ports delegate to children; unit ports bind operations to members.

Example fragment: `port api: Orders requires;`

Keyword: priority

Concern: Behavior and states. **Grammar:** `Transition`.

Nonnegative safe-integer tie-break among matching transitions for the same state and stimulus; larger wins, default 0. It is not an operation scheduling priority.

Example fragment: `on receive commands.begin priority 2 {}`

Keyword: process

Concern: Timing and deployment. **Grammar:** `Node`.

Execution-node kind for a logical process. It supplies no hidden IPC access, shared memory or task scheduling policy.

Example fragment: `node worker process;`

Keyword: provides

Concern: Communication. **Grammar:** `Port`.

Provider role of a service interface. Handles incoming RPC, returns correlated results/failures and publishes its declared events in reverse connector direction.

Example fragment: `port api: Orders provides;`

Keyword: publish

Concern: Communication. **Grammar:** `Send`, `ScenarioStep`.

Emits an outgoing event through a publisher/provides port. In a case, describes a routed event step. Arguments are evaluated once and copied per route; no free event syntax exists.

Example fragment: `publish api.changed(current);`

Keyword: publisher

Concern: Communication. **Grammar:** `Port`.

Outgoing role for a publishSubscribe contract. One port may fan out through several connectors; every outgoing member is tied to an emitting operation.

Example fragment: `port events: Notices publisher;`

Keyword: publishSubscribe

Concern: Communication. **Grammar:** `Interface`.

Event-only contract kind with publisher/subscriber ports. Subscriptions are explicit static connectors. Runtime subscribe/unsubscribe and wildcard topics are outside 1.1.

Example fragment: `interface publishSubscribe Notices version "1.0.0" { event changed(); }`

Keyword: qos

Concern: Communication. **Grammar:** `Qos`.

Introduces connector delivery, ordering, capacity and overflow, followed by optional latency and age in the grammar's order. Each connector must supply the block.

Example fragment: `qos { delivery atMostOnce; ordering fifo; capacity 4; overflow reject; }`

Keyword: queued

Concern: Communication. **Grammar:** `DataMember`.

SenderReceiver data mode representing a sequence of retained values under queue policy. It does not imply losslessness or unbounded history.

Example fragment: `data text: String queued;`

Keyword: Real

Concern: Values and calculation. **Grammar:** `TypeRef`.

Finite IEEE 754 binary64 type. Arithmetic is nearest-even; non-finite results and division by zero fail. Scalar Int widens, but fixed arrays do not implicitly widen.

Example fragment: `var temperature: Real = 20.0;`

Keyword: realtime

Concern: Timing and deployment. **Grammar:** `RootComponent`.

Root component profile requiring operation timing, connector latency and sampled freshness contracts. Those declarations require target evidence before any hard realtime claim.

Example fragment: `component Heating realtime { ... }`

Keyword: receive

Concern: Behavior and states. **Grammar:** `Trigger`, `Transition`.

Consumes an incoming data/event member through a receiver/subscriber/requires port. An ordinary operation receives copied parameters; a transition exposes event parameters or data payload in guard/effect.

Example fragment: `on receive commands.begin`

Keyword: receiver

Concern: Communication. **Grammar:** `Port`.

Incoming senderReceiver role. Every data member has one handler owner; at most one route feeds a closed receiver port. Multiple sources require an explicit merge/adaptor unit.

Example fragment: `port input: Samples receiver;`

Keyword: record

Concern: Values and calculation. **Grammar:** `RecordLiteral`.

Constructs a nominal datatype value by assigning each declared field exactly once. Field order in the literal does not change record equality; fields remain copied values.

Example fragment: `record Point { x = 1.0; y = 2.0; }`

Keyword: reentrant

Concern: Timing and deployment. **Grammar:** `Timing`.

Allows overlapping operation activations with independent dispatch-time unit snapshots. Direct or helper writes to unit state are statically invalid. Local variables remain mutable; one state reaction cannot overlap itself.

Example fragment: `concurrency reentrant;`

Keyword: reject

Concern: Communication. **Grammar:** `Qos`.

On full queue, rejects the offer. Data/event producers fail after all fanout offers; requests take otherwise with Rejected. Previously accepted offers and effects are preserved.

Example fragment: `overflow reject;`

Keyword: reply

Concern: Cases and evidence. **Grammar:** `ScenarioStep`.

Describes a response step in a case, reversing an existing connector/member request. It must have one unanswered causal request ancestor. Executable handlers reply by returning, not by a reply statement.

Example fragment: `reply backend.api.get -> client.api.get via link after asked label answered;`

Keyword: request

Concern: Communication. **Grammar:** `Request`, `ScenarioStep`.

Awaits one RPC on a local client/requires port with fresh correlation, result name, finite timeout and explicit failure block. In cases it describes the corresponding routed exchange.

Example fragment: `request api.get() as value within 1 s otherwise { return; };`

Keyword: requirement

Concern: Cases and evidence. **Grammar:** `Requirement`.

Named natural-language obligation. It has descriptive text and can be referenced by cases; the core supplies no implicit requirements solver or verified checkmark.

Example fragment: `requirement Safe "Duty shall remain within the allowed range.";`

Keyword: requires

Concern: Communication. **Grammar:** `Port`.

Consumer role of a service interface. Issues RPC and handles reverse-direction provider events. It has at most one route in a closed root component.

Example fragment: `port orders: Orders requires;`

Keyword: return

Concern: Behavior and states. **Grammar:** `Return`.

Completes the current operation with a copied typed result or bare Void. Inside an invoked helper it resumes the caller; inside state behavior it terminates the owning operation immediately.

Example fragment: `return accepted;`

Keyword: s

Concern: Timing and deployment. **Grammar:** `Duration`.

Second duration unit; one s equals 1,000,000,000 ns. A duration must still fit the exact safe nanosecond range.

Example fragment: `within 2 s`

Keyword: sampled

Concern: Communication. **Grammar:** `DataMember`.

SenderReceiver data mode retaining the latest pending sample: capacity 1 and dropOldest are mandatory. Dispatch removes the sample; the core does not define a remote shared-variable read.

Example fragment: `data temperature: Real sampled;`

Keyword: satisfies

Concern: Cases and evidence. **Grammar:** `Scenario`.

Links a case to one or more requirement declarations as intended coverage. It is a claim requiring test/analysis/review evidence, not a proof by syntax.

Example fragment: `case Checkout for Shop satisfies Accepted, Audited { ... }`

Keyword: send

Concern: Communication. **Grammar:** `Send`, `ScenarioStep`.

Emits an outgoing senderReceiver data member through a sender port with exactly one value. In a case it describes a data exchange. It cannot publish events or bypass another unit's interface.

Example fragment: `send output.temperature(23.5);`

Keyword: sender

Concern: Communication. **Grammar:** `Port`.

Outgoing senderReceiver role. May fan out through multiple explicit connectors; each member has at least one syntactically associated emitting operation.

Example fragment: `port output: Samples sender;`

Keyword: senderReceiver

Concern: Communication. **Grammar:** `Interface`.

Data-only contract kind with sampled/queued members and sender/receiver roles. No RPC operations or event members may be declared inside it.

Example fragment: `interface senderReceiver Samples version "1.0.0" { data value: Real sampled; }`

Keyword: serial

Concern: Timing and deployment. **Grammar:** `Timing`.

Default activation mode: acquire the unit's one serial lock until completion/failure, including RPC waits. Local invokes share the activation lock. Independent unit activations may still interleave.

Example fragment: `concurrency serial;`

Keyword: server

Concern: Communication. **Grammar:** `Port`.

Provider role for clientServer contracts. Each operation signature has exactly one on call handler; results/failures follow the request correlation.

Example fragment: `port health: Health server;`

Keyword: service

Concern: Communication. **Grammar:** `Interface`, `RootComponent`.

As an interface kind, combines RPC operations and events with requires/provides roles. As a root component profile, makes timing contracts optional while retaining interfaces, explicit QoS and static validity; it is the default when no profile is written.

Example fragment: `interface service Orders version "1.0.0" { ... } / component Shop service { ... }`

Keyword: start

Concern: Behavior and states. **Grammar:** `Trigger`.

Once-per-run startup trigger for a parameterless Void operation. Stateful operations require it. All stores/routes/handlers exist before startup work runs; readiness protocols still need explicit ports.

Example fragment: `operation boot(): Void on start { ... }`

Keyword: state

Concern: Behavior and states. **Grammar:** `State`.

Declares a named state directly inside an operation after its initial reference. Optional entry/exit blocks contain ordinary operation statements. State names are operation-local; no separate state allocation or performer exists.

Example fragment: `state Idle { entry { active = false; } }`

Keyword: String

Concern: Values and calculation. **Grammar:** `TypeRef`.

Finite Unicode text with exact equality and no normalization. Strings support portable escapes; implicit concatenation, ordering and bounded transport size are absent.

Example fragment: `var lastError: String = "";`

Keyword: subscriber

Concern: Communication. **Grammar:** `Port`.

Incoming publishSubscribe role. Multiple publishers may feed one subscriber port through distinct queues. Each event has one handler owner; cross-source ordering is unspecified.

Example fragment: `port input: Notices subscriber;`

Keyword: SV

Concern: Graphical views. **Grammar:** `View`.

Sequence View of a case or a root component's selected case. Lifelines are occurrences; arrows show port/member/connector and request correlation. Independent steps must remain visibly unordered.

Example fragment: `view checkout SV of Checkout;`

Keyword: timing

Concern: Timing and deployment. **Grammar:** `Timing`.

Operation contract containing wcet, deadline and concurrency in that order. Required on used operations in realtime/mixed roots; stateful operation timing applies per reaction.

Example fragment: `timing { wcet 1 ms; deadline 5 ms; concurrency serial; }`

Keyword: transition

Concern: Behavior and states. **Grammar:** `Transition`.

Named state edge owned by one operation, with source/target, stimulus, optional guard/priority and effect. Execution orders exit, effect, state update, entry, then completion processing.

Example fragment: `transition beginRun: Idle -> Running on receive commands.begin {}`

Keyword: true

Concern: Values and calculation. **Grammar:** `Literal`.

The true Bool literal, with no implicit equivalence to numeric one or other nonempty values.

Example fragment: `assert true;`

Keyword: unit

Concern: Structure and names. **Grammar:** `Unit`, `Instance`.

Smallest interface-protected software boundary. With def, defines private data/functions/operations; a typed usage creates an occurrence inside a nested or root component. Each occurrence has distinct storage and handlers.

Example fragment: `unit def Worker { ... } / unit worker: Worker;`

Keyword: unordered

Concern: Communication. **Grammar:** `Qos`.

No source-language dispatch ordering guarantee beyond other causal rules. A deterministic backend may still choose FIFO; this choice must not be mistaken for a model guarantee.

Example fragment: `ordering unordered;`

Keyword: us

Concern: Timing and deployment. **Grammar:** `Duration`.

Microsecond duration unit; one us equals 1,000 ns. The source spelling is ASCII us, not the micro sign.

Example fragment: `wcet 100 us;`

Keyword: var

Concern: Values and calculation. **Grammar:** `Variable`.

Declares private per-unit mutable state with a typed pure initializer independent of mutable variables. Each unit occurrence owns separate storage; other units cannot read it directly.

Example fragment: `var count: Int = 0;`

Keyword: version

Concern: Communication. **Grammar:** `Interface`.

Required interface semantic-version string major.minor.patch. Compatibility is nominal declaration identity, not automatic version-range matching or structural subtyping.

Example fragment: `interface clientServer API version "1.0.0" { ... }`

Keyword: via

Concern: Cases and evidence. **Grammar:** `ScenarioStep`.

Names the root-level connector used by a case exchange. Endpoint-member paths must resolve on its immediate child component/unit ports with a legal orientation.

Example fragment: `via orderLink label submitted;`

Keyword: view

Concern: Graphical views. **Grammar:** `View`.

Declares a named projection of an existing semantic subject with GV/IV/CV/SV/GRV/FV kind. It carries no executable behavior or diagram geometry.

Example fragment: `view wiring IV of Product;`

Keyword: Void

Concern: Values and calculation. **Grammar:** `TypeRef`.

No result value. Operation implementations and interface signatures default to Void when : Type is omitted; an explicit : Void is equivalent. Forbidden for parameters, variables, fields, arrays and pure function results. A Void RPC completion label is not readable data.

Example fragment: `operation notify(): Void { return; }`

Keyword: wcet

Concern: Timing and deployment. **Grammar:** `Timing`.

Positive active-execution-time bound including local helpers and state reaction microsteps, excluding request wait. Must not exceed deadline; physical evidence belongs to a target binding.

Example fragment: `wcet 1 ms;`

Keyword: when

Concern: Behavior and states. **Grammar:** `Transition`.

Optional pure Bool transition guard evaluated against the unit snapshot/live environment and stimulus payload as applicable. False consumes/ignores the stimulus when no other transition matches.

Example fragment: `on receive commands.begin when (enabled) {}`

Keyword: while

Concern: Behavior and states. **Grammar:** `While`.

Pretest Bool loop with mandatory positive bound. Each iteration has a lexical body scope. A still-true condition after the bound fails instead of silently exiting.

Example fragment: `while (i < 3) bound 3 { i = i + 1; }`

Keyword: within

Concern: Communication. **Grammar:** `Request`.

Required positive RPC timeout measured from issuance. Only a response observed strictly before expiry can win; equal-time expiry wins and late server effects are not canceled.

Example fragment: `within 100 ms otherwise { return; };`

Punctuation and operator reference

Token	Meaning
`;`	Declaration/statement terminator where the production requires it.
`{ }`	Owned declarations, block scope, record assignments, QoS or timing block according to context.
`( )`	Arguments, parameters, conditions or precedence grouping.
`[ ]`	Fixed type extent, array literal or one final index read.
`:`	Name/type separator, operation result or connector/transition header separator.
`,`	List separator; no implicit trailing comma.
`::`	Package qualification; imports append `::*`.
`.`	Structural/member path, local record field, enum literal separator or real-literal decimal point.
`->`	Connector, delegation, state transition or case exchange direction.
`=`	Initialization, assignment, pure function body or record field assignment; never equality.
`== !=`	Typed recursive equality/inequality; numeric scalar promotion permitted.
`< <= > >=`	Numeric ordering; no String ordering.
`+ - * / %`	Checked numeric arithmetic; slash always Real, remainder Int only; minus also unary.
`// /* */`	Line / non-nested block comments outside strings.

Precedence, lowest first: or; and; equality; ordered comparison; addition/subtraction; multiplication/division/remainder; unary not/minus; primary. Binary operators associate left. Only and/or short-circuit. Identifiers, strings, numerics and durations have the exact lexical rules in chapter 1; the complete grammar is authoritative for concrete syntax.

NORMATIVE / LANGUAGE CONTRACT

SWAML language specification

Version: 1.1.0 · **Source:** headerless packages; parser/host pinned to 1.1 · **Status:** complete 1.1 specification · **Date:** 2026-10-14

SWAML means Software Architecture Modeling Language. This document and its normative chapters define a software architecture language over a KerML-based semantic model. SWAML is independently authored; “unofficial UML 3.0” describes its motivation and does not denote an OMG standard or compatibility level.

1. Normative document set

Read these chapters together. Their requirements have equal standing:

1. Structure, names and types.
2. Interfaces and communication.
3. Operation and state semantics.
4. Timing, deployment, cases and conformance.
5. KerML semantic mapping.
6. Graphical notation.
7. Concrete grammar.
8. Operational contract and target bindings.
9. Normalized semantic model.
10. Conformance obligations and evidence.
11. Formal core and simulation relation.
12. Machine-readable mapping inventory.

The user guide, architecture comparison, examples and design decisions are informative. The implementation matrix reports current tool support; a missing implementation does not waive a language rule.

2. Normative terms and precedence

MUST and MUST NOT state conformance requirements. SHOULD states a recommendation with an explicitly documented justification needed for deviation. MAY grants permission. “Error” means a model is not valid under this version. An implementation may retain an invalid model for editing but MUST NOT report it as fully valid.

The grammar determines tokens and parse structure. Prose determines validity and meaning. If grammar accepts text that violates a semantic rule, that text is invalid. If a discovered conflict makes the two impossible to satisfy, record a specification defect; do not silently choose a runtime meaning. Unknown syntax and unsupported requested language versions MUST be rejected. The host selects a pinned 1.1 parser before loading source; a file does not guess its own edition. See the 1.1 migration guide for the intentional syntax migration from 1.0.

The language is closed at version 1.1: constructs not listed in the grammar are not implicitly inherited from UML, SysML or KerML. Definitions may be reused by instantiation, but inheritance, generics, redefinition, unrestricted KerML embedding and dynamic creation are not part of this version. This constraint makes the whole v1.1 language implementable without pretending to reproduce three other standards.

3. Design invariants

- **I1 - Boundary:** every exchange between distinct unit instances crosses port instances typed by the same interface contract and a permitted connector/delegation path.
- **I2 - Ownership:** components own composition; units own private data, functions and operations; operations own statement flow or states. No performer element, behavior allocation, free action, free signal, or global variable exists.
- **I3 - Event provenance:** every outgoing data/event member of a unit port has at least one syntactically associated emitting operation. Every incoming member has exactly one handling operation, including a stateful operation where applicable.
- **I4 - Type safety:** payloads and results conform to nominal contracts. Messages transfer values, never aliases to another unit's private state.
- **I5 - Explicit communication policy:** every connector declares delivery, ordering, capacity and overflow. Time-sensitive connections add latency/freshness bounds.
- **I6 - Views preserve meaning:** views reference existing semantic elements. Drawing a message cannot create a hidden communication channel.

Local calls, calculation values, timer expiration, initial activation and state completion do not constitute exchanges between units. They are the only exceptions needed for local computation and lifecycle. An external actor or device that exchanges model data MUST be represented through a component/unit port; a case actor label cannot bypass I1.

4. Abstract model inventory

Element	Owns	References / meaning
Model	packages	language version supplied by the pinned parser/host
Package	imports, named elements	namespace
DataType / EnumType	fields / literals	immutable value shape
Interface	data, event or operation members	nominal versioned contract
Component definition	ports, child instances, connectors, delegation	reusable composition type
Unit definition	ports, private variables, operations, functions	reusable behavior/data boundary
Contained instance	no inline definitions	one component or unit definition
Port	role and optionality	one interface
Connector	QoS	source and target port paths
Delegation	no data conversion	outer port and immediate child port
Operation	parameters, result, trigger, timing; flow **or** owned states	executable behavior owned by a unit
Function	parameters, result, expression	pure local calculation
State body	states, transitions, initial reference	syntax grouping of one operation; not an independent model element

Element	Owns	References / meaning
State	entry and exit blocks	one operation-local condition
Transition	guard, effect, priority, trigger	source and target states
Root component	inline unit/component occurrences and connectors, or one component definition; execution nodes/deployment	named closed occurrence; default service profile
Requirement	text	named obligation
Case	exchange steps, trace links	one root component and requirements
View	kind	one subject, no runtime semantics

All named elements have stable identity within a model version. Tools SHOULD persist a separate UUID identity map for editing; the language's normative textual identity is its qualified declaration/instance path. A rename is an identity-preserving edit only when the tool updates that map and all references atomically.

5. Conformance summary

A parser supports concrete syntax. A semantic validator additionally enforces static rules. A KerML mapper additionally produces the normalized representation in chapter 5. An execution backend additionally realizes chapter 3 and communication policies. A graphical editor additionally satisfies the view/editing contract. A tool MUST state which levels it implements; success at one level MUST NOT imply the others.

This repository supplies a parser, LSP foundation, documented static validation and a reference simulator. The specification defines more than that implementation, especially runtime delivery, idempotency, target snapshot realization, timing proof, interchange and graphical editing. See the exact support matrix.

NORMATIVE / LANGUAGE CONTRACT

1. Structure, names and types

Normative chapter of the SWAML 1.1 specification.

1.1 Lexical model

Source files MUST use UTF-8 and the `.swaml` extension. Whitespace and `//` line comments and `/* … */` block comments are insignificant outside strings. Block comments do not nest. Identifiers match `[_a-zA-Z][a-zA-Z0-9_]*` and are case sensitive. Every quoted word in the grammar is reserved. This version has no escaped identifiers.

Strings use double quotes and backslash escapes. A conforming implementation MUST support `\"`, `\\`, `\n`, `\r`, and `\t`; other escape sequences are invalid in conforming source. Raw control characters U+0000-U+001F, including line breaks and tabs, MUST NOT occur inside strings; use the permitted escapes. The lexer enforces this portable subset.

Unsigned decimal integers and decimal real literals are supported; a minus sign is a unary operator. Decimal real literals require digits on both sides of the point. Hexadecimal, exponent and suffix forms are absent. Duration literals consist of a positive integer followed by `ns`, `us`, `ms` or `s`. A duration MUST normalize to an exact integer nanosecond count no larger than 2^53-1. Time has no implicit unit.

The document contains zero or more packages, with no language header. The parser and its host MUST select SWAML 1.1 explicitly before loading the workspace; the CLI in this edition supports exactly 1.1. A host MUST NOT silently select whichever language version is newest. Historical `swaml "1.0";` headers are rejected by this grammar. An empty model is valid as an editing/library artifact, but contains no root component to execute.

1.2 Namespaces and resolution

Packages are top-level and non-nested in 1.1. A package name MUST be unique in the loaded model workspace. A package is declared in one file, although one file may contain several packages. Named elements in the same namespace MUST be unique; overload resolution is not supported.

`A::Type` identifies a package declaration. `import A::*;` exposes A's top-level declarations by short name in the importing package. Imports are not transitive. A short reference MUST resolve to exactly one declaration across the local package and its direct imports. Ambiguity is an error even when one candidate is local. Fully qualified references do not require an import.

Periods express containment/feature paths, not package qualification. A structural instance path follows actual child instances. A behavior communication path has exactly two identifiers: `localPort.member`. Directly following another unit's data or operation is forbidden. Pure record field paths follow only values already in the operation's local environment.

Local operation and function references resolve only in the owning unit. State names resolve only inside the owning operation's state body. Deployment nodes resolve only in the owning root component. No lookup fallback may expose a similarly named declaration from another unit.

1.3 Definitions and occurrences

`component def C` and `unit def U` declare reusable types. `component c: C;` and `unit u: U;` declare contained occurrences. The keyword MUST agree with the referenced definition kind. An instance has multiplicity exactly one in this version. Repeated instances require separate declarations.

A component definition owns ports, unit/component instances, connectors and delegation. A unit definition owns ports, private variables, functions and operations. `def` is required for each reusable component/unit type; it is never inferred from the presence of a body.

A package may also declare a **root component occurrence**: `component Demo { … }` supplies inline unit/component instances, connectors, execution nodes and deployments; `component demo: Assembly;` instantiates a reusable component definition. A typed root may use a body instead of its terminating semicolon, but that body MUST contain only execution nodes and deployments; its architecture comes entirely from its definition. For example, `component demo: Assembly realtime { node ecu mcu; deploy controller on ecu; }` binds existing descendants without adding, replacing or hiding them. Names declared in a typed root binding MUST NOT collide with the referenced definition's member names. Direct unit children are allowed in an inline root; an extra component wrapper is unnecessary.

A root has optional `service`, `realtime` or `mixed` profile following its name or type. Omission means `service`. It is a named occurrence, never a definition, and cannot be used as a child's type. Inline roots have no outward ports/delegations: they are closed execution contexts. For a typed root, all nonoptional definition ports and their aliases remain subject to closure; a required outward boundary port cannot acquire an invented external route.

The containment graph of component definitions MUST be acyclic. Every occurrence of a unit receives distinct variable storage and operation activations. Two instances of the same definition are distinct communication boundaries and still require ports between them.

Definitions are not runtime singletons. Defining an unused component or unit creates no runtime participant. Library definitions may leave their nonoptional ports unconnected until instantiated. A closed root component MUST connect every nonoptional port, including boundary aliases, after resolving delegation. An optional leaf cannot weaken a required boundary, nor the reverse.

Unit state is private to its instance. A variable has a declared type and a pure initializer. Initializers MUST be independent of other mutable variables and all communication. Operation/function parameters and local variables MUST NOT shadow unit variables or other live values in their scope. Blocks create nested lexical scopes; a variable declared in a branch does not exist after that branch. Parameters are copied input values; assignment changes the local copy only.

1.4 Value types

Type	Domain / semantics
`Bool`	`true` or `false`
`Int`	exact signed integers between -(2^53-1) and 2^53-1
`Real`	finite IEEE 754 binary64 values, with nearest-even arithmetic
`String`	finite Unicode text; no implicit encoding or transport size guarantee
`Void`	no result value; the default when an operation/signature omits `:` Type, never a variable, parameter, field, array or pure-function result
`datatype`	a finite record with named typed fields
`enum`	a nominal set of one or more unique literals
`T[n]`	a fixed array of exactly n values, n a positive safe integer

Datatypes and enums are nominal: two declarations with identical members are distinct types. Recursive records, including recursion through arrays, MUST be rejected. There are no pointers, references, nullable values, dynamic collections, aliases, generics or inheritance in 1.1. Bounded strings and machine integer widths may be added by future target profiles; they are not inferred from these primitives.

`record T { field = expression; … }` MUST assign every field of T exactly once and no others. `T.Literal` MUST name a declared enum literal; a package-qualified form is `Package::T.Literal`. `[e1, e2, …]` constructs a

nonempty fixed array; its elements MUST have one type, allowing numeric Int-to-Real promotion inside the literal. Arrays of differing lengths are incompatible.

Enum and local-value paths resolve deterministically. If the first segment of an unqualified dotted path is a live lexical/private value accessible in the current expression environment, the whole path is a local value read; a missing field is an error and MUST NOT fall back to an enum. Otherwise a two-segment `T.Literal` path resolves T among declarations in the current package and direct imports, checking ambiguity across all declaration kinds before requiring an enum. `Package::T.Literal` explicitly selects a loaded package enum without using the local-value environment. A wrong declaration kind, unknown literal, ambiguous name or array index applied to an enum is invalid. Pure functions expose only parameters to that environment, and variable initializers expose no mutable values. A unit variable therefore does not shadow an enum where that variable is inaccessible. No variant permits cross-unit private-value lookup.

Record fields may be read by `value.field.subfield`. `arrayValue[index]` reads one array element, with an Int index. Only one final index suffix is available in 1.1; a local variable can hold an intermediate array value. Bounds are zero inclusive to length exclusive. Out-of-bounds access fails the current operation at runtime and MUST NOT produce an undefined value. Whole-value assignment is supported; mutating a record field or array element in place is not part of 1.1.

Assignment, argument passing and return require the same nominal type and array length, with one exception: a scalar Int may widen to Real. Arrays do not implicitly widen. No numeric narrowing or conversion between enums and integers is implicit.

Numeric overflow, non-finite results, division by zero and invalid indexing fail execution. The parser/validator rejects out-of-range literal integers, but proving arithmetic ranges requires execution or additional analysis.

1.5 Expression rules

From lowest to highest precedence: `or`, `and`, `== !=`, `< <= > >=`, `+ -`, `* / %`, unary `not -`, primary expressions. Binary operators associate left. Parentheses override precedence. `and` and `or` short-circuit left-to-right. Other operands and call arguments are evaluated left-to-right.

Boolean operators require Bool operands. Ordered comparison and arithmetic require numeric operands. Equality compares equal nominal value types recursively (records by declared fields, arrays by ordered values), or two numerics after promotion. String equality is exact code-point equality with no normalization; no ordering or concatenation operator is defined. `%` is integer remainder with the sign of the dividend and requires Int operands. `/` always returns Real. Other arithmetic returns Real if either input is Real and Int otherwise.

A pure function has the syntax `function name(parameters): Type = expression;`. Its environment consists only of parameters, literals, record/enum/array construction and other functions in the same unit. It MUST NOT read unit variables, call operations, or communicate. Local call graphs MUST be acyclic in 1.1. `name(arguments)` evaluates the expression once with copied arguments and returns its value. The function's non-Void result annotation is mandatory. Parentheses distinguish a pure local function call from a value reference; no `compute` marker is required.

1.6 Requirements for tools

A reference that cannot be resolved, has the wrong target kind, or is ambiguous is an error. Tools MUST distinguish declaration paths, instance paths and local value paths even when their spelling is similar. A tool supporting rename MUST update cross-references and string-valued path references together; generic Langium rename alone is not sufficient for the latter in this prototype.

NORMATIVE / LANGUAGE CONTRACT

2. Interfaces and communication

Normative chapter of the SWAML 1.1 specification.

2.1 Contract identity and members

An interface declaration has a kind, package-qualified name, explicit `major.minor.patch` version string, and at least one member. Its identity is its declaration plus that version. Member names are unique across the entire interface; operation overloading is absent.

Kind	Permitted members	Port roles
`senderReceiver`	`data name: T sampled;` or `queued;`	`sender`, `receiver`
`clientServer`	operation signatures	`client`, `server`
`publishSubscribe`	event signatures	`publisher`, `subscriber`
`service`	operation and event signatures	`provides`, `requires`

An operation signature declares ordered input parameters, an optional result annotation, and optionally `idempotent`. Omitting `: Type` means Void, exactly as an explicit `: Void`. An event signature declares zero or more ordered payload parameters and no result. Data is one typed value. A notification without payload uses a zero-parameter event. No event may be declared directly in a package, unit, component or state.

A service field is modeled explicitly as a getter operation, optional setter operation and changed event. This avoids a second implicit communication mechanism. A setter returns an application result; its implementation owns the decision whether and when to publish the changed event. Property storage never becomes remotely writable data.

Version strings document compatibility intent: a breaking change SHOULD increase major; additive changes SHOULD increase minor; a meaning-preserving correction SHOULD increase patch. However, 1.1 performs **exact nominal matching**, not automatic version negotiation. Use a differently named declaration for another version in the same package and an adapter between versions. Adding an event may require a new receiver and is not automatically compatible with an existing complete implementation.

2.2 Ports and boundary exposure

A port belongs to one unit or component and references exactly one interface. Its role must be legal for that interface kind. `optional` permits the port to have zero connections in a closed root component; it does not relax type, direction, handler or emission requirements.

A connector joins immediate child ports of its owner: `connect name: child.port -> child.port qos { … };`. Its owner is a component definition or inline root component. A typed root uses the connectors of its definition. Both paths MUST have exactly two segments and reference ports, not members, operations, variables, states or grandchildren. The interface declarations MUST be identical.

Legal connector orientations are sender → receiver, client → server, publisher → subscriber, and requires → provides. The last two roles support provider-to-consumer event traffic over the same service contract; the arrow denotes the request orientation, not a unidirectional transport restriction.

A component port MUST have exactly one delegation to an immediate child port: `delegate outer -> child.port;`. Delegation MUST preserve role and nominal interface. It is a forwarding boundary, with no implicit queue, payload conversion, delivery retry or additional source. Child components can delegate again. To cross a hierarchy, expose a port at each boundary; no connector may tunnel to a grandchild.

At root closure, recursively expand instances, resolve each delegation to a leaf unit port, and inspect actual connector endpoints. Every nonoptional port, including boundary aliases, MUST resolve to a connected leaf. Receiver ports have exactly one source; client/requires ports have exactly one provider. Sender, publisher, server and provides ports can participate in multiple connectors. Subscriber ports may receive from multiple publishers. A duplicate connector resolving to the same directed leaf pair is an error; it must not create an accidental duplicate subscription. A component may not expose the same child port under two aliases in 1.1.

2.3 Emission and reception ownership

A unit's outgoing data/event member MUST occur in at least one `send`/`publish` statement inside one of its operations, including a state entry, exit or transition effect. This is a static provenance relationship, not a proof of reachability or a promise that the operation always emits.

Each incoming data/event member MUST have exactly one handler owner: an ordinary operation triggered by that member, or one stateful operation containing one or more receiving transitions for it. An operation signature on a server/provides port MUST have exactly one ordinary operation with `on call port.member`. Several transitions in one stateful operation count as one owner. Two different operations receiving the same member are invalid. To intentionally fan out processing, use a local coordinating handler that invokes helpers, or explicit subscriber units.

An ordinary handler MUST reproduce the member's ordered parameter types. Names may differ. A data handler has one parameter of the data type; an event handler has the event's parameter list. Both return Void. An RPC handler MUST also reproduce the signature's result type.

2.4 Exchange semantics

`send p.data(value);` emits one typed data value through a sender port. `publish p.event(args);` emits one typed event through a publisher/provides port. Both paths are exactly `localPort.member`. The transfer copies the value; a target implementation may use zero-copy storage only if it preserves isolation and observable value semantics.

For each outgoing permitted connector, the communication system independently offers the message to that connector's destination queue. Emission does not execute a remote handler on the caller's stack. All fanout offers are attempted. Any `reject` fails the emitting activation after those decisions; accepted offers cannot be rolled back. No interconnector offer order is implied. Programs needing atomic broadcast must model a protocol explicitly.

`request p.operation(args) as result within duration otherwise { … };` issues one correlated RPC to the single provider selected by its connector. The result name becomes available in the current block only after successful completion. The caller is suspended until a result or failure; this is an awaited request in the model, independent of whether generated code uses a blocking call or a future.

The communication system creates a fresh correlation identity for each request and carries its response back through the same interface/port/connector route. That response is the operation's declared result, not an independent event or unmodeled channel. A server return and a service event are distinct occurrences and have no implicit order beyond their operation's control flow.

The `otherwise` block executes on timeout, unavailable optional provider, admission rejection, transport failure or server `fail`. Its local `failure: String` is respectively `Timeout`, `Unavailable`, `Rejected`, `Transport`, or `RemoteFailure`. Remote implementation error strings are not automatically disclosed. Every path of the failure block MUST return from or fail the enclosing operation. No result is bound on failure. The implicit failure name must not shadow a live value. A Void success name labels completion only and cannot be used as data.

The timeout starts at issuance and includes admission, queueing, service execution and return transport. Timeout does not cancel committed server effects. A late response to that correlation is discarded. Automatic application-level retry is absent. A backend using retries to satisfy delivery policy MUST preserve

correlation and timeout budgets. Application-specific unsuccessful results remain typed results, not communication failures.

An outgoing optional unbound data/event port accepts an emission with no receivers; an unbound request fails as Unavailable. Incoming optional ports retain their handlers even if no external source is connected.

2.5 Queue and delivery policy

Every connector MUST specify all four base QoS fields, in the grammar's order:

Field	Meaning
`delivery bestEffort`	messages may be lost, no retransmission obligation; accepted unique transmissions are not deliberately duplicated
`delivery atMostOnce`	delivery may be absent, but a given logical message is presented to a handler no more than once
`delivery atLeastOnce`	admitted messages may be retried and duplicated; eventual delivery is required only under a live connection, available receiver and unexpired lifetime/budget
`ordering fifo`	per connector and per member, presented messages preserve sender emission order; duplicates may repeat an item before the next item
`ordering unordered`	presentation order is unspecified
`capacity n`	positive bound on buffered messages per connector, direction and member; executing activations are not in this queue
`overflow dropOldest`	remove the oldest buffered message and admit the new one
`overflow dropNewest`	discard the incoming message
`overflow reject`	refuse admission and signal the emitting activation

Overflow and expiration govern admission/retention and may intentionally discard messages even when delivery is atLeastOnce. This policy is therefore not a promise of lossless application execution. Reliability is separate from handler success; processing that fails is not silently repeated. AtLeastOnce RPC is permitted only when every operation on the interface is declared idempotent. A runtime/implementation MUST substantiate idempotency (including event side effects); the declaration alone is not a proof.

Exactly-once delivery is deliberately not a QoS option. Application-level deduplication and transactional processing can be modeled as operations and state, but depend on storage and failure assumptions outside the connector.

Sampled data retains only the latest pending value for that member. Any connector on an interface containing sampled members MUST use capacity 1 and dropOldest; this single connector policy also applies to other members of that interface. Use separate interfaces if mixed buffering policies are needed. Queued data preserves individual samples subject to explicit capacity/overflow. Receiving a sampled value is an activation with that value; there is no hidden cross-unit shared variable or implicit read operation.

Subscriptions are active while their endpoint units are alive. Pub/sub does not require a central broker; a logical connector describes the allowed subscription. Broker placement and behavior can be modeled explicitly. Runtime discovery, topic wildcards, content filters, retention persistence and dynamic subscription management are outside 1.1 and MUST NOT be assumed. Use explicit alternate root components or adapter components to document such deployment variants.

2.6 Timing of messages

`maxLatency` is a bound from accepted emission to availability for destination dispatch, excluding handler execution. `maxAge` is a maximum age since the producing emission timestamp; expired messages are discarded before dispatch. Clock translation across nodes is an implementation obligation. If a backend cannot establish a sufficiently bounded common time basis, it cannot claim the corresponding bound.

Request and response each have their own directional queue. A service event has its provider-to-consumer event queue. The connector QoS applies to each independently. Capacity is not shared between request, response and event traffic. A timed-out correlation does not keep its response queue alive indefinitely.

`maxLatency` and `maxAge` are obligations, not simulated delay values. Violations must be observable as runtime contract violations by a conforming checking executor. A declaration is not a measured proof. Security, authorization, integrity, serialization, encryption and transport binding belong to a target implementation profile and MUST preserve these interface boundaries. Chapter 6 defines admission, timeout races, lifetime and failure details.

NORMATIVE / LANGUAGE CONTRACT

3. Operations, flow and states

Normative chapter of the SWAML 1.1 specification.

3.1 Operation identity and activation

An operation belongs directly to a unit definition and is instantiated per unit instance. It has a name, ordered parameters, optional result annotation, optional trigger, optional timing contract, and exactly one body form: a statement block or directly owned states. Omitting `: Type` means Void; an explicit `: Void` has identical meaning. A component does not execute operations; execution lives in its contained units.

An **activation root** is one dispatched operation activation or state reaction and its synchronous local-call stack. It owns the execution lock/snapshot context inherited by helpers. This runtime term is distinct from a **root component**, which selects architecture and never executes statements itself.

Trigger	Activation and permitted signature
omitted	local `invoke`; the callable operation must return Void
`on manual`	parameterless Void activation activated by a test/administrative harness, carrying no model data
`on start`	parameterless Void operation, once after unit initialization and connector readiness
`on every D`	parameterless Void operation released at unit start + kD, for positive integer k
`on receive p.m`	copied data/event arguments, Void result
`on call p.m`	copied RPC arguments and exact contract result

Manual activation is a control of the modeling environment, not a way to inject an external data exchange without a port. A modeled UI command, interrupt payload, device sample, network event or external client MUST use an interface. Timers and initial activation are local lifecycle mechanisms.

Several ordinary operations may exist in one unit. There is no UML-style separate “method implementation of action definition” layer. The interface signature describes an RPC contract, while the unit operation triggered by that member is its implementation. A stateful operation directly contains its states. Reuse of algorithms is through functions or local operations, and reuse of architecture is through definitions and instances.

3.2 Flow execution

An activation holds its copied parameters, locals, a program position, its unit reference and any request correlation. Execute statements in source order. Blocks introduce lexical scope and discard block-local variables on leaving the block.

Statement	Semantics
`let x: T = e;`	evaluate e, typecheck, create a local mutable value
`x = e;`	replace an existing local/parameter/unit-variable value; no field or remote assignment
`invoke op(args);`	execute an untriggered local Void operation synchronously; return resumes the caller
`return;` / `return e;`	complete the current operation; return a copied value when required

Statement	Semantics
`if (e) { … } else { … }`	evaluate a Bool once and execute the selected block; else is optional
`while (e) bound n { … }`	evaluate the Bool before each iteration; at most n body executions
`assert e;`	require a Bool true; false fails the current activation
`fail "reason";`	terminate the activation with a local failure reason
`send` / `publish` / `request`	execute chapter 2's interface exchanges

Reaching the end of a Void operation completes it. A value-returning operation MUST return a value or fail on every path. If the loop condition remains true after n iterations, execution fails with a bound violation; the nth iteration is allowed. Local recursion is forbidden for both operations and functions. `invoke` cannot call a triggered operation, a stateful operation, or a non-Void operation. Use functions for side-effect-free returned values, and a local Void operation plus owned data for local imperative work.

Failure unwinds the current activation and its synchronous local callers. It does not undo earlier variable assignments, published messages, responses, or external effects. For an RPC handler, failure produces a RemoteFailure response through the originating interface. For a periodic/start/manual/receive activation it produces a local diagnostic occurrence, not an implicit broadcast. To notify another unit, model a fault event and publish it explicitly on a declared interface in an explicit branch before failing.

Conditionals, bounded loops, expression evaluation and sequential local calls are the full flow language in 1.1. There is no generic fork/join statement, exception hierarchy, arbitrary host-language body or independent action node. Concurrent architectural behavior is expressed by separately activated operations and connected units. A later fork/join extension must define data-race and cancellation semantics before entering the grammar.

3.3 Scheduling and concurrency

All ordinary operations and state-machine reactions are serial by default. The unit has one logical serial dispatch lock. A serial activation holds this lock until completion, including while suspended on a request; the logical lock is not required to consume a physical thread. A local invoke borrows its caller's lock and does not acquire a second lock.

A callback to the same unit while a serial request is pending cannot execute until the serial activation ends. Cycles of requests can therefore time out or deadlock absent timeouts; all requests have finite timeouts. The language does not silently release the lock and allow reentrant state mutation. Authors SHOULD avoid synchronous request cycles.

`concurrency reentrant` permits overlapping activations with separate local frames. Such an operation MUST NOT mutate unit data directly or through local helpers, and MUST NOT invoke an operation that mutates shared state. It may read the unit's state snapshot captured at dispatch, and functions receive copied parameters. A conforming backend must preserve that snapshot or reject the model if its target cannot realize it. The reference validator follows local helper calls to check this prohibition. Snapshot realizability remains a backend obligation.

Input queues are per connector/member. FIFO constrains order within one queue. Across independent ready queues, the scheduler may choose any ready activation consistent with deadlines and the target profile; no global total order is inferred. Periodic activations released while the serial unit is busy remain pending; a target profile MUST bound that activation backlog; exceeding the bound records ActivationOverflow and stops the selected root execution, as defined in chapter 6. Deadlines are measured from release, not eventual dispatch. Merely listing WCET/deadline does not define thread priorities.

The core semantics is therefore a set of allowed execution traces. An execution backend may add a deterministic scheduling policy that narrows this set but must document it. A model requiring determinism across sources must explicitly order exchanges or provide a coordinating operation.

3.4 Stateful operations

A stateful operation MUST have no parameters, result Void (explicit or omitted), and `on start`. Its body begins `{ initial StateName;`, followed by one or more states and zero or more transitions. There is no `machine` declaration or keyword. The grammar's `StateBody` only groups the owned syntax; it has no separate authored name, definition, allocation or runtime identity. `initial` MUST be first, so the body is unambiguously distinguished from a statement block. The behavior is mathematically a flat state machine with exactly one initial reference. State and transition names are unique within that operation's state body. Hierarchical/orthogonal states, history pseudostates, entry/exit points and state inheritance are absent in 1.1.

Starting the operation enters its initial state and executes that state's entry block once. The operation remains alive between reactions without holding the serial unit lock. Each entry/exit/transition reaction is scheduled as a serial activation unless its timing contract explicitly says reentrant; a stateful operation nevertheless MUST have at most one reaction in progress and cannot reenter its own state machine.

A state owns an optional entry block and optional exit block. A `final` state completes the operation after its entry finishes and MUST have no outgoing transition. Ordinary completion is not an event emitted to another unit. A notification on completion must be published explicitly before completing.

3.5 Transition triggers and selection

A transition names source and destination states in the same operation. It has one trigger and optional Bool guard and nonnegative safe integer priority. Missing priority means 0; larger values take precedence.

- `on receive port.member` consumes one incoming data/event for the machine. Event parameter names are available as copied values in the guard and effect. Data has the local name `payload`. These names MUST NOT collide with unit variables. RPC operations are not allowed as state transition triggers.
- `on after D` becomes eligible after D elapsed time since completing the source state's entry block. Leaving the source cancels its timer. Reentering starts a fresh timer.
- `on completion` is eligible after the state's entry and any prior reaction have finished, before waiting for a new external message.

For one stimulus, first select transitions from the current state with matching trigger, then evaluate guards, then choose the highest-priority enabled transition. Potentially competing transitions with the same source and identical trigger MUST have distinct priorities even when an author believes their guards are mutually exclusive. A missing guard means true. Unrelated queued stimuli are dispatched one at a time, in the scheduler order permitted by 3.3.

A received message with no enabled transition is consumed and ignored. A timer with no enabled transition expires once and is not retriggered until reentry. Completion transitions are reevaluated after each successful state entry and each machine reaction, including an ignored stimulus. Models MUST NOT contain unconditional completion cycles. Guarded cycles must reach quiescence; a checking executor MUST fail a reaction after 1,024 completion microsteps rather than spin forever. This limit is a semantic protection, not a proof of useful behavior.

3.6 Transition execution

A selected transition executes atomically with respect to this unit's other serial reactions:

1. Consume the matching stimulus, if any, and cancel all timers for the source state's entry generation.
2. Run the source state's exit block.
3. Run the transition effect with the stimulus payload in scope.
4. Update the current-state identity to the destination.
5. Run the destination state's entry block.
6. Complete if final; otherwise evaluate completion transitions or await a stimulus.

Entry and exit blocks do not inherit event parameters; only the guard/effect does. Copy a needed payload to unit data in the effect if entry behavior needs it. A self-transition exits and reenters, including entry/exit effects and resetting timers.

`return;` in any state block terminates the owning stateful operation immediately and cancels pending timers; remaining transition phases are skipped. `fail` terminates it with failure and also cancels timers. Neither rolls back prior side effects. Other queued messages still belong to the unit, but a terminated machine no longer dispatches its members; a target runtime must record these as undeliverable until unit restart.

Every state MUST be reachable from the initial state in the transition graph, ignoring guards for this static check. Reachability through satisfiable guards and eventual completion are stronger analysis obligations, not consequences of graph reachability. No state, operation or component may use an unbound global event to change another unit's behavior.

The operational contract further defines initialization barriers, snapshots, timer generations, stimulus ties, failure and resource bounds.

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4. Root components, timing, cases and conformance

Normative chapter of the SWAML 1.1 specification.

4.1 Root components and profiles

A root component is the closed occurrence selected for validation, deployment and execution. It uses the same component concept as reusable architecture: an inline `component Demo { … }` owns unit/component children and connectors, while `component Demo: Assembly;` obtains them from a component definition. Several named roots in one package may instantiate the same definition independently or describe alternative assemblies. Root selection is explicit in the simulator/host by package-qualified name, and is not inferred from file order.

A typed root's optional body is a deployment binding only: nodes and deployments may reference existing descendants; instances, connectors, ports and delegation MUST NOT be added there. An inline root may contain direct unit instances because the root itself supplies their component boundary. All paths in cases, deployment and simulation are relative to the selected root; the root name is not an extra child segment.

Instantiation recursively creates children and local variable storage, resolves delegation and connectors, checks closure, and then enables startup/periodic activation. No unit may publish during a partially connected startup phase.

The root profile is `service`, `realtime` or `mixed`; omission means `service`. Only the service profile defaults; QoS, interface versions, timeouts and realtime obligations remain explicit:

Obligation	service	realtime	mixed
Explicit connector QoS	required	required	required
Operation WCET/deadline/concurrency	optional	required for all used operations	required for all used operations
Connector maxLatency	optional	required	required
Sampled data maxAge	optional	required	required
Interface isolation and handler checks	required	required	required
Production scheduling evidence	target dependent	required for hard realtime claims	required for realtime claims

The mixed profile is intentionally conservative: all used operations and connectors carry timing contracts. It does not infer that a cloud-facing operation has no timing requirement. A future subassembly profile can relax this with explicit partitioning; no such implicit relaxation exists in 1.1.

A timing contract declares positive `wcet`, positive `deadline`, and `concurrency serial|reentrant`. WCET MUST be no larger than deadline. For periodic operations, deadline MUST be no larger than the period. A stateful operation's timing contract applies separately to each run-to-completion reaction, including entry/exit/effect and any completion microsteps, not to the lifetime of the machine.

WCET is active CPU execution time for the activation including synchronous local helpers. Deadline is elapsed time from activation release to completion, including waiting and interference. Connector maxLatency excludes destination dispatch wait. For a chain of operations, the end-to-end bound must

account for release delay, queue delay, interference, transport, WCET and response delay; simply summing WCET is insufficient.

A conforming validator checks arithmetic consistency. A conforming target backend must additionally prove or measure its declared timing assumptions with an appropriate execution model. The core specifies no rate-monotonic/EDF policy, task mapping, interrupt priorities, memory size, hardware access mechanism or clock synchronization algorithm. Such a backend MUST supply a named external deployment profile, and MUST NOT claim timing conformance without evidence for those obligations.

4.2 Deployment

An execution node is named and has kind `mcu`, `process` or `container`. A deployment maps an existing immediate or descendant unit/component instance path onto a node declared in the same root component. It does not reference a definition. The same explicit instance path may occur in only one deployment.

A component deployment applies to all its descendants by default. A deployment of a more specific descendant overrides that inherited location for its subtree. If no deployment exists for an instance or ancestor, the instance is logically modeled but unallocated. An unallocated architecture can be valid; an executable target configuration cannot silently pick a node and call it specified.

Deployment does not change interfaces, connector identity, callback ownership or private-data visibility. Interprocess or intermachine implementations must preserve value and correlation semantics. Serialization formats, discovery identifiers, service addresses, credentials, broker topics and physical channels belong to a separate target binding document. Any exported backend MUST record the SWAML element identity for every generated mapping.

4.3 Requirements and cases

`requirement Name "text";` declares a named natural-language obligation. `case Name for RootComponent satisfies Requirement, … { … }` defines a scenario and optional trace links. A `satisfies` link is an assertion of intended coverage, not verified fulfillment. Verification evidence belongs to a test, analysis or review artifact linked by an implementation tool.

A case step has a kind (`send`, `publish`, `request`, `reply`), two endpoint-member paths, a named connector at the selected root composition level, an optional earlier-step predecessor and a unique label. Case endpoints must be visible ports of immediate child units/components. A typed root resolves those children and connectors through its definition. Internal interactions at a deeper level require a separate root component assembled at that abstraction level. This limitation keeps case paths subject to the same hierarchy rule as structural connectors.

Both endpoints MUST identify the same member of the connector's interface. Sender-receiver steps use data, publish steps use events, and request/reply steps use operation signatures. A request follows client → server or requires → provides orientation; a reply reverses the matching request. A service publish goes from provides → requires; ordinary pub/sub follows publisher → subscriber.

`after label` imposes a happens-before dependency on an earlier declared step. No predecessor means no ordering is imposed relative to other steps. Source order is only a stable display tie-break, not an implicit total order. Thus unrelated notifications can remain concurrent in SV. A reply MUST have exactly one unanswered matching request among its transitive `after` ancestors, with reversed endpoints and the same connector/member. Zero or multiple candidates is an error (SW083); each request has at most one reply. A request may remain unanswered in a partial trace. A prior reply closes a candidate only when that reply is itself an ancestor of the current step. A reply on an independent causal branch cannot resolve ambiguity merely by appearing earlier in the file. Source order never chooses between ambiguous correlations. Realization by operation bodies is a separate analysis claim requiring a witness; route-valid cases remain unverified until that evidence exists.

Cases describe representative traces, not exhaustive behavior. They do not execute a second program, introduce independent event definitions, or loosen operation guards. A complete behavioral verification must check traces against operation semantics and environmental assumptions.

4.4 View declarations

A view declaration names a view kind and a dot-separated subject in its own package. GV and GRV may reference any resolvable named semantic declaration or the owning package, except another view; IV requires a root component, a component definition or a nested component occurrence; CV references a case or root component, SV references a case or root component (for case selection), and FV references an operation. A view of an instance resolves through its definition while retaining the occurrence path and identity.

View declarations carry no executable behavior. Layout, color, filters and collapsed regions are presentation data. The full graphical contract defines representation and source-edit invariants.

4.5 Static validity rule groups

The following groups collectively define semantic validity. Diagnostic numbers identify the present implementation's related checks, not a license to omit unimplemented clauses.

Group	Required checks	Prototype family
Names	pinned language edition, uniqueness, imports, reference target kind, ambiguity, lexical scopes	SW002-SW004, parser/linker
Contracts	interface kind/member compatibility, role compatibility, exact interface identity	SW010-SW013, SW031-SW032
Values	nominal typing, arity, field completeness, enum membership, array shape, literal ranges, purity, no recursive value containment	SW020-SW028
Architecture	acyclic composition, immediate-child boundaries, one delegation, no duplicate links/aliases, closure/cardinality	SW030-SW035
Time/QoS	positive/bounded values, explicit policy, sampled buffering, WCET/deadline/period, realtime completeness	SW040-SW045
Behavior	local-only calls/data, outgoing provenance, signatures, unique handlers, return coverage, terminating error paths, bounded loops, no recursion	SW050-SW059
States	legal stateful signature, final-state rules, priority, reachability, completion-cycle restrictions	SW060-SW064
Allocation	existing instances and nodes, no duplicate exact deployments	SW070
Presentation/cases	subject kind, connector/member paths, happens-before and uniquely correlated replies	SW080-SW083

4.6 Conformance levels

Syntax conformance (S): the complete concrete grammar, pinned host/parser version, tokenization and AST construction. Recovery during editing must retain syntax diagnostics; an incomplete AST is not a valid model.

Static conformance (V): S plus every static validity rule in this specification, classified explicitly in chapter 8. A partial validator MUST enumerate checks that it lacks. It must not hide unresolved links or a validator crash as success.

KerML mapping conformance (K): V plus chapter 5's normalized elements and preserved SWAML semantic constraints, with a tested mapping against KerML 1.0 and its selected libraries. Export must retain a reverse trace map and stable identity. K does not automatically imply SWAML execution.

Execution conformance (E): V plus the operational, correlation, queue, timing-observation, failure and concurrency semantics. Target binding assumptions must be explicit. A simulator may abstract physical time while still reporting which timing obligations it does not check.

Graphical conformance (G): S and the graphical specification's projection and edit rules. G may present invalid models with diagnostics, but must not invent valid connectivity from layout. FV projects operation-owned statement flow and state behavior.

A tool claiming `SWAML 1.1 S, partial V` accurately describes this repository's current core. It must not advertise K, E or G until those implementations and checks exist. A language host must require an explicit supported edition, and incompatible changes require documented migration. Version 1.1 is an intentional syntax revision; it is not backward-compatible with 1.0 despite the minor version number. Migration tooling must not change the meaning of models silently.

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5. KerML semantic mapping

Normative SWAML mapping contract, with implementation status stated explicitly. Part of the SWAML 1.1 specification.

5.1 Foundation and boundary

SWAML defines software-specific specializations and constraints over a normalized KerML model. Its concrete syntax is independent: a `.swaml` file is not automatically valid `.kerml` text. The Langium AST is an authoring representation, not itself the KerML abstract syntax.

The selected foundation is OMG KerML 1.0. KerML supplies structural classifiers/features, behavior classifiers/steps, connectors, successions and expressions. The relevant normative kernel concepts are in clauses 7.3-7.4 and their abstract-syntax/semantic counterparts. SWAML's mapping below is this project's proposed specialization contract, not an OMG-approved extension. KerML specification.

The mapping inventory is a machine-readable design contract. This release does not implement or validate a KerML exporter, does not bundle a kernel semantic library, and does not claim K conformance. Implementers must test the normalized representation against the selected KerML metamodel and library before making that claim.

5.2 Semantic specializations

SWAML element	Proposed KerML base/representation	SWAML restriction or additional meaning
Package	Package	flat authoring namespace
Component definition	Structure	only composition, interfaces and connectors
Unit definition	Structure	encapsulated state and local executable features
Root component	composite Feature typed by a referenced or synthetic Structure	selected named occurrence, closed topology and effective execution profile; typed binding cannot extend topology
Component/unit instance	composite Feature typed by its Structure	exactly-one multiplicity and acyclic composition
Value datatype / enum	DataType	finite value semantics; enum domain restriction
Interface contract	Classifier	nominal version, legal members and communication mode
Port	Feature typed by contract Classifier	direction/role projection and boundary ownership
Operation signature	Behavior plus typed parameter Features	no implementation; copied request/result contract
Operation implementation	Behavior and an owning-unit executable Step typed by it	direct ownership, activation, serialization and timing
Pure function	Function	parameter-only environment and no side effects
Connector	Connector between port Features	permitted interface exchange and QoS obligations
Delegation	Connector plus identity-preserving forwarding constraint	same role and contract; no extra queue

SWAML element	Proposed KerML base/representation	SWAML restriction or additional meaning
Ordered flow	contained Steps and Successions	structured branches/loops, bounded execution
Message emission	Step typed by a SWAML-library communication Behavior	source activation, route and correlation metadata
State / transition	contained behavior steps, state predicate and succession relations	SWAML state-selection and reaction semantics
Requirement / case / view	annotated named elements and trace memberships	SWAML-specific interpretation

The table establishes a mapping target, not a claim that names such as “SWAML state” already exist as native KerML metaclasses. A full exporter needs a versioned SWAML semantic library for the additional predicates, activation and communication constraints. Chapter 7 completes the information model, reference edges and lowering rules for every syntax family; the inventory covers all 80 grammar rule/type names.

5.3 Normalization algorithm

A mapper MUST perform the following in dependency order:

1. Validate the complete workspace and resolve every textual reference. Allocate stable IDs for package declarations and each named semantic element. Retain file URI and source range as provenance, not as identity.
2. Create value types and interface classifiers. Materialize every data/event/operation parameter and result as a typed feature with copied-value semantics. Preserve interface version and member order.
3. Create component/unit structures and owned instance features, plus synthetic structures for inline root composition, with membership and typing links. Each root is a named occurrence of its referenced or synthetic structure; a typed root has no extra wrapper child. Preserve source profile omission in provenance and effective service in semantics. Keep definition identity separate from every instance-path identity. A reused definition must not merge the states of its occurrences.
4. Create each port feature in its owning structure. Apply role-specific direction to each member projection: RPC inputs/results have opposite directions; service events run provider-to-consumer. Do not assign a single simplistic “out” bit to an entire bidirectional service contract.
5. Create operation behaviors within their unit's ownership. Add a synthetic owned executable step typed by that behavior and a unit-instance context binding. This lowering artifact gives execution a structural context; it is not a user-visible performer or a second editable allocation model.
6. Lower statement blocks to structured steps with explicit sequencing. A branch has a Boolean selection predicate and branch result control; a bounded loop has condition, body, count and failure edge. Lower expressions as typed expressions/functions. A mapper must preserve source order and short-circuit evaluation.
7. Lower each stateful operation to one state identity feature per activation, nested state-entry/exit/effect behaviors, trigger predicates, priority selection and succession relationships. Preserve finality, timer start/cancel and the run-to-completion relation. A bare succession graph without these constraints is not an equivalent mapping.
8. Lower communication statements to message occurrences linked to the emitting activation, sending port, interface member, connector/delegation route and destination port. Requests additionally carry correlation, timeout and response/error continuation. An event cannot become an unowned signal merely because a kernel target permits it.
9. Materialize queues, policy constraints, timing/deployment annotations and case happens-before relations in the SWAML semantic library vocabulary. Keep queue identity per connector/direction/member.

10. Produce a reverse mapping from every generated element to its authoring element, including synthetic lowering elements. Validate kernel typing/membership constraints and SWAML constraints separately. Report failures at the source element, not only at generated IDs.

5.4 Required extension predicates

The semantic library MUST define, or a mapper MUST carry as explicit constrained metadata, at least:

- `ownsExecutable(unit, operation)` and instance-local `contextOf(activation)`;
- `contractOf(port)`, `roleOf(port)`, `memberOf(contract, member)`;
- `route(sourcePort, destinationPort, connectorPath)`;
- `emits(activation, port, member, copiedPayload)` and `handles(operation, port, member)`;
- `correlates(request, response)`, `expiresAt(request)` and `lateResponseDiscarded(response)`;
- `queueOf(connector, direction, member)`, capacity, overflow, ordering and delivery policy;
- `currentState(activation)`, `enabled(transition, stimulus)`, priority and final-state completion;
- `wcet(operation)`, `deadline(operation)`, `period(operation)` and instance allocation;
- `happensBefore(caseStepA, caseStepB)` and `tracesTo(case, requirement)`.

These predicates are part of SWAML meaning even if an exchange format stores them as metadata. A consumer lacking their interpretation MUST report a partial import and MUST NOT execute a “best guess.” A generic KerML consumer can preserve structural content while lacking SWAML runtime semantics.

5.5 Interchange and round-trip obligations

A future exporter SHOULD use the standard KerML interchange representation supported by its chosen backend, with a declared SWAML semantic library version. No new unregistered MIME type, XMI metamodel URI, or fake OMG namespace is assigned by this document. SWAML library identifiers must use a project-controlled namespace when published.

Round-trip tests MUST compare semantic identities, typing, owned instances, routes, triggers, guards, flow ordering, queue policies, timing and trace dependencies. Byte-for-byte textual equality is not required. Comments, formatting and diagram layouts are separate preservation requirements for an authoring tool.

Dropping a guard, converting a queued event to a sampled value, merging per-subscriber queues, or omitting a request's response path is a semantic loss and must be reported as an error for K conformance. The mapping must not infer a performer allocation that changes the operation's direct owning unit.

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6. Operational contract and target bindings

Normative chapter of the SWAML 1.1 specification. This chapter makes the allowed traces precise; it does not prescribe an implementation thread model. The reference simulation binding makes a deterministic logical-time choice within this contract; production backends require their own binding and evidence.

6.1 Configuration and observable occurrences

A running configuration consists of: the selected root component instance tree; private variable stores per unit instance; live activation frames and their local call stacks; one serial lock per unit; the state and timer generation of each live stateful operation; message queues keyed by connector occurrence, direction and member; pending request correlations; released lifecycle activations; and a monotonic logical time.

Each message has a logical identity, original emission time, producing activation identity, typed copied payload, source/destination port occurrences, interface member, connector occurrence and delegation route. A retry keeps the logical identity and original timestamp. A transport attempt has a separate identity. An RPC response additionally references its request correlation. Occurrence identities MUST be unique within a run and cannot be declaration IDs reused for every emission.

The minimum observable trace alphabet is initialization, release, dispatch, statement completion, emission, admission, drop, delivery, request completion, timeout, state entry/exit, activation completion/failure, and contract violation. Each record includes logical time, occurrence identity and relevant source identities. A backend may omit detailed statement traces in normal operation, but MUST provide sufficient checking instrumentation to diagnose violated obligations. Trace records and diagnostics are observations, not implicit interface messages.

A step of execution is either a local statement/expression step, communication admission/delivery, dispatch, reaction microstep, or time advancement. Independent units may step in either order. No text-file order between units creates a happens-before edge. Elapsed physical execution time must be represented when checking deadlines; treating each statement as zero-time cannot establish production timing compliance.

6.2 Initialization and release

Initialization has a barrier with the following phases:

1. Expand the acyclic instance tree, establish effective deployments, create private stores from pure initializers, resolve all routes and install all queues/handlers.
2. Install each stateful operation's initial state identity and arm its initialization reaction. It is not yet ready to consume a stimulus. Register all ordinary startup activations.
3. Establish one run epoch `t0`. Release every startup/initialization activation at `t0`; activate periodic releases at `t0 + k * period`, with k starting at 1. All ports may now admit traffic.

The scheduler may interleave startup activations across units and within a unit subject to its lock. Messages arriving before a machine finishes initial entry remain in their connector queue; they are not discarded merely because initial entry is pending. Timers start when entry finishes, not when the initialization reaction was released. A startup operation may request a service whose unit's startup activation has not run: handlers and initialized data already exist. If an application needs an additional readiness protocol, it MUST model that protocol through an interface.

Each startup activation is released exactly once per root run. The core defines no automatic component restart. A deployment implementation restarting a failed unit starts a new run epoch for the affected binding and MUST report its reset, queue-retention and correlation policy; it cannot imply uninterrupted E conformance for a reset it has not modeled.

Periodic releases are independent of earlier completions. Serial dispatch does not merge or silently skip them. Manual activations are parameterless local harness releases. Input-triggered work remains in the input queue until dispatch is possible; it is not copied into an unbounded second activation queue. Admission of an RPC response resumes its suspended frame without acquiring another unit lock.

6.3 Dispatch, backlog and snapshots

A serial activation root acquires its unit's lock at dispatch and releases it on completion/failure. Local invokes execute on the activation root's stack and inherit its concurrency mode and snapshot, regardless of a helper's own timing declaration. A reentrant activation root may overlap other activation roots, but neither it nor any reachable helper may assign unit data. Assigning its own parameters/locals is allowed.

A reentrant activation root captures an atomic value snapshot of all unit variables at dispatch, before its first expression. All reads by that activation and its helpers use that snapshot throughout suspension and resumption. An assignment by a serial activation is atomic at the whole-value level; snapshots observe some consistent store between assignments. A machine's next reaction captures a fresh snapshot if it is reentrant. Reentrant does not allow overlapping reactions of the same machine.

Between independent ready queues and lifecycle releases, the scheduler chooses any enabled activation root. For a FIFO queue, dispatch preserves queue order; completion order of reentrant handlers need not match dispatch order. A target backend MUST specify fairness and scheduling policy, and MUST NOT introduce starvation while claiming eventual at-least-once delivery under continuously enabled receivers.

An executable target binding MUST bound pending start/manual/periodic activations separately from connector queues. On exceeding that bound it MUST record `ActivationOverflow` and stop the affected root execution; silently dropping or coalescing releases is not an implementation of this core. A simulator may expose an explicitly unbounded mathematical backlog, but cannot use it as evidence for a bounded production deployment. The binding also bounds simultaneous reentrant activation roots; when full, subsequent work remains pending rather than exceeding the limit.

WCET includes local helpers and active reaction microsteps; request wait consumes deadline budget but no active CPU budget. The activation root's deadline is measured from release. For input reactions release occurs when the message is ready in its destination queue. A timer reaction is released at its due time even if dispatch is later. Local helpers have no independent release; a backend checks their declared execution bounds over the invocation and also includes them in the activation root's bounds.

6.4 Queue offers and fanout

`send` and `publish` evaluate all arguments once and snapshot the matching active connector set. They offer a distinct per-connector logical message to **every** connector in that set, even if an earlier offer rejects. Offers are independent and have no specified interconnector order. The statement finishes after all local admission decisions. If any offer rejects, the activation fails with an admission failure; successful offers remain committed. Loss through dropOldest/dropNewest is not rejection and does not fail the sender. Atomic broadcast requires a modeled protocol.

For a given queue, offer and dispatch are atomic decisions. Capacity counts retained messages waiting for dispatch, including ones whose transport availability is pending; it excludes running handlers. Removing an old message or discarding a new one MUST be observable to a checking executor. An optional port with no connectors produces no offers and succeeds for data/events.

Condition	Data/event emission	RPC request	RPC response
No live endpoint	optional unbound succeeds; failed bound route reports transport failure	`Unavailable` if unbound; otherwise `Transport` when detected	requester gets `Transport` if detectable, otherwise timeout

Condition	Data/event emission	RPC request	RPC response
Full queue with reject	producer activation fails after all fanout offers	caller executes `otherwise` with `Rejected`	requester gets `Rejected` if the binding can report it; otherwise timeout; server's completed effects remain
dropOldest/dropNewest	dropped occurrence is observable; statement can succeed	dropped request waits until timeout unless transport reports failure	dropped response leaves request waiting until timeout
Age expired	discard before dispatch	no service dispatch; caller eventually times out unless already failed	discard; no successful completion
Receiver handler fails	receiver records local failure; sender is not retrospectively failed	correlated `RemoteFailure`	not a second handler; resume request failure path

Control failures such as admission rejection use the same request route/correlation. They are transport outcomes of that interface exchange, not arbitrary model event channels. A binding unable to signal a failure cannot fabricate a successful result.

An admitted message becomes ready no later than emission plus maxLatency when such a bound is claimed. Deadline miss is a violation even if the message is later delivered; it is not itself a new delivery policy. `maxAge` eligibility is inclusive: age <= maxAge may dispatch; age > maxAge is expired. Discarded/expired messages have no eventual-delivery obligation. Without maxAge, asynchronous messages have no core expiry before connection teardown; a backend must maintain its delivery obligation or explicitly report loss of availability. Retry policy, finite memory and resource exhaustion behavior belong in the target binding, not undocumented defaults.

6.5 RPC completion and races

A request correlation has states Pending, Succeeded or Failed. Creation precedes the first request offer. Its expiry is issuance time plus timeout. At most one terminal completion may win. A response can succeed only if it is observed at the requester at a time **strictly earlier** than expiry and the correlation is Pending. At expiry timeout wins ties. Terminal state is immutable; later responses and duplicate replies are discarded and observable.

AtLeastOnce may dispatch multiple handler activations with the same logical request/correlation. Idempotency must cover storage writes and published side effects; returning the same value alone is insufficient. The first eligible response completes the caller. Retrying must never create a new caller-visible correlation or extend its timeout. A binding may deduplicate repeated requests while preserving required behavior.

The mandatory `as name` on a Void RPC labels successful completion without introducing an inhabitable value. It reserves that name in the enclosing lexical scope but it MUST NOT be read or assigned as data. A Void handler uses bare `return;`. The failure block has an implicit copied `failure: String`; an existing live value with that name is a static error. Failure blocks cannot fall through to the successful continuation.

A timeout does not retract messages already sent or cancel a server frame. Pending requests cannot be used as detached futures; each statement awaits its own outcome. Synchronous dependency cycles remain expressible but may time out. A static validator is not required to prove their absence; an execution/timing claim must account for them.

6.6 State reaction ordering

A machine has one initialization reaction and then a sequence of reactions. Each reaction includes all immediately enabled completion transitions until quiescent/final/failure. Completion selection precedes dispatch of the next external stimulus. If no completion guard holds, the machine becomes quiescent. A write by another operation does not itself awaken completion transitions; they are reconsidered after the next machine reaction, including one that consumed an unhandled message or expired a timer with no enabled transition.

On completing a state's entry, create one timer stimulus per distinct `after` duration on its outgoing transitions, tagged with the current state-entry generation. All transitions with that duration compete using guard/priority. Distinct durations produce distinct stimuli. Leaving a state cancels every timer from that generation. A queued timer from an obsolete generation is discarded. Reentering, including a self-transition, creates a new generation. Entry code time is outside the residence timer; the timer measures time since entry completion.

A due timer and a ready input are independent stimuli. Their selection order is unspecified unless the target scheduler constrains it. Transition priority selects among transitions for the **same** stimulus; it is not a task/queue priority. A consumed stimulus is not retained merely because all guards were false. Entry/exit blocks have no event payload scope. Exit and effect run before state identity changes; destination entry follows that change.

The completion microstep counter is zero at the start of a reaction and increases once per taken completion transition. Taking a 1,025th completion transition fails the reaction before executing that transition. External dispatch does not occur between those microsteps. Guards are pure expressions and cannot send, mutate, or consume another message. A guard evaluation failure fails the owning activation.

Final entry completion, bare return or failure terminates the machine, cancels all its timers and disables its dispatch. Future messages to its members become undeliverable occurrences; they are not rerouted to another operation. Other operations in the same unit may remain alive. No implicit exit behavior runs on return/failure unless it has already been reached in the specified transition phases.

6.7 Execution binding contract

Every backend claiming E MUST identify its backend version, supported SWAML version, selected root component, model digest and target configuration. It MUST supply explicit values or a justified symbolic abstraction for:

- effective deployment of every unit; clock basis, precision and cross-node error bounds;
- scheduler, fairness, serial locking, snapshot implementation and reentrant limits;
- lifecycle backlog limits and overflow observation;
- transport, serialization, copying/isolation, queue storage and retry/deduplication policy;
- deadline/latency/age measurement points and what happens after a reported violation;
- external adapter behavior and assumptions, startup readiness, shutdown/reset and queue retention;
- resource failure handling, correlation uniqueness, observability and evidence artifacts.

Timing/resource failure policies may stop execution after recording a violation. Continuing is permitted only with the violation visible, and cannot count as a satisfying execution for the violated contract. An execution trace is conforming when all observed steps obey core transitions; satisfying a particular model additionally requires its declared assertions and contracts to hold. A runtime that correctly reports a failed assertion is not thereby a broken interpreter.

Security and target-specific bus/OS policies can strengthen this contract, but cannot add hidden model exchanges. AUTOSAR Classic task/RTE mappings, Adaptive service bindings and broker configurations must preserve the same identities, member directions and copied-value boundaries.

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7. Normalized semantic model

Normative SWAML contract extending chapter 5. This chapter defines the complete information a mapper must preserve. It specifies SWAML normalization, not a new OMG metamodel or an implemented serializer.

7.1 Identity, ownership and source trace

Normalization consumes a resolved workspace and emits a finite declaration graph plus instance expansions for selected root components. Each semantic node has an immutable identity within that normalized artifact, a semantic kind, at most one owning node, ordered owned children where order matters, typed reference edges, and source provenance. Except for model roots, every semantic node has exactly one owner. Reference edges never imply containment.

The deterministic textual identity of a declaration is its package plus owned-name path, for example `Settings::SettingStore.update`. Anonymous syntax nodes use the owning declaration ID plus a structural slot path such as `/body/statements/2/condition`; these IDs are stable for an unchanged syntax tree, not across arbitrary edits. Synthetic elements append a reserved role segment such as `/@execution` or `/@failure`. No source identifier contains `/` or `@`, so generated roles cannot collide with source names.

An occurrence identity consists of the root component declaration ID and the child-instance path. A port occurrence adds the owned port name. An operation occurrence additionally refers to its shared operation declaration and unit occurrence. Runtime activations add a run ID and unique activation ordinal. A mapper MUST NOT collapse occurrence paths merely because they share a definition. Persistent editor UUIDs are a separate identity map with explicit rename/move history; source locations and in-memory AST addresses are never persistent IDs.

A reverse trace relation maps every generated element to one or more authoring node identities and their URI/ranges. Synthetic nodes retain their source owner and generation role. No generated constraint may be untraceable. An import can have an import-declaration source location while referencing a package located in another document.

7.2 Declaration graph and links

Normalized family	Required information and constraints
Workspace/package	language version, packages, direct import edges, unique declaration namespaces
Record/enum	nominal identity, ordered fields or ordered unique literal names, finite domain/containment
Type use	primitive or nominal target, optional array extent; never encode a type solely by a display label
Interface	kind, version, ordered data/event/RPC members, signatures, data mode and idempotency claims
Structure definition	component/unit kind, owned ports, contained instance features, private variables, owned executable declarations as applicable
Instance feature	component/unit kind, exactly one definition target, composite ownership, multiplicity one
Port feature	interface target, role, optionality and derived member directions
Connection/delegation	owner, endpoint feature references, explicit path provenance, interface identity, queue policy for a connection; no queue for delegation

Normalized family	Required information and constraints
Variable/local/parameter	declared type, ownership/scope, initializer where present, mutable copied-value semantics
Executable	signature, trigger, concurrency/timing, one statement or state body, context bound to the owning unit
Root component/binding	named occurrence with inline composition or component-definition reference, execution nodes, explicit/effective deployment and effective service/realtime/mixed profile
Requirement/case/view	text, trace edges, scenario step graph, view kind and resolved subject with definition/occurrence distinction

All textual references MUST resolve before normalization. A `TypeRef` is a use, not a new nominal declaration. Array extents are carried as shape constraints and do not create element aliasing. Empty records have one value. Void results create no output-value feature. Parameters preserve order, independent of parameter names at a handler implementation.

A typed root expands its component definition, and an inline root expands its authored children; no extra wrapper occurrence is inserted. Its normalized profile records service even when the source omits that default, while provenance preserves the omission. A typed binding body contributes only nodes/deployment, never topology. Every component child expands its definition's child features recursively. Expansion preserves each connector's owner path, so two uses of the same component produce distinct connector occurrences and queues. Effective deployment is computed by the nearest explicitly deployed ancestor, including the occurrence itself. An absent ancestor yields an explicit Unallocated value, not an invented node.

7.3 Contract projection and route normalization

For every port occurrence and interface member, derive a member projection with contract/member identity and role-specific directions:

Role	Data/event projection	RPC request	RPC result/failure
sender / receiver	out / in data	absent	absent
publisher / subscriber	out / in event	absent	absent
client / server	absent	out / in	in / out
requires / provides	in / out event	out / in	in / out

Contract signatures remain shared; these projections belong to specific port occurrences. One bit on a service port cannot replace this table. Every receiving projection references exactly one handling operation declaration within its unit context. Every outgoing data/event projection references the nonempty set of syntactic emission sites. Those links do not assert reachability.

Resolve a connection by following each boundary port's unique delegation chain to a unit port. Preserve the ordered original boundary chain as provenance. The normalized route is `(connectorOccurrence, sourceLeafPort, destinationLeafPort, interface)`. Delegation aliases do not add storage, queues, dispatch, retries or observable messages. Distinct connectors resolving to the same directed leaf pair are invalid. Optionality is checked at every port along a boundary chain; one optional declaration cannot erase a required declaration.

A queue key is `(connectorOccurrence, direction, memberIdentity)` with direction relative to the connector's declared orientation. RPC responses and requests use opposite directions. Service events use the response direction but their distinct event member identities prevent queue sharing with results. Correlation state is separate from queue storage.

7.4 Executable normalization

Every operation has a behavior declaration and one synthetic executable feature owned by its unit structure. The feature is typed by that behavior and binds an activation to a unit occurrence. It MUST NOT create a user-visible performer, an independently editable implementation allocation, or permission to call it across a unit boundary.

A statement body normalizes to an entry, ordered steps, exit/failure terminals and lexical scopes. Every step records its source construct and ordered operands. The following lowerings are required:

Source construct	Normalized meaning
let / assignment	evaluate expression, then create/replace copied local or private value
send / publish	operand evaluation followed by per-route message offers, emission provenance and success/rejection continuation
request	issuance/correlation creation, suspended frame, timeout, typed success binding or terminating failure branch
invoke	synchronous same-unit call edge, copied arguments, borrowed activation context/snapshot and caller continuation
return / fail	terminate the current frame; unwind callers on failure; correlated result/failure for RPC activations
if	evaluate Bool once, mutually exclusive selected branch and merge of control only
while	repeated condition, body, iteration counter and bound-failure edge; no implicit unbounded loop
assert	Bool evaluation and failure edge when false
function	expression tree with parameter-only environment and explicit acyclic function call edges

Expressions carry inferred type and evaluation order. Binary short-circuit nodes conditionally skip their right operand. A record value maps fields to declared identities, not input order; an array preserves value order and fixed extent; enum values retain enum and literal identity. Numeric failures and invalid index edges are part of the expression contract. A mapper MUST NOT silently substitute mathematical unbounded integers for SWAML bounded Int execution.

The directly owned state body is syntax grouping, not an independent behavior declaration. It contributes to its operation's normalization: state identities, initial reference, entry/exit/effect step groups, trigger matchers, pure guards, priorities, finality, current-state feature, timer-generation features and the chapter 6 dispatch/reaction rules. A transition is not merely an unconstrained edge: its priority selection, source-state condition, payload environment and ordered exit/effect/update/entry phases are required semantics.

7.5 Kernel representation and SWAML library responsibilities

The `kernel` field of the mapping inventory lists proposed KerML bases; the `semantic` field names the SWAML meaning to preserve. Empty kernel lists mean a lexical/grouping node contributes to its owner's representation, not that its meaning may be discarded. The inventory covers every authored grammar rule and type alias; punctuation and reserved keywords are covered by their productions.

Kernel ownership, membership, typing, parameter, feature-value and reference relationships MUST be instantiated according to the selected KerML backend. The inventory is not a recipe to instantiate abstract metaclasses blindly. A backend must choose suitable concrete subclasses for abstract bases such as Expression and validate all resulting relationships. Source grouping rules do not generate duplicate nodes for the same AST element.

The SWAML library supplies typed semantics for activation/context, copied values, versioned port contracts, queue offers, request correlation, failure outcomes, bounded flow and machine reactions. A communication lowering uses a Step typed by the library's send/publish/request behavior, with typed features for its ports, member and payload. It MUST NOT serialize a made-up kernel class called `SWAML-library-message-interaction`. The kernel Step is the base; the library behavior is the additional meaning.

Requirements, timing, QoS, deployment, case and view annotations use project-owned metadata definitions with typed fields and explicit references. Cases additionally carry a scenario behavior/step graph and happens-before edges; they do not become an independently executing program. A view's subject uses semantic identity and occurrence context; geometry/layout stays in a separate sidecar.

A K-conforming artifact MUST include or resolvably identify its exact SWAML library version and declare which constraints are checked. Carrying uninterpreted metadata is preservation, not proof that a generic kernel tool understands or executes it. The project has not yet selected/published a library namespace or implemented this backend; these obligations define its acceptance criteria.

7.6 Case and view normalization

Normalize each case label to a distinct step identity, resolve its route/member and orientation, and translate `after` to an explicit happens-before edge. The transitive closure determines causal ancestry. A reply links to its unique unanswered request ancestor with reversed endpoints and the same connector/member. A request is already answered for candidate selection only if its reply is also a causal ancestor; an independent branch cannot settle the choice. Ambiguous candidates and multiple replies to one request are invalid. Source position contributes display order only. An unanswered request is allowed in a partial trace; no synthetic reply is inserted.

A case realization claim records a witness execution and assumptions, including payload values absent from the diagram, dispatch choices and operation paths. Mere route validity does not supply that witness. A graphical renderer can display an unverified case but must not show it as proven.

Normalize views to `(kind, subjectDeclaration, occurrenceContext?)`. GV/GRV can inspect any resolvable named semantic declaration or the owning package, except another view. IV requires a root component, a component definition or a nested component occurrence; CV/SV require a case or root component; FV requires an operation. A package name colliding with a top-level declaration makes that root ambiguous and is rejected for view lookup. The dot path begins at a top-level declaration in the owning package or is the package's name alone. Descents follow component/unit instances, their structural members and interface members exposed through ports. State-level navigation is Browser/FV selection within an operation, not a second view subject syntax.

7.7 Equivalence and completeness

Two normalized artifacts are equivalent when an identity-respecting graph correspondence preserves ownership, types, value shapes, interface versions, instance contexts, routes, evaluation order, triggers, state selection, policies, deployments, case dependencies/correlations and view subjects. Whitespace, comments, source offsets and diagram geometry are excluded. Requirements text and failure reasons remain observable documentation/diagnostic values and must be preserved.

A round trip may rename anonymous lowering IDs while retaining their owner/role trace. It may not remove guarded branches, reorder effects, merge queues, weaken timing contracts, convert copied payloads into remote references, or drop SWAML metadata because a receiver lacks support. Unsupported meaning requires an explicit partial-import result. This information-preservation contract is complete for the closed 1.1 syntax; implementing and validating a KerML backend is a separate milestone.

NORMATIVE / LANGUAGE CONTRACT

8. Conformance obligations and evidence

Normative classification of the SWAML 1.1 specification. This chapter separates static language validity, runtime behavior and proof claims. It prevents acceptance by one tool from being mistaken for evidence supplied by another.

8.1 Static acceptance

Static validity is determined from the complete loaded source workspace without executing a model. The grammar plus the following obligations define the static core. The reference core implements these groups, but remains advertised as partial V pending an independent clause-by-clause conformance suite.

ID	Static obligation	Reference-core evidence
V01	Pinned edition, headerless package syntax; portable tokens, escapes and safe literal domains	parser/SW025; lexical and migration tests
V02	Unique package/scoped names; imports exist; references resolve uniquely before kind filtering	SW002/SW004, owner-aware linker; multi-file tests
V03	Finite nominal records/enums/arrays; no Void data; complete typed constructions	SW020-SW028; values/flow and audit fixtures
V04	Pure initializer/function environments, no cross-unit value lookup, compatible operands/arguments/results	SW022-SW024; isolation and type tests
V05	Legal interface members/port roles, exact nominal matching and signature implementations	SW010/SW011/SW031/SW032/SW052
V06	Acyclic composition, correct occurrence kinds, immediate-child routes, one delegation per boundary, no duplicate aliases/routes	SW012/SW013/SW030/SW033/SW035
V07	Required-port closure through every alias; single source/provider where required	SW034/SW035; closure fixtures
V08	Every outgoing member has syntactic emission provenance; every incoming member has one handler owner	SW050/SW051/SW055
V09	Positive safe durations, capacities, array extents and loop bounds; nonnegative safe priorities; complete QoS; sampled buffering and idempotent RPC declaration	SW020/SW040/SW041/SW043/SW059/SW062

ID	Static obligation	Reference-core evidence
V10	WCET <= deadline; periodic deadline <= period; realtime/mixed required timing/latency/age metadata	SW042/SW044/SW045
V11	Local acyclic calls; legal invoke targets; copied input parameters; no live-name shadowing	linker, SW002/SW056/SW058
V12	Value returns and request failure paths terminate; Void completion is not a value	SW020/SW022/SW023/SW053/SW057
V13	Reentrant activation roots do not write unit data directly or through any reachable helper	SW054; transitive-effect audit tests
V14	Stateful operation signature, initial reference, graph reachability, final-state rules, competing priorities and no unconditional completion cycle	SW060-SW064
V15	Valid occurrence deployment/node targets; no duplicate exact allocation	SW070
V16	View subject kind, case member/route/direction, earlier predecessor and uniquely correlated reply	SW080-SW083

“Every path returns” in V12 is structural definite termination: a sequential block contains return/fail or an if with both branches definitely terminating. A loop is never sufficient by itself, because it may run zero times or hit its bound. A request failure block obeys the same rule. This rule is intentionally conservative and does not require solving guard predicates. A request's success continuation is checked independently from its failure block.

Literal array indices, including unary sign expressions, are checked statically. General expression-derived bounds and arithmetic overflow are execution checks; a validator may additionally reject a proven inevitable failure with an explained analysis diagnostic. Syntactically present but unreachable emission still satisfies provenance; proving that an interface is useful is a stronger claim.

Imported declarations and string-valued semantic paths MUST be revalidated after a dependency change, addition or deletion. Correctness includes negative lookups: a newly added declaration can make an existing short reference ambiguous. A conservative implementation may rebuild all loaded language documents. Source recovery may suspend semantic checks on a syntactically invalid document, but syntax errors and unresolved references remain visible and the document cannot pass validation.

8.2 Runtime obligations

ID	Obligation	Required evidence for E
E01	Distinct instance stores, copied payloads and activation frames	isolation traces for reused definitions and transports
E02	Startup barrier, release epochs, serial lock, snapshots and bounded activation resources	scheduler traces including suspended RPCs and reentrant readers

ID	Obligation	Required evidence for E
E03	Per-route queues, sampled replacement, overflow, ordering, age and delivery policies	adversarial queue/duplicate/drop tests
E04	Unique correlations, response route, timeout race, failure outcome and late-response suppression	concurrent request/timeout/reply traces
E05	Structured statement order, bounded loops, failure unwinding and numeric/index checks	reference evaluator cases and failure traces
E06	State reaction order, timer generations, priority, completion bound and termination	competing input/timer and self-transition traces
E07	Timing observation and explicit deployment/resource assumptions	target binding plus measured/proved scheduling evidence

The runtime must detect dynamic out-of-bounds access, non-finite arithmetic, integer overflow, division by zero and loop/microstep-limit violations. V does not have to prove that these failures never occur. A static accepted model can describe a failing execution. Likewise V checks the idempotent declaration where required; a deployment claim that it is true requires implementation evidence.

8.3 Analysis claims

A **case realization** claim requires at least one operation-level witness trace consistent with all case dependencies, route/member identities and model assumptions. A route-valid case is only a candidate until such evidence exists. Invalid correlation is statically rejectable; guard feasibility is not implied by correlation validity.

A **requirement satisfied** claim links named requirements to cases plus test/analysis/review evidence; `satisfies` syntax alone denotes intent. A **schedulable** claim identifies the target scheduler, release assumptions, execution costs, transport/queue bounds and proof or measurement envelope. A **deterministic** claim accounts for independent stimulus ordering and reentrant completion. A **deadlock-free** claim accounts for synchronous request dependency cycles and retained serial locks.

These analyses may be conservative, bounded or incomplete, but MUST state assumptions and limitations. Failure to find a witness is not automatically proof that no witness exists. Runtime snapshot realizability is a backend obligation, not something syntax validation can establish.

8.4 Kernel, editor and graphical acceptance

K requires the complete normalized model, an implemented versioned semantic library/exporter, kernel validation and round-trip evidence. The production inventory checks coverage of the mapping design; it does not validate generated KerML because no such exporter exists yet.

G requires Browser and all six view projections, instance-aware identity, accessible notation, validated source edits, layout separation and atomic undo/redo. An LSP with generated TextMate tokens is an editor foundation, not G. Full IDE refactoring additionally requires path-aware rename/reference/navigation and persistent identity handling.

A tool's conformance report MUST identify language version, implementation version, levels claimed, rule groups checked, unsupported capabilities and evidence references. It MUST NOT promote S or partial V to K/E/G by implication. Implementation status is this repository's report.

8.5 Scope and future evolution

Completeness of this specification means every 1.1 construct has syntax, ownership, reference/type rules, static constraints, operational or descriptive meaning, a normalization contract and a conformance category. It does not mean every future software architecture concern is a built-in keyword or that all backends are implemented.

The finite boundary remains explicit: no inheritance, generics, reference-valued payloads, dynamic object creation/subscriptions/discovery, unrestricted fork/join, hierarchical/orthogonal state machines or arbitrary host-language code. Model adapters and protocols can express supported interactions; a new construct needs a versioned proposal with the same obligations. Patch clarifications preserve valid-model meaning for the pinned edition. Incompatible changes require an explicitly selected new edition and migration notes. The 1.1 release intentionally migrates syntax from 1.0 despite using a minor version; it provides no silent legacy aliases.

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9. Formal core and simulation relation

Normative chapter of the SWAML 1.1 specification. These judgments constrain implementations alongside chapters 1-8. They make explicit the distinction between static validity, allowed executions, and a particular target's choice among them. They do not assert a mechanically proved metatheory.

9.1 Domains and judgments

Let D be the resolved declaration graph; O the finite expanded occurrence tree; Γ a lexical environment mapping names to types; σ an occurrence-indexed store of copied values; ρ an activation's parameter/local store; Q the connector queues; A the live activations; M the machine states and timer generations; R the correlation table; L the unit serial locks; B the released lifecycle/timer work; and t an integer nanosecond time. A runtime configuration is:

```text
C = (D, O, σ, Q, A, M, R, L, B, t)
D ⊢ O valid                  closed assembly validity
D; Γ ⊢ e : T                expression typing
D; Γ ⊢ s ⇒ Γ'               statement typing and resulting lexical scope
D; σ; ρ ⊢ e ⇓ v : T         successful pure expression evaluation
D; σ; ρ ⊢ e ⇑ f             expression failure
C  - label→ C'                one permitted observable/local step
trace(C) = label₀ label₁ …   finite or infinite execution trace
```

A value is a Bool, bounded Int, finite Real, String, nominal enum literal, finite nominal record of values, or fixed array of values. Void has no inhabitants. Record identity includes its qualified declaration identity; arrays include element type and extent. Values never contain references into O, σ, A or Q. An environment binding also carries its declared type; widening an Int into a Real binding preserves Real type even when its mathematical value is integral.

The only assignment compatibility relation is T ≤ T and Int ≤ Real for scalars. Numeric common type is Real if either operand is Real, otherwise Int. There is no array covariance. A scope extension Γ[x:T] requires x absent from the live environment. Block-local extensions are discarded at block exit; assignments to existing cells remain.

9.2 Static rule schemas

The schemas below apply recursively, together with the name, ownership, route and contract restrictions in chapters 1-4. A failed premise rejects the construct, rather than adding an implicit conversion or exchange.

Construct	Premises	Conclusion
Reference x	x:T in Γ	Γ ⊢ x:T
Enum T.literal	no first-segment value T in the current expression environment Γ; one unambiguous accessible enum T; literal in its domain	nominal value:T
Qualified enum P::T.literal	loaded package P declares enum T and its literal	nominal value:P::T, independent of lexical T
Field e.f	e has nominal record T; T declares f:U	e.f:U
Index e[i]	e:T[n], i:Int	e[i]:T; runtime requires 0 ≤ i < n

Construct	Premises	Conclusion
Record T	one value for every declared field; each value assignable to that field	constructed value:T
Array	nonempty values have one nominal type or numeric common type T	value:T[n]
Arithmetic	numeric operands, % additionally Int operands	/:Real; %:Int; other operators:numeric common type
Boolean/comparison	chapter 1 operand conditions	result:Bool
f(args)	f owned by current unit; signature-compatible copied args; pure acyclic graph	result has f's declared non-Void type
let x:T=e	e:U, U ≤ T, fresh x	Γ ⇒ Γ[x:T]
x=e	x:T in Γ, e:U, U ≤ T; activation effect rules permit write	Γ ⇒ Γ
invoke f(args)	same-unit untriggered Void operation; signature-compatible args; acyclic graph	Γ ⇒ Γ
send/publish p.m	p owned by current unit; correct outward member and verb; compatible args	Γ ⇒ Γ
request p.m as x	p owned by unit; outward RPC; compatible args; positive timeout; fresh x and failure name; failure block cannot fall through	Γ ⇒ Γ[x:result] on success; Void x is reserved but uninhabited
return e	e assignable to owning operation result	current operation terminates
if / while / assert	condition:Bool; loop bound positive safe integer	each nested block checked in a fresh child scope
transition	states in same operation; legal stimulus; Bool guard; effect typed with payload; distinct competing priorities	transition belongs directly to its operation

For root validity, an inline root expands only its authored component/unit children and connectors; a typed root expands exactly its referenced component definition. Both require an acyclic expansion, distinct occurrence stores and port closure. A typed root binding contains only nodes/deployments and introduces no topology. An omitted profile normalizes to service. An omitted operation/signature result normalizes to Void; parameter/data/function result types cannot be omitted. These are defined elaborations, not inference from body behavior.

Let F(op) be the set of unit-variable assignments in op and recursively in every locally invoked helper. For every reentrant activation root, F(op) = ∅. Pure functions have no unit-variable environment and no operation edges. These finite graphs are acyclic, so both checks terminate.

For every normalized connector (p,q), contracts(p) = contracts(q) by declaration identity and roles(p,q) are complementary in the orientation defined in chapter 2. Delegation preserves both identity and role and introduces no queue. Every cross-unit effect is an offer to a member projection of such a connector. Thus a successful path lookup into a remote private store is never an execution rule.

9.3 Local execution rules

Expressions evaluate operands left to right, except short-circuit `and`/`or`. A checked arithmetic operation either produces a value in its result domain or raises a local Arithmetic failure. Integer arithmetic is mathematically exact before checking the bounded result; implementations MUST NOT let floating-point rounding turn an out-of-range integer into a valid one. Remainder uses truncation toward zero. Real arithmetic uses binary64. Equality recursively compares typed values; Int/Real numerics compare after scalar promotion.

A let creates a new local cell initialized by a copied, widened value. An assignment evaluates first, then atomically replaces the target cell. An invoke pushes a copied parameter frame on the existing activation root, preserving its lock and snapshot. A helper return pops only that frame. Returning from the outermost activation frame completes the activation. A failure unwinds all frames of that activation root without rollback. A loop tracks completed body iterations: after n executions, it evaluates the condition once more and fails if still true.

A reentrant dispatch snapshots σ for its unit. All unit reads by that activation, including helpers after suspension, use that snapshot. A serial dispatch reads the live store and holds its lock through requests. Local statements may interleave with other enabled activation roots at statement boundaries, but no competing serial activation root can acquire the same lock.

9.4 Communication transition schemas

Let q be a route/direction/member queue, capacity k, m a copied message, and c a fresh correlation. `offer(q,m)` is atomic. Each accepted logical message retains its original timestamp through all attempts.

Rule	Preconditions	State change and observation
Admit	retained(q) < k	append/admit m; emit admission
Reject	full(q), overflow=reject	q unchanged; report rejection to producer/correlation
Drop newest	full(q), overflow=dropNewest	q unchanged; emit drop(m); producer admission succeeds
Drop oldest	full(q), overflow=dropOldest	discard oldest retained logical message; admit m; observe both
Ready	message has reached destination transport boundary	set ready time; check maxLatency against original emission
Expire	t - emitted(m) > maxAge	discard before dispatch; no receiver effect
Dispatch	m ready, live handler, lock/limit available; FIFO head when required	remove dispatchable occurrence; copy args into an activation frame; record delivery/release/dispatch
Request	outward RPC statement, no other awaited statement in that frame	create Pending c with expiry=t+timeout; offer request; suspend caller
Respond	handler returned/failed	offer copied result/RemoteFailure on reverse route referencing c
Complete	response ready, R(c)=Pending, t < expiry(c)	R(c)=Succeeded/Failed exactly once; resume caller
Timeout	R(c)=Pending, t ≥ expiry(c)	R(c)=Failed(Timeout); resume failure block
Late response	R(c) terminal	discard; no caller state change

Fanout applies offer independently to every selected route before propagating any rejection. Running handlers are not queue entries. A dispatch or response cannot fabricate another member, route or

correlation. Lifecycle releases, observations and local failures are not message offers. All queue loss and duplicate dispatch must be visible in a checking trace. Delivery policies constrain allowed traces, not source-language branches.

9.5 Stateful operation transition schemas

M(op) contains (state, generation, ready, live, reacting). One reaction at a time may execute. Initialization runs initial entry before setting ready. For stimulus z in state s, form candidates with matching source and trigger, evaluate their guards in the appropriate copied payload environment, and select the enabled candidate of maximum priority. No candidate means consume/ignore z. Selection is not applied across unrelated stimuli.

For selected transition s→u: invalidate s's timers; execute exit(s); execute effect with z's payload; change state to u and increase generation; execute entry(u); install one timer per distinct outgoing duration. Any return/failure stops these phases immediately. Final entry terminates the machine. Completion selection then repeats until no transition is enabled or the machine terminates, with at most 1,024 taken completion edges per reaction. Entry/exit lack payload bindings. Completion reconsideration also follows an ignored external stimulus.

A timer stores its machine generation. A stale timer has only a discard transition. A valid timer is released at entry-completion time plus duration, and competes with independent work under the target scheduler. A terminated machine never regains liveness inside the same run.

9.6 Time, traces and refinement

Logical time never decreases. A target may advance to the next release, transport availability, deadline or timer event when its scheduling policy permits. Time addition MUST remain exact; a simulator encountering an unrepresentable time stops with an explicit resource/clock failure. Deadline elapsed time includes dispatch delay and suspension; WCET active time excludes suspension. A zero-cost logical simulation can check ordering, queues and elapsed waits but cannot certify physical WCET or schedulability.

An implementation binding refines this specification when every trace it produces can be matched by the core relation, with any internal bookkeeping steps hidden, and it satisfies its declared progress/fairness assumptions. A deterministic simulator chooses one such trace. Passing that run does not establish universal safety or liveness across all allowed schedules, nor realize every descriptive case automatically.

The reference simulation profile specifies a concrete, bounded choice of these rules. Reproducible inputs include all model source digests, selected root component, profile version, time horizon, harness releases, transport parameters and resource limits. No theorem of type preservation, kernel compatibility or trace equivalence is claimed without a separately checked proof artifact.

NORMATIVE / LANGUAGE CONTRACT

SWAML graphical notation, version 1.1

This is the normative notation and interaction contract for a future workbench. The current repository defines the model, grammar and language server; it does not claim to ship these diagram editors.

The rendered example catalog illustrates every view family against validated source models.

1. Views and navigation

SWAML defines the following view families. GRV is Grid View.

ID	Name	Primary subject	Contents
Browser	Model Browser	workspace / selected subtree	ownership tree, definitions/usages, errors, references
GV	General View	package, root component, definition, unit or declaration	owned elements, types, properties and relations
IV	Interconnect View	root or nested component occurrence, or component definition	immediate children, exposed ports, connectors, delegation
CV	Case View	root component or case	cases, requirements and intended coverage
SV	Sequence View	case, or case selected from a root component	interface exchanges and happens-before dependencies
GRV	Grid View	package or named semantic declaration	model inventory, interfaces, operations, timing, allocation or traces
FV	Flow View	operation	statement flow or states/transitions owned by that operation

GV/GRV subjects exclude other views. IV accepts a root component, a component definition or a nested component occurrence; a unit is inspected through GV/FV or its containing IV. A package subject is its own name alone. These categories are validated.

FV projects an operation's statement flow or owned states. Label grouping frames with their component, unit or operation paths.

The Browser shows packages → definitions/root occurrences → child occurrences → ports, variables, functions and operations → flow statements or states/transitions. A definition and a usage MUST have distinct badges. Opening a usage preserves its instance path even when the editor displays members inherited from its definition.

Selection updates Properties; **Open declaration** selects the shared declaration. **Diagram** opens the most appropriate view for the selected semantic element, and **Up** goes to its owning element. These commands are requirements for the future host, not implemented commands in this package.

2. Visual alphabet

Element	Shape and required labels
Root component	component frame, «component», root occurrence name and effective profile; typed roots show name : Type
Component	rectangle with double-tab component marker, name/type; children may be nested
Unit	plain rectangle with «unit» and name/type
Definition	«component def» / «unit def»; no occurrence underline
Instance	name : Type, occurrence badge; unique instance path in tooltip
Port	small boundary square labeled name : Interface @ version and role abbreviation
Interface	contract rectangle with kind and member signatures; optional lollipop/socket glyph supplements, never replaces, text
Function	rounded rectangle with ƒ, signature and purity badge
Operation	rounded rectangle with op, signature, trigger and optional timing
State	rounded rectangle with state, name and optional entry/exit compartments
Initial / final	filled circle / circle with inner filled circle, plus accessible labels
Requirement	rectangle with «requirement», name, text or synopsis
Case	rounded case node with name; links to requirements are dashed and labeled satisfies (claimed)
Execution node	framed box with node name and mcu/process/container badge

A unit is not shown as a UML actor or a performer. Physical quantities, devices and hardware drawings may appear as named external software-facing adapters, not as geometry elements. Color MAY distinguish types but MUST NOT carry meaning without shape/text redundancy.

Port role labels are S, R, C, Svr, Pub, Sub, Prov, and Req. The full role appears in Properties and accessible text. An optional unconnected port is shown with a hollow center and optional, never hidden by default.

3. Lines and direction

Relation	Line
Sender-receiver	solid arrow from sender to receiver; «data» and interface label
Publish-subscribe	solid arrow with event marker from publisher to subscriber; one branch per connector
Client-server / combined service	solid connector from client/required to server/provided, labeled «rpc» or «service»; opposite event/reply direction indicated in member inspection
Delegation	solid elbow from boundary port to child port, labeled delegate; roles remain unchanged
Definition typing	dashed open-triangle reference labeled type; not runtime communication
Deployment	dashed arrow labeled deploy to execution node

Relation	Line
State transition	arrow labeled `trigger [guard] / effect`, with priority when set
Flow edge	directed line for local control succession; conditional branches labeled true/false
Requirement trace	dashed arrow labeled `satisfies (claimed)`

Every communication line MUST terminate at a visible or explicitly elided port, never at an operation box that conceals a cross-unit bypass. An elided path must expose the full route on inspection. A connector label includes QoS at expanded detail. Invalid connectors are shown with a diagnostic badge and text; they are not restyled into apparently valid ones.

This Mermaid figure illustrates topology only; a production IV uses the boundary-square port glyphs above:

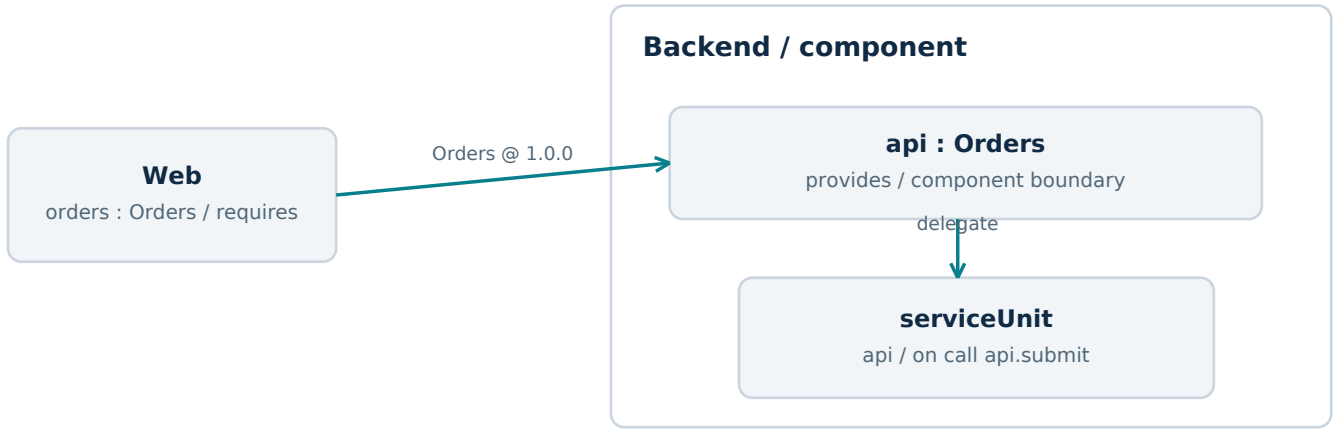


Figure 1. Delegated service contract and direct operation ownership.

The handler association inside the unit is not a second communication connector. Only the external `req-bp` line denotes an intercomponent exchange.

4. Flow View

FV has two projections under one view ID:

- **Statement body:** start, ordered statement nodes, decisions, bounded-loop back edges, request success/failure branches, local invoke nodes and end/failure terminals. A send/publish/request node carries its exact `port.member` path. Data values are labels or optional local edges, not global event nodes.
- **Stateful body:** initial state, state nodes, entry/exit compartments, transitions, timer/event labels and final states. State rectangles remain inside the owning operation frame. A state is not allocated later to a performer.

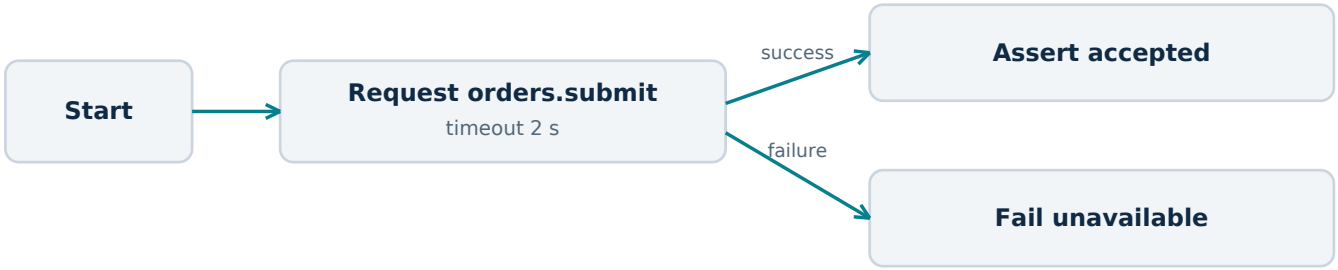


Figure 2. Request flow with explicit success and failure continuations.

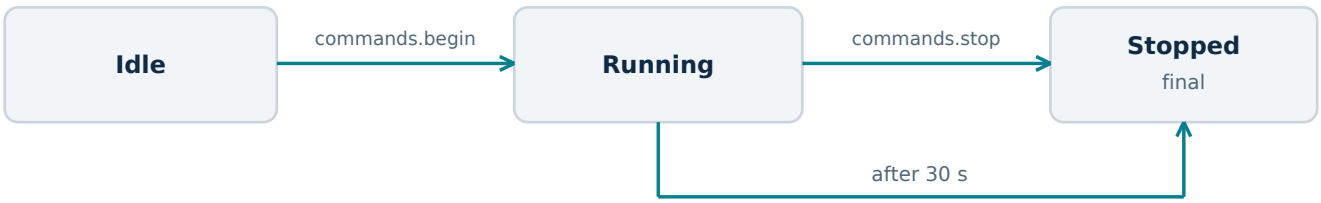


Figure 3. State behavior owned by one operation.

The state diagram is the FV of `Device.lifecycle`. In a full renderer, the frame and port-trigger labels link back to that exact operation and its interface contract.

5. Sequence and case views

SV lifelines represent component or unit **instances**, with the visible port path on each message. A request has a solid arrow and correlation label; its reply is dashed with the same correlation. Data/events use asynchronous arrows. Provider events on a combined service go in the reverse direction to client requests.

An SV renderer MUST preserve `after` dependencies and show unrelated steps as unordered/concurrent rather than inventing semantic ordering from vertical placement. It MAY choose a stable topological arrangement for readability, but must mark independent groups and expose the dependency relation. A case cannot create an operation, free event, implicit connector or arbitrary actor bypass merely by drawing a message.

CV shows the root component's named cases and explicit requirement links. A filled verification checkmark requires linked evidence from a verification tool; a `satisfies` assertion alone is shown as claimed coverage.

6. Grid View

GRV is a table projection, not geometry. Available grids SHOULD include:

Grid	Minimum columns
Inventory	name, kind, owning path, definition/usage, diagnostic count
Interfaces	interface/version, member, parameters/result, port roles, emitter/handler
Connections	source, target, interface, delivery, capacity, overflow, latency, age
Operations	unit path, signature, trigger, WCET, deadline, period, concurrency
Deployment	instance, effective node, explicit/inherited allocation
Trace	requirement, case, evidence status, source location

Sorting and filtering do not change semantic order. Editing a value performs a validated source edit with undo; computed effective values must reveal the declaration that owns them.

7. Synchronization, layout and editor safety

Text and diagrams project one semantic model. A layout file stores view ID, subject identity, node positions, edge bends, collapsed regions, filters and styling only. It MUST NOT store a competing definition of ports, triggers, interfaces or connections. Source remains authoritative for semantics.

A graphical mutation MUST resolve selected semantic IDs, prepare a source edit, parse/link/validate the proposed result, report new errors, apply the edit atomically, and preserve undo. A tool may intentionally

apply an incomplete editing step only if it remains visibly invalid. Moving a node changes layout only; reparenting an operation changes semantics and must validate references and interface access.

Deleting a port must reveal impacted connectors, statements and handlers. Renaming must update every typed cross-reference and path string, including cases and view subjects. Diagram nodes need stable IDs and source ranges; generated AST object identity is not stable across reparses.

The Browser, property panes and diagrams MUST support keyboard navigation and text alternatives. Error states need icons and labels in addition to color. Large models may elide details, but hidden exchanges must remain discoverable through the Browser and interface grid.

NORMATIVE APPENDIX

A. Complete concrete grammar

Authoritative Langium grammar, reproduced in full. Line wraps are typographic continuations; only numbered rows begin original source lines.

LANGIUM GRAMMAR

```
grammar Swaml

entry Model: packages+=Package*;
Package: 'package' name=ID '{' (imports+=Import | elements+=Element)* '}';
Import: 'import' namespace=QualifiedName '::' '*' ';';
Element: DataType | EnumType | Interface | Component | Unit | RootComponent | Requirement | Scenario |
View;
DataType: 'datatype' name=ID '{' fields+=Parameter* '}';
EnumType: 'enum' name=ID '{' literals+=ID (',' literals+=ID)* '}';
Parameter: name=ID ':' type=TypeRef ';';
TypeRef: (primitive=('Bool'|'Int'|'Real'|'String'|'Void') | type=[ValueType:QualifiedName]) ('['
size=INT ']')?;
type ValueType = DataType | EnumType;
Interface: 'interface' kind=('senderReceiver'|'clientServer'|'publishSubscribe'|'service') name=ID
'version' version=STRING '{' members+=InterfaceMember* '}';
InterfaceMember: DataMember | EventMember | Signature;
DataMember: 'data' name=ID ':' type=TypeRef mode=('sampled'|'queued') ';';
EventMember: 'event' name=ID '(' (parameters+=Argument (',' parameters+=Argument)*)? ')' ';';
Signature: 'operation' name=ID '(' (parameters+=Argument (',' parameters+=Argument)*)? ')' (':'
result=TypeRef)? (idempotent?='idempotent')? ';';
Argument: name=ID ':' type=TypeRef;
Component: 'component' 'def' name=ID '{' members+=ComponentMember* '}';
ComponentMember: Port | Instance | Connect | Delegate;
Unit: 'unit' 'def' name=ID '{' members+=UnitMember* '}';
UnitMember: Port | Variable | Operation | FunctionDef;
type Definition = Component | Unit;
Instance: kind=('component'|'unit') name=ID ':' definition=[Definition:QualifiedName] ';';
Port: 'port' name=ID ':' interface=[Interface:QualifiedName]
role=('sender'|'receiver'|'client'|'server'|'publisher'|'subscriber'|'provides'|'requires')
(optional?='optional')? ';';
Variable: 'var' name=ID ':' type=TypeRef '=' value=Expression ';';
RootComponent: 'component' name=ID (':' definition=[Component:QualifiedName]
(profile=('realtime'|'service'|'mixed'))? (';' | '{' members+=RootMember* '}') |
(profile=('realtime'|'service'|'mixed'))? '{' members+=RootMember* '}');
RootMember: Instance | Connect | Node | Deployment;
Node: 'node' name=ID kind=('mcu'|'process'|'container') ';';
Deployment: 'deploy' target=Path 'on' node=[Node:ID] ';';
Connect: 'connect' name=ID ':' source=Path '->' target=Path qos=Qos? ';';
Delegate: 'delegate' outer=[Port:ID] '->' target=Path ';';
Qos: 'qos' '{' 'delivery' delivery=('bestEffort'|'atMostOnce'|'atLeastOnce') ';' 'ordering'
ordering=('fifo'|'unordered') ';' 'capacity' capacity=INT ';' 'overflow'
overflow=('dropOldest'|'dropNewest'|'reject') ';' ('maxLatency' maxLatency=Duration ';')? ('maxAge'
maxAge=Duration ';')? '}';
Operation: 'operation' name=ID '(' (parameters+=Argument (',' parameters+=Argument)*)? ')' (':'
result=TypeRef)? trigger=Trigger? timing=Timing? (body=Block | stateBody=StateBody);
Trigger: 'on' (kind=('start'|'manual') | kind='every' period=Duration | kind=('receive'|'call')
endpoint=Path);
Timing: 'timing' '{' 'wcet' wcet=Duration ';' 'deadline' deadline=Duration ';' 'concurrency'
concurrency=('serial'|'reentrant') ';' '}';
FunctionDef: 'function' name=ID '(' (parameters+=Argument (',' parameters+=Argument)*)? ')' ':'
result=TypeRef '=' value=Expression ';';
Block: '{' statements+=Statement* '}';
Statement: Let | Assign | Send | Request | Invoke | Return | If | While | Fail | Assert;
Let: 'let' name=ID ':' type=TypeRef '=' value=Expression ';';
Assign: target=ID '=' value=Expression ';';
Send: kind=('send'|'publish') endpoint=Path '(' (args+=Expression (',' args+=Expression)*)? ')' ';';
Request: 'request' endpoint=Path '(' (args+=Expression (',' args+=Expression)*)? ')' 'as' name=ID
'within' timeout=Duration 'otherwise' failure=Block ';';
```

LANGIUM GRAMMAR / CONTINUED

```
Invoke: 'invoke' operation=[Operation:ID] '(' (args+=Expression (',' args+=Expression)*)? ')' ';';
Return: 'return' value=Expression? ';';
If: 'if' '(' condition=Expression ')' then=Block ('else' otherwise=Block)?;
While: 'while' '(' condition=Expression ')' 'bound' bound=INT body=Block;
Fail: 'fail' reason=STRING ';';
Assert: 'assert' condition=Expression ';';
StateBody: '{' 'initial' initial=[State:ID] ';' states+=State+ transitions+=Transition* '}';
State: 'state' name=ID (final?='final')? '{' ('entry' entry=Block)? ('exit' exit=Block)? '}';
Transition: 'transition' name=ID ':' from=[State:ID] '->' to=[State:ID] 'on' (kind='completion' |
kind='after' delay=Duration | kind='receive' endpoint=Path) ('when' '(' guard=Expression ')')?
('priority' priority=INT)? effect=Block;
Requirement: 'requirement' name=ID text=STRING ';';
Scenario: 'case' name=ID 'for' root=[RootComponent:QualifiedName] ('satisfies'
requirements+=[Requirement:QualifiedName] (',' requirements+=[Requirement:QualifiedName])*)? '{'
steps+=ScenarioStep* '}';
ScenarioStep: kind=('send'|'publish'|'request'|'reply') source=Path '->' target=Path 'via' connector=ID
('after' predecessor=ID)? 'label' name=ID ';';
View: 'view' name=ID kind=('GV'|'IV'|'CV'|'SV'|'GRV'|'FV') 'of' subject=Path ';';
Duration: value=INT unit=('ns'|'us'|'ms'|'s');
Path returns string: ID ('.' ID)*;
QualifiedName returns string: ID ('::' ID)*;
ValuePath returns string: QualifiedName ('.' ID)*;
type Expression = BinaryExpression | UnaryExpression | Literal | FunctionCall | ReferenceExpression |
RecordLiteral | ArrayLiteral;
Expression returns Expression: OrExpression;
OrExpression returns Expression: AndExpression ({infer BinaryExpression.left=current} operator='or'
right=AndExpression)*;
AndExpression returns Expression: EqualityExpression ({infer BinaryExpression.left=current}
operator='and' right=EqualityExpression)*;
EqualityExpression returns Expression: RelationalExpression ({infer BinaryExpression.left=current}
operator=('=='|'!=') right=RelationalExpression)*;
RelationalExpression returns Expression: AdditiveExpression ({infer BinaryExpression.left=current}
operator=('<'|'<='|'>'|'>=') right=AdditiveExpression)*;
AdditiveExpression returns Expression: MultiplicativeExpression ({infer BinaryExpression.left=current}
operator=('+'|'-') right=MultiplicativeExpression)*;
MultiplicativeExpression returns Expression: UnaryExpression ({infer BinaryExpression.left=current}
operator=('*'|'/'|'%') right=UnaryExpression)*;
UnaryExpression returns Expression: {infer UnaryExpression} operator=('not'|'-') operand=UnaryExpression
 | PrimaryExpression;
PrimaryExpression returns Expression: Literal | FunctionCall | ReferenceExpression | RecordLiteral |
ArrayLiteral | '(' Expression ')';
Literal: {infer Literal} (number=Numeric | text=STRING | boolean=('true'|'false'));
FunctionCall: function=[FunctionDef:ID] '(' (args+=Expression (',' args+=Expression)*)? ')';
ReferenceExpression: path=ValuePath ('[' index=Expression ']')?;
RecordLiteral: 'record' type=[DataType:QualifiedName] '{' fields+=FieldValue* '}';
FieldValue: name=ID '=' value=Expression ';';
ArrayLiteral: '[' (values+=Expression (',' values+=Expression)*)? ']';
Numeric returns number: NUMBER | INT;
hidden terminal WS: /\s+/;
terminal ID: /[_a-zA-Z][\w]*/;
terminal NUMBER returns number: /[0-9]+\.[0-9]+/;
terminal INT returns number: /[0-9]+/;
terminal STRING: /"([^"\\\x00-\x1F]|\\["\\nrt])*"/;
hidden terminal ML_COMMENT: /\/\*[\s\S]*?\*\//;
hidden terminal SL_COMMENT: /\/\/[^\n\r]*/;
```

NORMATIVE DESIGN INVENTORY

B. Grammar-to-semantics inventory

Every grammar rule/type name is represented. Proposed kernel bases identify the design target; they do not constitute an implemented or validated exporter. Empty bases indicate syntax normalized through its owner.

Rule / kind	Proposed kernel base	Meaning to preserve
Model element	Package	Workspace source container under a host-pinned 1.1 language edition; no file version header.
Package element	Package	Flat unique declaration namespace with ordered owned members.
Import element	NamespaceImport	Direct wildcard visibility edge; no transitive import or implicit copying.
Element grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
DataType element	DataType	Finite nominal value domain: ordered record fields or unique enum literals.
EnumType element	DataType	Finite nominal value domain: ordered record fields or unique enum literals.
Parameter element	Feature	Typed field/input feature; preserve declared position and copied values.
TypeRef reference	FeatureTyping	Reference to primitive/nominal type, plus optional fixed-array shape.
ValueType grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
Interface element	Classifier	Nominal versioned member contract; legal member set depends on kind.
InterfaceMember grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
DataMember element	Feature	Typed payload contract with sampled/queued delivery interpretation.
EventMember element	Behavior	Contract-only payload signature; only port-projected operation emission can instantiate an exchange.
Signature element	Behavior	Contract-only RPC signature with copied parameters/result and idempotency assertion; omitted result is Void.
Argument element	Feature	Typed field/input feature; preserve declared position and copied values.
Component element	Structure	Component composition or unit private-state/executable boundary; distinct occurrence contexts.
ComponentMember grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.

Rule / kind	Proposed kernel base	Meaning to preserve
Unit element	Structure	Component composition or unit private-state/executable boundary; distinct occurrence contexts.
UnitMember grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
Definition grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
Instance element	Feature	Composite owned occurrence typed by a definition, multiplicity one.
Port element	Feature	Interface-typed role projection with optionality and instance-local endpoints.
Variable element	Feature, FeatureValue	Private variable with pure initializer and per-unit-occurrence storage.
RootComponent element	Feature, Structure	Named root component occurrence: typed composite Feature, or inline composition lowered through a synthetic Structure; effective profile defaults to service. Binding nodes/deployments cannot extend typed topology. Preserve root and child occurrence identity without an extra wrapper.
RootMember grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.
Node element	Structure	Logical execution-node declaration with mcu/process/container binding metadata.
Deployment element	MetadataFeature	Explicit occurrence-to-node allocation; nearest ancestor determines inherited allocation.
Connect element	Connector	Port route or queue-free forwarding alias; preserve hierarchy and nominal interface identity.
Delegate element	Connector	Port route or queue-free forwarding alias; preserve hierarchy and nominal interface identity.
Qos element	MetadataFeature	Per-route/member/direction queue capacity, overflow, delivery, order, latency and age constraints.
Operation element	Behavior, Step	Owned implementation plus synthetic executable feature; typed unit context and exactly one statement or directly owned state body; omitted result is Void.
Trigger element	MetadataFeature	Local release rule or port-member handler binding; not a free event.
Timing element	MetadataFeature	WCET, elapsed deadline and concurrency contract; activation root and local helper interpretation.
FunctionDef element	Function	Pure parameter-only expression with acyclic local call graph.
Block element	Step, Succession	Ordered steps, lexical scope, success/failure exits and source trace.
Statement grouping	Owner representation	Grammar union; normalize the selected concrete alternative exactly once.

Rule / kind	Proposed kernel base	Meaning to preserve
Let element	Step, Feature	Evaluate then create/replace one copied value in the proper lexical/private environment.
Assign element	Step, Feature	Evaluate then create/replace one copied value in the proper lexical/private environment.
Send element	Step	SWAML library emission behavior with operation/port/member provenance and independent route offers.
Request element	Step	SWAML library awaited RPC behavior with correlation, bounded timeout and exclusive success/failure.
Invoke element	Step	Same-unit synchronous call preserving caller context, lock, snapshot and continuation.
Return element	Step	Frame completion or failure unwinding; no rollback or implicit event.
If element	Step, Succession	Guarded structured control with branch exclusivity, loop count and bound failure.
While element	Step, Succession	Guarded structured control with branch exclusivity, loop count and bound failure.
Fail element	Step	Frame completion or failure unwinding; no rollback or implicit event.
Assert element	Step	Pure Boolean check with explicit activation failure continuation.
StateBody grouping	Owner representation	Initial-first syntax grouping contributing directly to its operation: states, transitions, current-state, timer generations and ordered reactions. No independently authored/executable element.
State element	Step	State identity, entry/exit groups and finality in one operation context.
Transition element	Step, Succession	Source/target, stimulus matcher, guard, priority and exit/effect/update/entry sequence.
Requirement element	MetadataFeature	Named text obligation; coverage claims carry no automatic proof.
Scenario element	Behavior	Descriptive partial trace tied to a root component and requirements; inline or definition-derived root topology, never a second executable program.
ScenarioStep element	Step, Succession	Port-member exchange with connector, explicit causal predecessor and inferred unique reply correlation.
View element	MetadataFeature	Projection kind and resolved declaration/occurrence subject; presentation has no runtime effects.
Duration value	LiteralInteger	Exact positive integer nanoseconds with retained authoring unit.
Path reference	Owner representation	Resolve according to reference context; preserve authoring spelling and resolved identities.

Rule / kind	Proposed kernel base	Meaning to preserve
QualifiedName reference	Owner representation	Resolve according to reference context; preserve authoring spelling and resolved identities.
ValuePath reference	Owner representation	Value-reference syntax permitting local field paths and explicit package-qualified enum literals. An accessible value first segment wins over an unqualified enum with no field-failure fallback; cross-unit private lookup remains forbidden.
Expression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
OrExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
AndExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
EqualityExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
RelationalExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
AdditiveExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
MultiplicativeExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
UnaryExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
PrimaryExpression lowering	Expression	Typed expression tree; grouping/precedence, operand order and conditional short-circuit evaluation.
Literal value	Expression	Concrete typed literal expression: Bool, bounded Int, finite Real or portable String.
FunctionCall lowering	Expression	Typed pure local function invocation with ordered copied arguments.
ReferenceExpression lowering	Expression	Local/copied/private value lookup with record traversal and checked final indexing; alternatively a nominal enum value resolved by lexical-first then package/import rules. Qualified enum identity is retained.
RecordLiteral value	Expression	Nominal construction, declared field/literal identities, fixed ordered array values.
FieldValue value	Expression	Nominal construction, declared field/literal identities, fixed ordered array values.
ArrayLiteral value	Expression	Nominal construction, declared field/literal identities, fixed ordered array values.
Numeric lexical	Owner representation	Unsigned decimal value; unary minus is an expression; finite/safe literal range checks.

Rule / kind	Proposed kernel base	Meaning to preserve
WS lexical	Owner representation	Insignificant source trivia; retain only for authoring round trips.
ID lexical	Owner representation	Case-sensitive ASCII identifier; grammar keywords are reserved.
NUMBER lexical	Owner representation	Unsigned decimal value; unary minus is an expression; finite/safe literal range checks.
INT lexical	Owner representation	Unsigned decimal value; unary minus is an expression; finite/safe literal range checks.
STRING lexical	Owner representation	Unicode text with only quote, backslash, n, r and t escapes; no raw controls.
ML_COMMENT lexical	Owner representation	Insignificant source trivia; retain only for authoring round trips.
SL_COMMENT lexical	Owner representation	Insignificant source trivia; retain only for authoring round trips.

INFORMATIVE / COMPLETE MODELS

C. Model examples

01 hello

SWAML / 01-hello.swaml

```
package Hello {
    interface senderReceiver Greeting version "1.0.0" {
        data text: String queued;
    }
    unit def Speaker {
        port output: Greeting sender;
        operation greet() on start {
            send output.text("Hello, SWAML");
            return;
        }
    }
    unit def Listener {
        port input: Greeting receiver;
        var last: String = "";
        operation hear(message: String) on receive input.text {
            last = message;
            return;
        }
    }
    component def Conversation {
        unit speaker: Speaker;
        unit listener: Listener;
        connect greeting: speaker.output -> listener.input qos {
            delivery atMostOnce; ordering fifo; capacity 4; overflow reject;
        };
    }
    component Demo { component conversation: Conversation; }
    view overview GV of Demo;
    view wiring IV of Conversation;
    view greetingFlow FV of Speaker.greet;
    view inventory GRV of Demo;
}
```

02 realtime controller

SWAML / 02-realtime-controller.swaml

```
package Control {
    interface senderReceiver Samples version "1.0.0" { data temperature: Real sampled; }
    interface senderReceiver Commands version "1.0.0" { data duty: Real sampled; }
    unit def Sensor {
        port output: Samples sender;
        operation sample() on every 10 ms timing { wcet 1 ms; deadline 3 ms; concurrency serial; } {
            send output.temperature(23.5);
            return;
        }
    }
    unit def Regulator {
        port input: Samples receiver;
        port output: Commands sender;
        function clamp(x: Real): Real = x / 100.0;
        operation regulate(temperature: Real) on receive input.temperature timing { wcet 1 ms; deadline 3 ms; concurrency serial; } {
            let duty: Real = clamp(temperature);
            if (duty > 0.8) { send output.duty(0.8); }
            else { send output.duty(duty); }
            return;
        }
    }
    unit def Actuator {
        port input: Commands receiver;
        var applied: Real = 0.0;
        operation apply(duty: Real) on receive input.duty timing { wcet 1 ms; deadline 2 ms; concurrency serial; } {
            applied = duty;
            assert applied <= 0.8;
            return;
        }
    }
    component def Controller {
        unit sensor: Sensor;
        unit regulator: Regulator;
        unit actuator: Actuator;
        connect measured: sensor.output -> regulator.input qos {
            delivery atMostOnce; ordering fifo; capacity 1; overflow dropOldest; maxLatency 1 ms; maxAge 20 ms;
        };
        connect commanded: regulator.output -> actuator.input qos {
            delivery atMostOnce; ordering fifo; capacity 1; overflow dropOldest; maxLatency 1 ms; maxAge 20 ms;
        };
    }
    component Heating realtime {
        component controller: Controller;
        node ecu mcu;
        deploy controller on ecu;
    }
    requirement LimitedDuty "Applied duty shall be at most 0.8.";
    view overview GV of Heating;
    view wiring IV of Controller;
    view regulating FV of Regulator.regulate;
    view deployment GRV of Heating;
}
```

03 service pubsub

SWAML / 03-service-pubsub.swaml

```
package Commerce {
    interface service Orders version "1.0.0" {
        // Submitting publishes effects, so this contract does not claim idempotency.
        operation submit(orderId: String): Bool;
        event accepted(orderId: String);
    }
    interface publishSubscribe Audit version "1.0.0" { event recorded(orderId: String); }
    unit def OrderService {
        port api: Orders provides;
        port audit: Audit publisher;
        operation submit(orderId: String): Bool on call api.submit {
            publish api.accepted(orderId);
            publish audit.recorded(orderId);
            return true;
        }
    }
    unit def Frontend {
        port orders: Orders requires;
        var lastAccepted: String = "";
        operation checkout() on manual {
            request orders.submit("order-42") as accepted within 2 s otherwise { fail "checkout unavailable"; };
            assert accepted;
            return;
        }
        operation confirmation(orderId: String) on receive orders.accepted { lastAccepted = orderId; return; }
    }
    unit def AuditStore {
        port input: Audit subscriber;
        var last: String = "";
        operation storeRecord(orderId: String) on receive input.recorded { last = orderId; return; }
    }
    component def Backend {
        port api: Orders provides;
        port audit: Audit publisher;
        unit serviceUnit: OrderService;
        delegate api -> serviceUnit.api;
        delegate audit -> serviceUnit.audit;
    }
    component def Web {
        port orders: Orders requires;
        unit frontend: Frontend;
        delegate orders -> frontend.orders;
    }
    component def Archive {
        port input: Audit subscriber;
        unit store: AuditStore;
        delegate input -> store.input;
    }
    component Shop {
        component web: Web;
        component backend: Backend;
        component archive: Archive;
        connect orderLink: web.orders -> backend.api qos { delivery atMostOnce; ordering fifo; capacity 32; overflow reject; };
        connect auditing: backend.audit -> archive.input qos { delivery atLeastOnce; ordering fifo; capacity 128; overflow dropOldest; };
        node edge container;
        node application container;
        node storage container;
        deploy web on edge;
        deploy backend on application;
        deploy archive on storage;
    }
```

SWAML / 03-service-pubsub.swaml / CONTINUED

```
    requirement AcceptedOrder "Accepted orders shall be published to audit subscribers.";
    case Checkout for Shop satisfies AcceptedOrder {
        request web.orders.submit -> backend.api.submit via orderLink label submitOrder;
        publish backend.api.accepted -> web.orders.accepted via orderLink after submitOrder label notification;
        publish backend.audit.recorded -> archive.input.recorded via auditing after submitOrder label auditRecord;
        reply backend.api.submit -> web.orders.submit via orderLink after notification label response;
    }
    view overview GV of Shop;
    view wiring IV of Shop;
    view cases CV of Shop;
    view checkout SV of Checkout;
    view services GRV of Shop;
    view submitFlow FV of OrderService.submit;
}
```

04 stateful operation

SWAML / 04-stateful-operation.swaml

```
package Lifecycle {
    interface publishSubscribe Commands version "1.0.0" { event begin(); event stop(); }
    interface publishSubscribe Status version "1.0.0" { event changed(value: String); }
    unit def Operator {
        port commands: Commands publisher;
        operation startDevice() on manual { publish commands.begin(); return; }
        operation stopDevice() on manual { publish commands.stop(); return; }
    }
    unit def Device {
        port commands: Commands subscriber;
        port status: Status publisher;
        var cycles: Int = 0;
        operation lifecycle() on start {
            initial Idle;
            state Idle { entry { publish status.changed("idle"); } }
            state Running { entry { cycles = cycles + 1; publish status.changed("running"); } }
            state Stopped final { entry { publish status.changed("stopped"); } }
            transition beginRun: Idle -> Running on receive commands.begin {}
            transition stopRun: Running -> Stopped on receive commands.stop {}
            transition expire: Running -> Stopped on after 30 s { publish status.changed("timeout"); }
        }
    }
    unit def Monitor {
        port status: Status subscriber;
        var current: String = "unknown";
        operation observe(value: String) on receive status.changed { current = value; return; }
    }
    component def Assembly {
        unit operatorUnit: Operator;
        unit device: Device;
        unit monitor: Monitor;
        connect commands: operatorUnit.commands -> device.commands qos { delivery atMostOnce; ordering fifo; capacity 8; overflow reject; };
        connect status: device.status -> monitor.status qos { delivery atMostOnce; ordering fifo; capacity 8; overflow dropOldest; };
    }
    component Product { component assembly: Assembly; }
    view lifecycle FV of Device.lifecycle;
    view wiring IV of Assembly;
}
```

05 values and flow

SWAML / 05-values-and-flow.swaml

```
package Values {
    enum Health { Good, Failed }
    datatype Measurement { value: Real; health: Health; }
    interface senderReceiver Measurements version "1.0.0" { data sample: Measurement queued; }
    unit def Producer {
        port output: Measurements sender;
        function twice(x: Real): Real = x * 2.0;
        operation prepare() {
            let samples: Real[3] = [1.0, 2.0, 3.0];
            let i: Int = 0;
            while (i < 3) bound 3 {
                let measurement: Measurement = record Measurement {
                    value = twice(samples[i]);
                    health = Health.Good;
                };
                send output.sample(measurement);
                i = i + 1;
            }
            return;
        }
        operation boot() on start { invoke prepare(); return; }
    }
    unit def Consumer {
        port input: Measurements receiver;
        var latest: Real = 0.0;
        operation update(measurement: Measurement) on receive input.sample {
            if (measurement.health == Health.Good) { latest = measurement.value; }
            return;
        }
    }
    component def Pipeline {
        unit producer: Producer;
        unit consumer: Consumer;
        connect samples: producer.output -> consumer.input qos { delivery atMostOnce; ordering fifo;
capacity 3; overflow reject; };
    }
    component Demo { component pipeline: Pipeline; }
    view preparation FV of Producer.prepare;
}
```

06 mixed gateway

SWAML / 06-mixed-gateway.swaml

```
package Gateway {
    interface senderReceiver Samples version "1.0.0" { data value: Real sampled; }
    interface service Telemetry version "1.0.0" {
        operation read(): Real idempotent;
        event observed(value: Real);
    }
    unit def Sensor {
        port output: Samples sender;
        operation sample() on every 10 ms timing { wcet 1 ms; deadline 2 ms; concurrency serial; } {
            send output.value(21.0); return;
        }
    }
    unit def Adapter {
        port input: Samples receiver;
        port api: Telemetry provides;
        var latest: Real = 0.0;
        operation receiveSample(value: Real) on receive input.value timing { wcet 1 ms; deadline 3 ms;
concurrency serial; } {
            latest = value;
            publish api.observed(value);
            return;
        }
        operation readLatest(): Real on call api.read timing { wcet 1 ms; deadline 3 ms; concurrency
serial; } { return latest; }
    }
    unit def Dashboard {
        port telemetry: Telemetry requires;
        var displayed: Real = 0.0;
        operation refresh() on manual timing { wcet 1 ms; deadline 30 ms; concurrency serial; } {
            request telemetry.read() as measured within 25 ms otherwise { return; };
            displayed = measured; return;
        }
        operation update(value: Real) on receive telemetry.observed timing { wcet 1 ms; deadline 5 ms;
concurrency serial; } {
            displayed = value; return;
        }
    }
    component def Acquisition {
        port output: Samples sender;
        unit sensor: Sensor;
        delegate output -> sensor.output;
    }
    component def Bridge {
        port input: Samples receiver;
        port api: Telemetry provides;
        unit adapter: Adapter;
        delegate input -> adapter.input;
        delegate api -> adapter.api;
    }
    component def Display {
        port telemetry: Telemetry requires;
        unit dashboard: Dashboard;
        delegate telemetry -> dashboard.telemetry;
    }
    component Product mixed {
        component acquisition: Acquisition;
        component bridge: Bridge;
        component display: Display;
        connect sensing: acquisition.output -> bridge.input qos {
            delivery atMostOnce; ordering fifo; capacity 1; overflow dropOldest; maxLatency 1 ms; maxAge
 20 ms;
        };
        connect observing: display.telemetry -> bridge.api qos {
            delivery atMostOnce; ordering fifo; capacity 8; overflow reject; maxLatency 2 ms;
        };
```

SWAML / 06-mixed-gateway.swaml / CONTINUED

```
        node controller mcu;
        node edge process;
        deploy acquisition on controller;
        deploy bridge on edge;
        deploy display on edge;
    }
    view overview GV of Product;
    view connections IV of Product;
    view conversion FV of Adapter.receiveSample;
    view allocation GRV of Product;
}
```

07 client server

SWAML / 07-client-server.swaml

```
package Rpc {
    interface clientServer Health version "1.0.0" {
        operation check(): Bool idempotent;
    }
    unit def Server {
        port health: Health server;
        operation checkHealth(): Bool on call health.check { return true; }
    }
    unit def Client {
        port health: Health client;
        var healthy: Bool = false;
        operation poll() on every 1 s {
            request health.check() as statusValue within 100 ms otherwise { healthy = false; return; };
            healthy = statusValue;
            return;
        }
    }
    component def Service {
        port health: Health server;
        unit serverUnit: Server;
        delegate health -> serverUnit.health;
    }
    component def Caller {
        port health: Health client;
        unit clientUnit: Client;
        delegate health -> clientUnit.health;
    }
    component HealthCheck {
        component caller: Caller;
        component serviceUnit: Service;
        connect healthCheck: caller.health -> serviceUnit.health qos {
            delivery atLeastOnce; ordering fifo; capacity 4; overflow reject;
        };
    }
    case Poll for HealthCheck {
        request caller.health.check -> serviceUnit.health.check via healthCheck label sent;
        reply serviceUnit.health.check -> caller.health.check via healthCheck after sent label answered;
    }
    view interaction SV of Poll;
    view requestFlow FV of Client.poll;
}
```

08 service field

SWAML / 08-service-field.swaml

```
package Settings {
    // A remotely visible property is an explicit getter, setter and event contract.
    interface service TemperatureSetting version "1.0.0" {
        operation get(): Int idempotent;
        operation setPoint(desired: Int): Bool;
        event changed(current: Int);
    }
    unit def SettingStore {
        port api: TemperatureSetting provides;
        var currentValue: Int = 20;
        // Each reentrant query sees its dispatch-time snapshot.
        operation read(): Int on call api.get timing {
            wcet 1 ms; deadline 5 ms; concurrency reentrant;
        } { return currentValue; }
        operation update(desired: Int): Bool on call api.setPoint {
            if (desired < 10 or desired > 30) { return false; }
            currentValue = desired;
            publish api.changed(currentValue);
            return true;
        }
    }
    unit def Panel {
        port setting: TemperatureSetting requires;
        var displayed: Int = 20;
        operation adjust() on manual {
            request setting.setPoint(22) as accepted within 100 ms otherwise { fail "setting
unavailable"; };
            assert accepted;
        }
        operation refresh() on manual {
            request setting.get() as observed within 100 ms otherwise { return; };
            displayed = observed;
        }
        operation display(current: Int) on receive setting.changed { displayed = current; }
    }
    component def Service {
        port api: TemperatureSetting provides;
        unit store: SettingStore;
        delegate api -> store.api;
    }
    component def Client {
        port setting: TemperatureSetting requires;
        unit panel: Panel;
        delegate setting -> panel.setting;
    }
    component Cabin {
        component settings: Service;
        component front: Client;
        component rear: Client;
        connect frontLink: front.setting -> settings.api qos {
            delivery atMostOnce; ordering fifo; capacity 8; overflow reject;
        };
        connect rearLink: rear.setting -> settings.api qos {
            delivery atMostOnce; ordering fifo; capacity 8; overflow reject;
        };
        node controller process;
        node displays container;
        deploy settings on controller;
        deploy front on displays;
        deploy rear on displays;
    }
    requirement VisibleSetting "Both panels can observe accepted temperature setting updates.";
    case Change for Cabin satisfies VisibleSetting {
        request front.setting.setPoint -> settings.api.setPoint via frontLink label adjust;
        publish settings.api.changed -> front.setting.changed via frontLink after adjust label
```

SWAML / 08-service-field.swaml / CONTINUED

```
    frontNotice;
65          publish settings.api.changed -> rear.setting.changed via rearLink after adjust label rearNotice;
66          reply settings.api.setPoint -> front.setting.setPoint via frontLink after frontNotice label
    accepted;
67      }
68      view catalog GRV of Settings;
69      view overview GV of Cabin;
70      view wiring IV of Cabin;
71      view coverage CV of Cabin;
72      view exchanges SV of Change;
73      view updateFlow FV of Cabin.settings.store.update;
74  }
```

09 component roots

SWAML / 09-component-roots.swaml

```
package Roots {
    unit def Counter {
        var count: Int = 0;
        operation initialize() on start
            timing { wcet 1 us; deadline 1 ms; concurrency serial; }
        {
            count = count + 1;
        }
    }

    // A reusable type defines topology. A typed root creates a concrete occurrence.
    component def Pair {
        unit left: Counter;
        unit right: Counter;
    }
    component Demo: Pair;
    component Embedded: Pair realtime {
        node cpu mcu;
        deploy left on cpu;
        deploy right on cpu;
    }

    // An inline root may contain a unit directly, without a wrapper definition.
    component Minimal { unit counter: Counter; }

    view reusable GV of Pair;
    view deployed IV of Embedded;
    view initializeFlow FV of Minimal.counter.initialize;
    view inventory GRV of Demo;
}
```

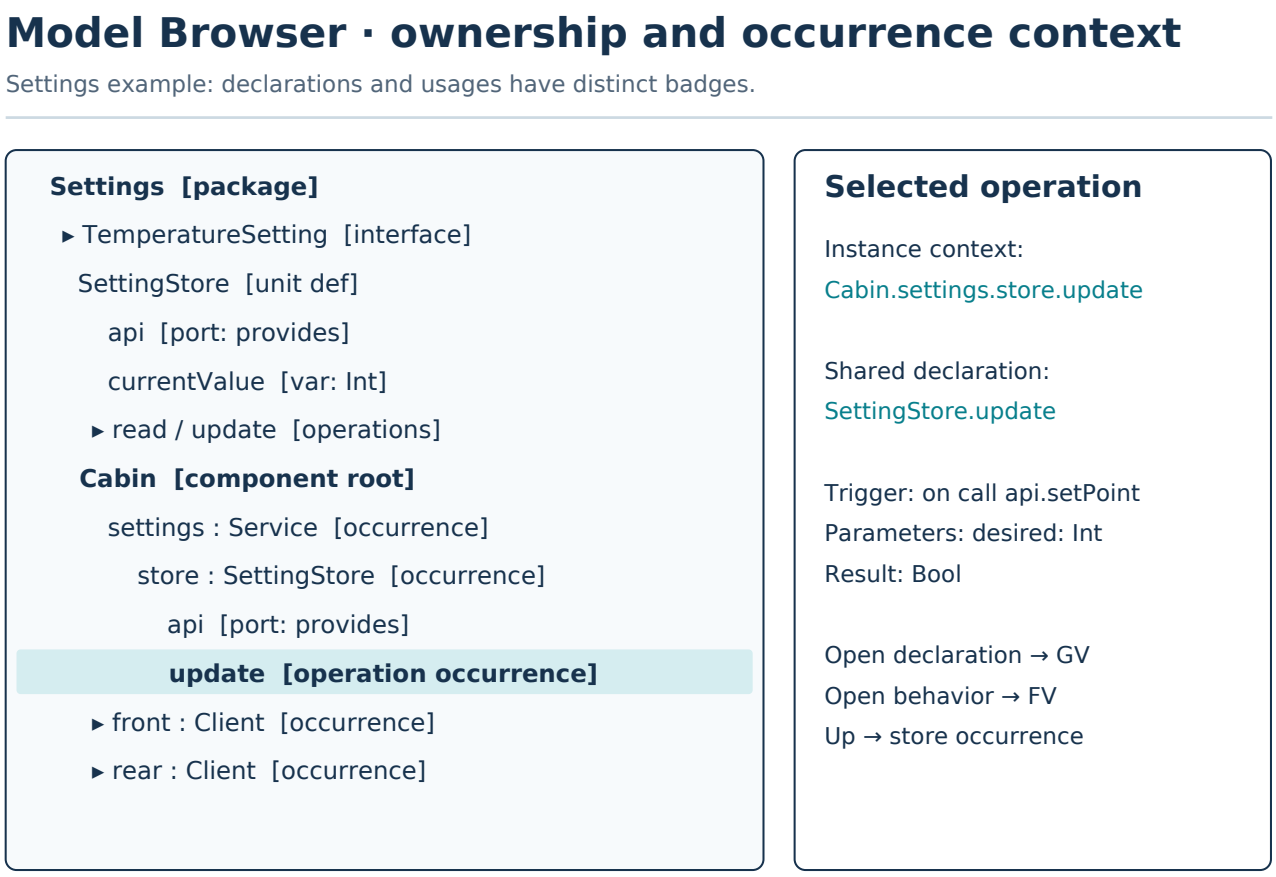
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Graphical notation examples

These rendered vector figures illustrate the normative view contract. The same vector scene sources are rendered into the PDF so labels and lines remain sharp. Every figure has accessible text and a specific model source; the source model remains authoritative.

The first seven figures use the service setting model. A provider owns the setting, two client components use its RPC/events, and private storage remains behind unit ports. The state figure uses the lifecycle model. The figure manifest links each illustration to its model and semantic subject.

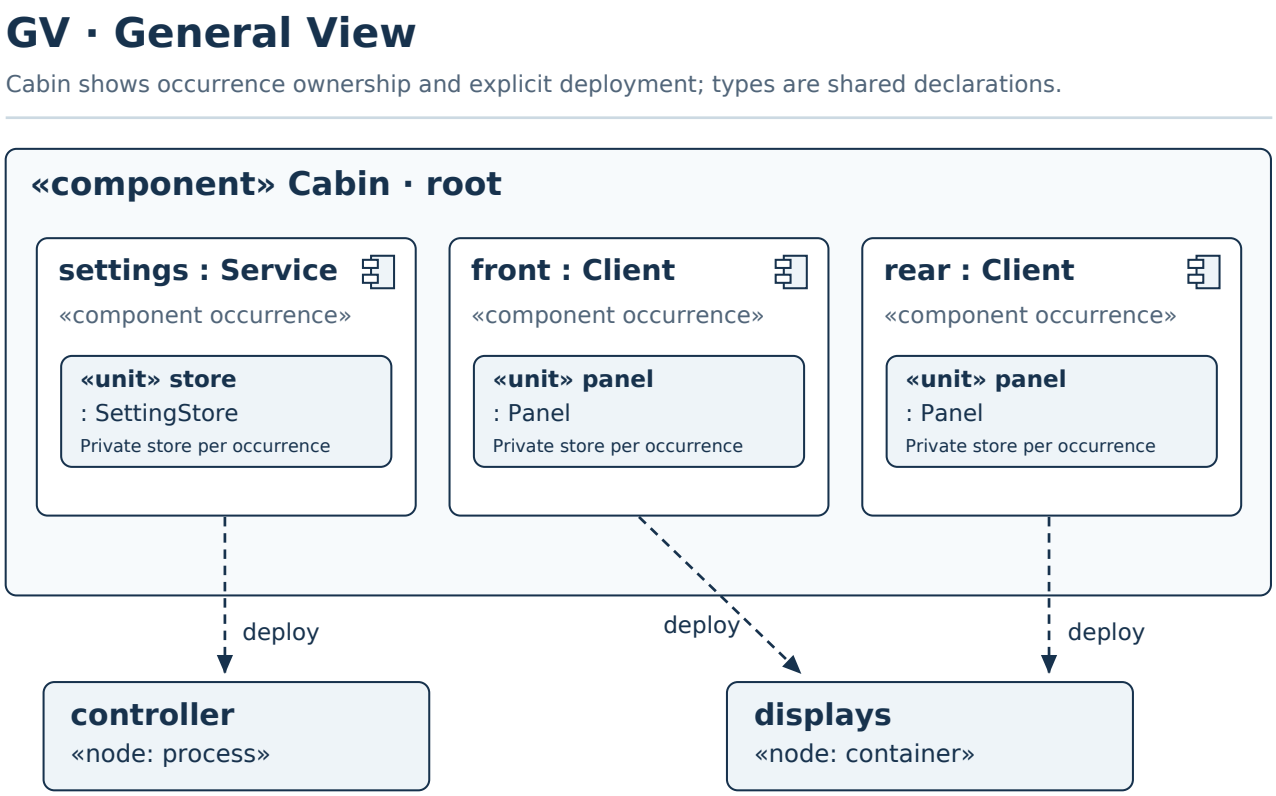
Model Browser: actual ownership



Model Browser showing Settings declarations, Cabin occurrences and selected settings.store.update operation

The tree distinguishes `SettingStore` as a unit definition from `Cabin.settings.store` as an occurrence. Selection of `update` retains its instance context. A declaration command navigates to the shared definition; an FV command opens the operation's body. There is no performer or separate action allocation node. Collapsed rows are presentation choices, not omitted semantics.

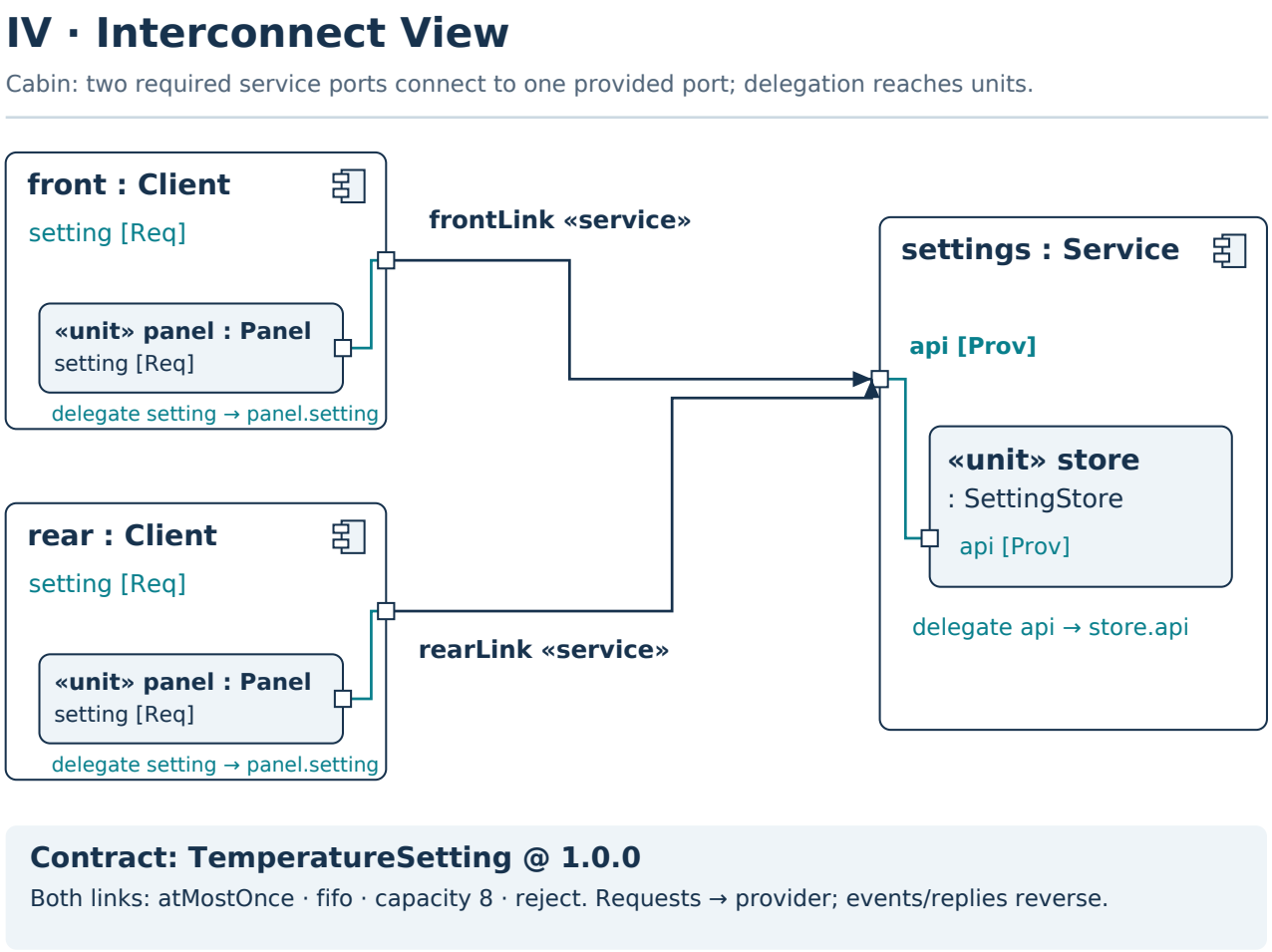
GV: general structure and allocation



General View of Cabin with settings, front and rear component occurrences and inherited deployments

`Cabin` is the root component occurrence and its profile is service. `settings`, `front` and `rear` are component occurrences. Each owns its unit occurrence. Dashed `deploy` edges assign settings to controller and the panels to displays; descendant units inherit those locations. These are allocation edges, not communication lines. The two Panel occurrences have distinct private stores despite sharing one definition.

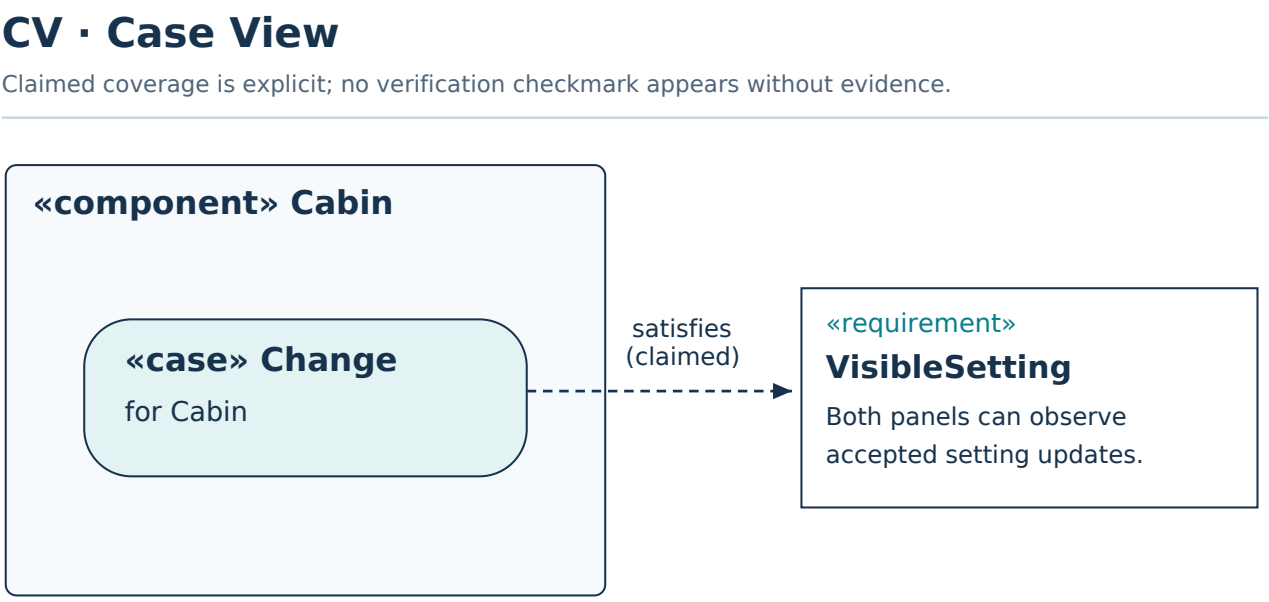
IV: explicit interconnection and delegation



Interconnect View showing required panel ports connected to the provided TemperatureSetting service and delegated unit ports

Both connectors are oriented from `requires` to `provides`. Each visible boundary port is a square and delegates to a child port with the same contract/role. The shared provided port can serve both clients. The arrows indicate connector orientation; member inspection exposes reverse provider-event and response directions. Every route has its own capacity-8 FIFO queue policy per member/direction. There is no line directly coupling panel data to store data.

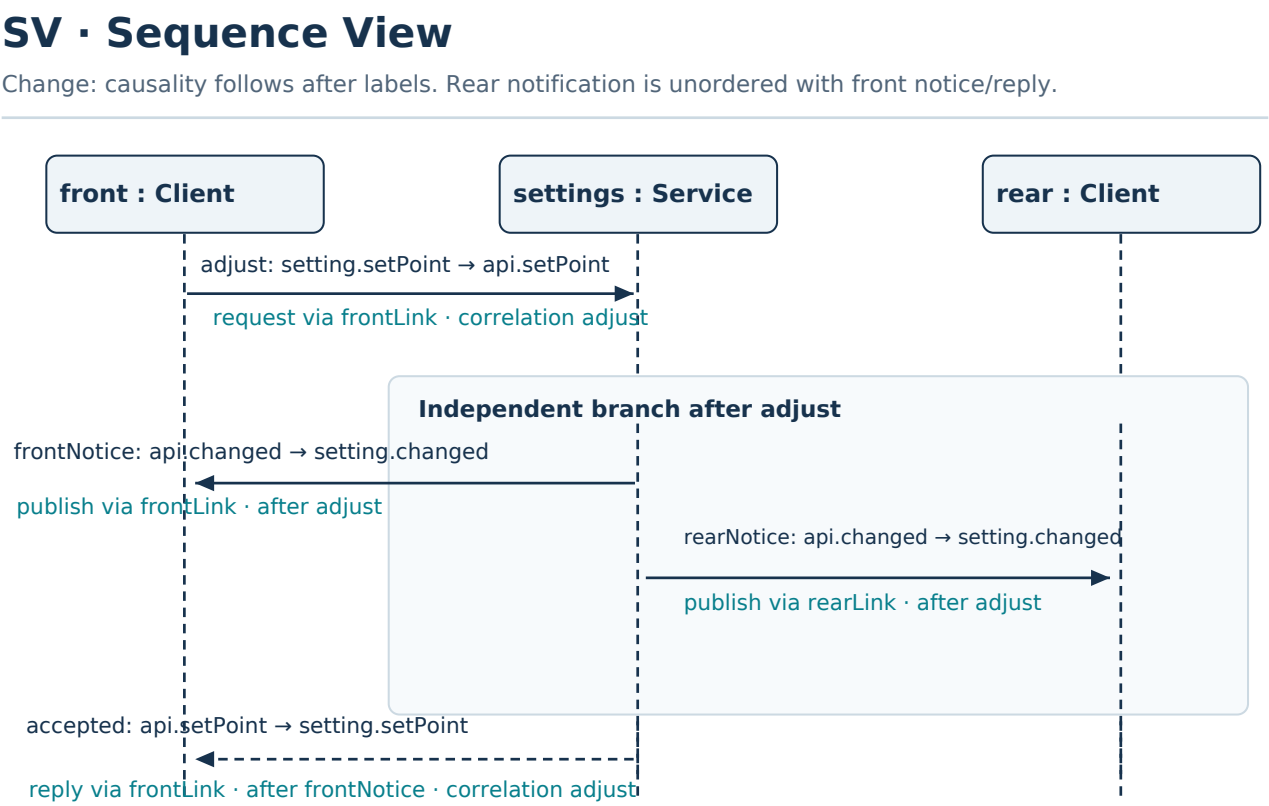
CV: cases and claimed requirement coverage



Case View of Change in Cabin linked to VisibleSetting with satisfies claimed

The dashed `satisfies (claimed)` link records intended coverage. It does not assert that the case is executable or verified. A future tool may attach actual analysis/test evidence; it must retain evidence identity and assumptions before showing verified status.

SV: interface exchanges and partial order



Sequence View of Change showing a setPoint request, two changed notifications and one correlated reply with explicit causal dependencies

Lifelines are component occurrences; every message names its exposed port/member and connector. `adjust` precedes both notifications; `frontNotice` precedes `accepted`. `rearNotice` is unordered relative to `frontNotice` and `accepted`, even though the drawing places its row between them. The pale branch and explicit causal summary prevent vertical placement from inventing a total order. The reply retains the `adjust` correlation. This diagram depicts the case's partial order, not a simulator's chosen trace.

GRV: inspect properties without changing semantics

GRV · Grid View

Operation inventory from Cabin; blank timing means no declared contract in service profile.

Occurrence	Operation	Trigger	Result	WCET	Deadline
settings.store	read()	call api.get	Int	1 ms	5 ms
settings.store	update(desired)	call api.setPoint	Bool	-	-
front.panel	adjust()	manual	Void	-	-
front.panel	refresh()	manual	Void	-	-
front.panel	display(current)	receive setting.changed	Void	-	-
rear.panel	(same operations)	distinct occurrence	Void	-	-

read is reentrant; other operations default to serial. Sorting this grid changes no execution order.

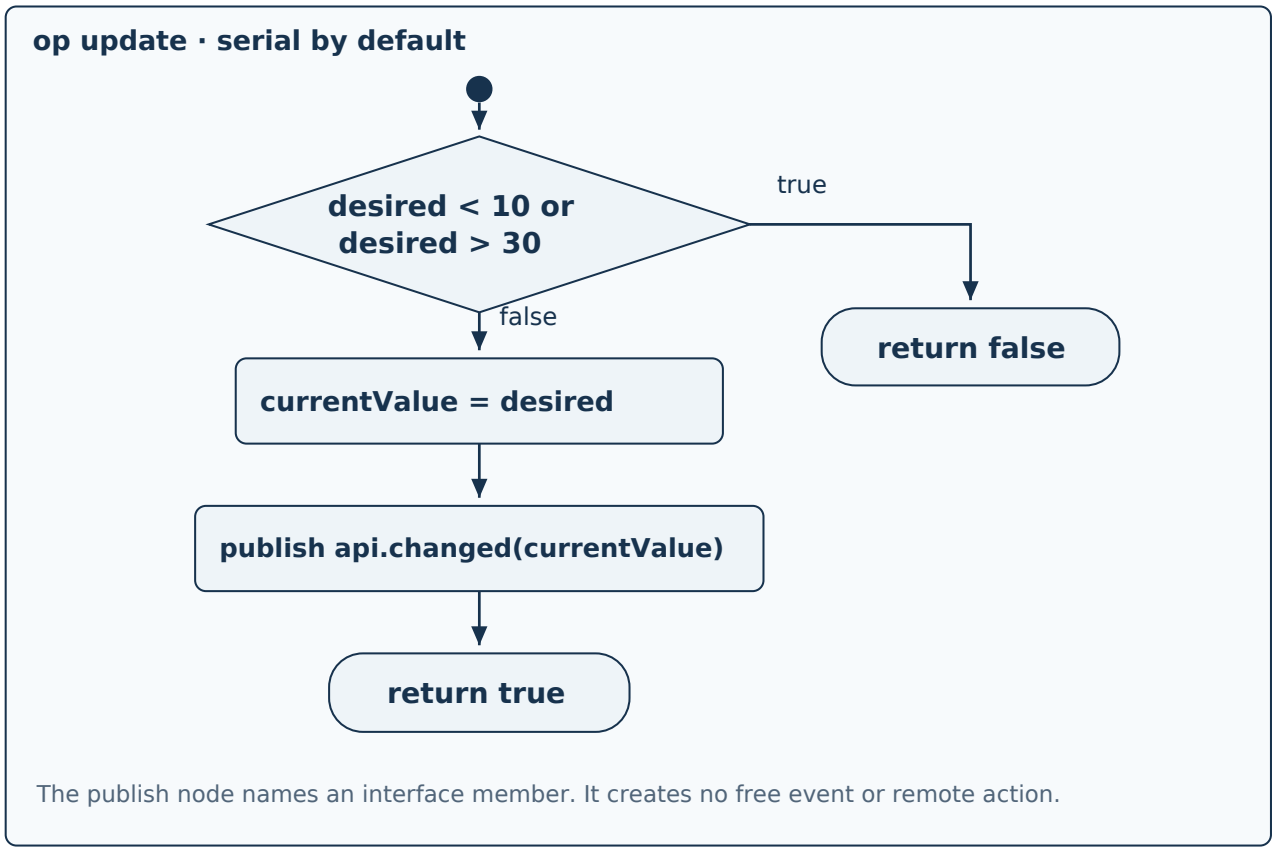
Grid View listing service and panel operation signatures, triggers, results and declared timing

The grid shows effective occurrence paths and source-declared contracts. A dash means the service-profile operation has no timing declaration; it never means zero physical execution time. `read` is reentrant; other activation roots use the serial default. The rear row summarizes the same three declarations in a different occurrence context. Sorting a grid does not reorder source statements, queues or case dependencies.

FV: operation statement flow

FV · Operation flow

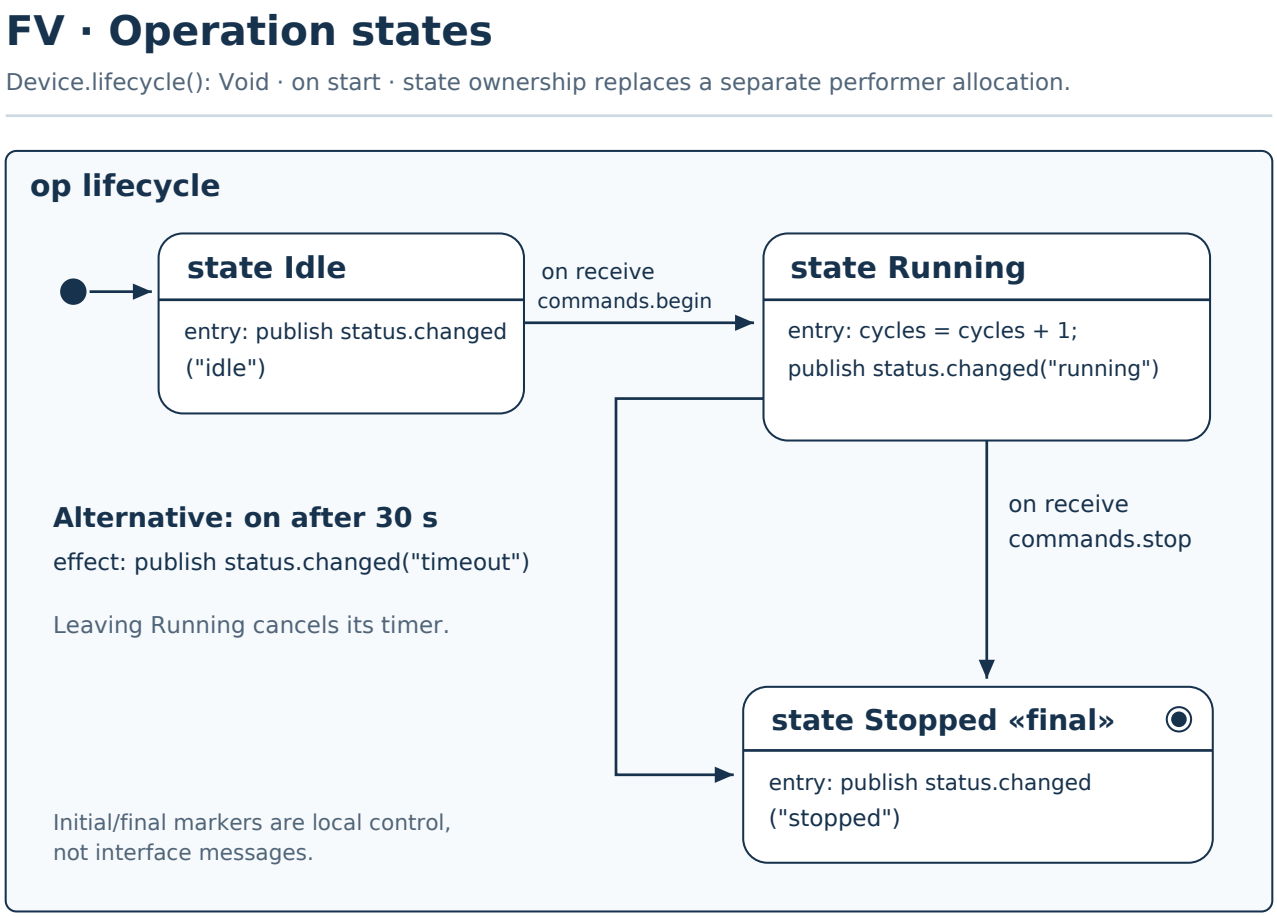
SettingStore.update(desired: Int): Bool · on call api.setPoint · directly owned by the unit.



Flow View of SettingStore.update with bounds check, false return, private assignment, interface publication and true return

The frame belongs directly to `SettingStore.update`. The decision tests the actual source condition. A valid value updates private state, emits `api.changed(currentValue)` through the provider port, then returns true to the correlated request. An invalid value returns false without mutation/publication. A connector is not drawn from this local flow node to another operation: the IV and port path describe the architectural exchange.

FV: operation-owned states



Flow View of Device.lifecycle with Idle, Running, Stopped final, interface event transitions and a 30-second timer

`Idle`, `Running` and `Stopped` belong directly to `Device.lifecycle`. Initial entry publishes idle status. The interface event `commands.begin` enters Running; entry increments cycles and publishes running. Either `commands.stop` or the local 30-second timer enters the final state. The timer edge first publishes timeout in its effect. Final entry publishes stopped and terminates its owning operation. Leaving Running cancels the old timer generation. FV projects both state behavior and statement flow from their owning operations.

Reproduction and semantic checks

Run `python3 scripts/build-graphical-examples.py` to regenerate the eight SVG/scene pairs, and use `--check` to detect stale figures. `node scripts/check-graphical-examples.mjs` resolves every manifest subject against its validated source model and verifies that all view families have an illustration. The semantic test establishes subject/reference validity; it is not a graphical editor conformance certification. Publication QA additionally renders and visually inspects the figures in the PDF.

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Reference simulation profile 1.1

This document specifies the **swaml-reference/1.1** execution binding. The portable TypeScript interpreter and value evaluator execute valid SWAML 1.1 root components. The profile implements logical behavior with explicit resource bounds. It does not model instruction duration, operating-system scheduling, hardware or network deployment.

Run a model

Build with `npm ci` and `npm run build`. Then run:

```sh
npm run simulate -- Commerce::Shop examples/simulation/03-service-pubsub.json
examples/03-service-pubsub.swaml
npm run simulate -- Lifecycle::Product examples/simulation/04-stateful-operation.json
examples/04-stateful-operation.swaml
node scripts/check-simulations.mjs
```

The command takes a qualified root component name, a JSON options file and all workspace model files. It validates the entire input before executing. JSON output contains the trace, final stores, machine states, effective deployments, exact source text and per-source SHA-256 digests. Redirect stdout to a file when collecting evidence. The direct `node out/src/cli/simulate.js ...` command emits only JSON to stdout on a valid invocation; `npm run` additionally prints its script banner.

The selected root may compose units/components inline or instantiate a component definition. A typed root's deployment body does not extend its topology. Instance paths, stores, manual release paths and deployment keys remain relative to the selected root without a root-name prefix. The output names this selection `root`, and identifies the binding as `swaml-reference/1.1`.

Exit status is 0 when no failure was observed, 1 for invalid models, a failed activation, a contract violation or a resource limit, and 2 for an invocation/configuration error. A bounded horizon with no observed failure is not proof of eventual success. Descriptive `case` declarations are not a second executable script and are not automatically proven by this command.

The run catalog names the published roots, options and expected final values. The hello trace is a complete reproducible observation artifact, including its exact model source. CI reruns every catalog entry, compares final values, checks repeatability and detects a stale published trace.

Binding parameters

All times and counts are safe integers. Times are nanoseconds. Unknown options and invalid numeric bounds are rejected.

JSON property	Default / domain	Meaning
`untilNs`	required; ≥ 0	Process events at times ≤ this horizon. Start epoch is zero.
`maxSteps`	100000; ≥ 1	Fuel for expansion, events, statements, expressions and transition work. Stops explicitly on exhaustion; it is a resource count, not elapsed CPU time.
`maxPending`	1024; ≥ 1	Global bound on pending lifecycle/manual/periodic/timer work, excluding active frames and connector queues.
`maxReentrant`	64; ≥ 1	Maximum simultaneous reentrant activation roots per unit. Serial activation roots have their separate one-lock rule.

JSON property	Default / domain	Meaning
`latencyNs`	0; ≥ 0	Constant one-way transport delay for accepted request, response, data and event offers.
`manual`	empty list	Parameterless harness releases: `{ "atNs": 0, "operation": "web.frontend.checkout" }`. Path is relative to selected root component. Only on `manual` operations may be selected.
`faults`	empty list	Fault schedule keyed by the one-based `offer` ordinal shown in the trace. Each offer has at most one entry.

Example options:

JSON

```
{
  "untilNs": 3000000000,
  "latencyNs": 1000,
  "maxSteps": 100000,
  "maxPending": 1024,
  "maxReentrant": 64,
  "manual": [{ "atNs": 0, "operation": "web.frontend.checkout" }],
  "faults": [{ "offer": 1, "kind": "drop" }]
}
```

Manual releases at the same time retain their list order. A release carries no model payload. Use modeled adapter operations and their typed ports to represent external commands or data. The simulator never allows a harness to call a remote handler directly.

Deterministic scheduling and time

The event ordering key is `(timeNs, phase, insertionOrdinal)`. At one timestamp, phase 0 processes request expiries; phase 1 processes lifecycle/timer releases and transport readiness; phase 2 advances execution; phase 3 checks deadlines/latency. Thus timeout wins an exactly simultaneous response, while an activation completing exactly at its deadline satisfies the bound. Each event is followed by a dispatch pass: pending lifecycle work in release order, then connector queues in first-offer order. Enabled work may acquire a lock during that pass; newly released work does not preempt an already held serial lock.

Execution advances between completed statements, local operation steps, loop iterations and state transitions. Pure expressions and individual assignments are atomic. Root activations interleave through the stable event ordinal. FIFO delivery is maintained per connector/direction/member. `unordered` permits this FIFO choice too; the simulator does not randomize it. Independent queues have a deterministic tie-break for reproducibility, not a source-language happens-before relation.

All local CPU costs are abstracted to zero. Transport delay, timeout, period, timer residence, suspension and queue delay advance logical time. Deadline observation begins at lifecycle/timer release or input readiness, including a message awaiting a busy unit. Helpers receive their own elapsed deadline watchdog at invocation. A detected deadline or latency violation stops this run and remains in the trace. Zero-cost execution cannot establish a physical WCET bound; deployment nodes retain their declared identities and do not become simulated processors.

Serial activation roots retain their lock while awaiting RPC. Reentrant activation roots read a deep dispatch-time snapshot; their helpers share that snapshot and mode. Reentrant activation roots may coexist with the serial activation root, subject to the reentrant limit. A machine allows one reaction at a time. Input stays in its connector queue until the relevant dispatch limit/lock permits execution. Pending lifecycle capacity is checked before adding another release; overflow stops with `ActivationOverflow`.

The finite-fuel policy also bounds non-progressing or Zeno model executions. No fairness or liveness result beyond the observed finite trace is claimed. There is no randomness, wall-clock dependency or automatic restart.

Transport and fault interpretation

Each connector occurrence, direction and member owns a separate queue. Delegation resolves boundary aliases to leaf unit ports without adding queues. Capacity includes messages waiting for transport readiness; dispatched handlers are excluded. A send/publish attempts every matching route before reporting rejection. Offers that succeeded remain committed.

Fault `kind`	**Meaning**
`drop`	For bestEffort/atMostOnce, discard the admitted message observably. For atLeastOnce, lose the first attempt, retain the logical identity/timestamp and retry once after an additional `max(1, delayNs)`; the retry succeeds unless overflow, age or availability terminates its obligation.
`duplicate`	Only legal on atLeastOnce. Dispatch two attempts of the same logical message/correlation. The retained logical queue entry reserves capacity until both attempts have been dispatched, with both preceding the next FIFO message. Response duplicates are observed and ignored after terminal completion.
`transport`	Detect a failed route immediately at emission. Report `Transport` to the relevant producer/correlation. This is an interface transport outcome, not a free event.

An entry may set `delayNs` to override the normal delay for that offer. Default transport is reliable with one attempt for every admitted offer; this is a permitted trace of each delivery policy. The finite injected loss/retry model exercises failure rules but does not simulate arbitrary networks or prove eventual delivery. An entry targeting an offer beyond the chosen horizon remains unused; inspect `emission.offer` to verify that the intended fault was exercised.

Queue reject fails a data/event producer after fanout. RPC admission rejection enters `otherwise` with `Rejected`. Drop policies do not fail the producer. Age is measured from original emission, inclusive at the declared maximum, with expiry on the next nanosecond. A timed-out request does not cancel server effects. Remote handler failure returns `RemoteFailure`; late/duplicate replies cannot resume a caller again. An optional unbound RPC reports `Unavailable`; an unbound optional data/event send succeeds without offers.

Values, flow and state coverage

Every expression and statement in the grammar is interpreted: primitives, enums, nominal records, fixed arrays, field/index reads, pure functions, local operations, copied parameters, lexical scopes, assignment, branches, bounded loops, assertions, returns, explicit failure, send, publish and awaited request. Int arithmetic uses exact intermediate integer operations before range checks. Real uses binary64. Scalar Int-to-Real widening preserves the declared Real type. Payloads, parameters and results are copied values.

State initialization, entry/exit, event payloads, guards, priority, timer generations, self transitions, completion reactions and finality are interpreted. Entry/exit lack event payload scope. Stale timers and inputs to a terminated machine are observable discards. A return or failure terminates the machine without completing later transition phases. A guarded completion cycle fails before taking edge 1,025.

These rules have regression tests for both success and failure, including callback blocking, snapshots across suspension, fanout partial rejection, freshness boundaries, retry identity, duplicate RPC, exact-time timeout races, arithmetic/bounds failures, timer cancellation, backlog limits and queued-input deadlines. They exercise observable semantics rather than only mirroring parser implementation.

Evidence and limitations

`status` reports why scheduling stopped: `quiescent` (no scheduled events), `horizon` (future events remain), `violation` (root contract/resource failure), or `limit` (fuel exhausted). `outcome` is `failed` when a failure/violation occurred and `no-failure-observed` otherwise. A failed ordinary activation need not stop unrelated units, so a quiescent run may still have a failed outcome. `timeNs` is the last processed event time, including inert expiry/deadline bookkeeping; it is not necessarily the horizon or last visible effect.

Trace records have monotonically increasing `ordinal`, nondecreasing `timeNs`, `kind`, and `subject`. Activation (`a`), correlation (`r`) and logical message (`m`) IDs are unique within a run; pair them with the evidence artifact's run identity when combining runs. Emissions carry producing activation, leaf endpoints, member, route, offer number and copied payload; source text preserves the boundary delegation provenance. Deliveries carry attempts and correlation where relevant. Statement records include source lines. Final stores and machine states remain available after limits and violations.

The CLI supplies SHA-256 per source; the portable library returns exact sources and leaves hashing/storage to its host. Keep those sources, digests, selected root, complete options, profile version and output together. Absolute source URIs may identify the local workspace; the published hello artifact uses stable repository-relative file URIs.

The implementation provides **reference simulation evidence with an explicit logical-time abstraction**. It is not a blanket E certification, model checker, production runtime, KerML exporter, AUTOSAR generator or graphical editor. Physical WCET/schedulability, distributed clock error, real serialization, middleware outages and independent conformance assessment still require separate evidence. A host can invoke the portable API after building and validating the entire referenced workspace.

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Migrating from SWAML 1.0 to 1.1

SWAML **1.1.0** is the compact authoring edition. It retains explicit `def` type declarations, the component → unit → operation ownership model and every interface, behavior, timing and graphical requirement. It intentionally changes source syntax despite the minor version number. Do not treat 1.1 as a drop-in parser for 1.0 models.

Published 0.1 and 1.0 tags, PDFs and references remain historical artifacts. Keep an original model copy and its pinned tool version when preserving old simulation evidence. Interface versions such as `version "1.0.0"` are independent contract identities; syntax migration does not change them.

Select the language edition in the host

Remove `swaml "1.0";` from source files. A 1.1 file begins with `package`, or is empty during editing. Use the 1.1 language-core package/CLI; an editor host must pin the supported edition before loading its workspace. No project configuration file format is invented by the model language, and the current CLI does not search for one. The 1.1 parser rejects old headers rather than silently accepting or guessing a version.

Keep definitions explicit; make roots component occurrences

`component def Assembly` and `unit def Controller` remain unchanged. `def` is required for a reusable type. A type declaration never starts a singleton or selects a simulation root.

Replace a 1.0 assembly wrapper with a named root component while preserving its child names and connections:

TEXT

```
// 1.0
system Product realtime {
    component controller: Controller;
    node ecu mcu;
    deploy controller on ecu;
}

// 1.1  -  same child path, routes and allocation
component Product realtime {
    component controller: Controller;
    node ecu mcu;
    deploy controller on ecu;
}
```

The profile can be `service`, `realtime` or `mixed`. Omitting it means service. Do not remove a realtime/mixed profile: doing so would weaken the model's required timing contracts.

An inline root can contain units directly. This avoids introducing a definition used only to hold one unit. Alternatively a root can instantiate a reusable assembly:

TEXT

```
component def Assembly {
    unit controller: Controller;
}
component Product: Assembly;
component DeployedProduct: Assembly {
    node application process;
    deploy controller on application;
}
```

The typed root's body is a deployment binding, containing only `node` and `deploy`. It cannot add units, components, ports, connectors or delegation. Its architecture comes entirely from `Assembly`. Binding names must not collide with members inherited from that definition. An inline root has its own composition, but no outward ports/delegation. Required boundary ports on a reused definition retain their closure obligations.

Selecting `Package::Product` executes that root occurrence. Simulation, deployment and case paths are relative to it: a child is `controller`, not `Product.controller`. Replacing a wrapper with `component Product: ExistingType;` may shorten the old child paths; make that structural refactoring separately and update every case, view, deployment, manual release and expected store path. Merely changing `system` to `component` preserves paths.

Put states directly in their operation

Remove the `machine` marker. Keep the `initial` declaration first; states follow, then transitions. The operation still has no parameters, has a Void result, and starts with its unit.

TEXT

```
// 1.0
operation lifecycle(): Void on start machine {
    initial Idle;
    state Idle {}
}

// 1.1
operation lifecycle() on start {
    initial Idle;
    state Idle {}
}
```

This is still stateful behavior with the same entry/exit, guard, priority, timer, completion and finality rules. `StateBody` is a grammar grouping, not a second authored executable element. Runtime documentation may use the mathematical term “state machine” without reintroducing a source declaration or performer.

Trim expressions and no-result annotations

1.0 source	1.1 source	Meaning retained
`operation stop(): Void`	`operation stop()`	missing result means Void; explicit `: Void` remains valid
`set count = count + 1;`	`count = count + 1;`	whole-value assignment to an existing permitted cell
`compute clamp(value)`	`clamp(value)`	pure function in the same unit
`array [1, 2, 3]`	`[1, 2, 3]`	nonempty fixed array value
`enumValue Status.Ready`	`Status.Ready`	nominal enum value, subject to lexical resolution
`enumValue Contracts::Status.Ready`	`Contracts::Status.Ready`	explicit package-qualified enum value

Only operation implementations and interface signatures may omit result annotations. Function results remain explicit and non-Void. Variable, parameter, field and local types remain explicit. No arithmetic, copying, scope or communication behavior is inferred from these shorter spellings.

A local or unit value named `Status` that is accessible in the current expression environment makes `Status.Ready` a field read, even if an enum has that name. A missing field is an error; the resolver does not

fall back to the enum. Qualify `Contracts::Status.Ready` when the enum is intended. Unqualified nominal names retain the usual local-package/direct-import ambiguity checks across declaration kinds. Pure functions expose only parameters, and variable initializers expose no mutable values; inaccessible unit variables do not shadow enums in those contexts. Qualified enum syntax never grants access to another unit's private values.

The removed words `swaml`, `system`, `machine`, `set`, `compute`, `array` and `enumValue` are no longer reserved. The current grammar has 122 reserved keywords, including type names, units, roles and policy values. Remaining words have defined modeling work: `record` identifies nominal construction, `invoke` distinguishes imperative local calls, and `request`/`send`/`publish` retain communication semantics.

Preserve architecture and check evidence

No 1.1 simplification removes interface versions, typed ports, handler ownership, emission provenance, explicit connectors/delegation, mandatory QoS, RPC timeout/failure branches or bounded loops. All exchange between distinct units/components still follows interfaces. States and operations retain direct ownership. Browser, GV, IV, CV, SV, GRV and FV remain.

Run `npm run check` after migrating. It checks current examples, executable documentation, reserved-word coverage, grammar mapping, simulation values and trace freshness. The simulation binding is now `swaml-reference/1.1`; its selector and result field are named `root`, replacing `system`. Rebuild any host using generated AST types: `RootComponent`/`RootMember` replace `System`/`SystemMember`, `StateBody` replaces `Machine`, and enum values resolve from `ReferenceExpression` using `ValuePath` rather than an `EnumLiteral` node.

A successful migration should preserve expected stores and observable effects after accounting for intentional occurrence-path edits. Exact trace source lines and source digests change when text changes. Do not overwrite historical evidence to imply it was generated by the new parser. Logical-time simulation continues to require a finite horizon and explicit resource bounds; it is not a physical scheduling proof or a KerML-conformance claim.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

SWAML 1.1 user guide

SWAML describes what software contains, how it communicates, and how operations and state evolve. The model remains independent of C++, Rust, an RTOS, a cloud runtime, or a particular network protocol.

1. Install and check a model

From the repository root:

```sh
npm ci
npm run build
node out/src/cli/main.js examples/01-hello.swaml
```

A successful result is `Checked 1 model(s): 0 error(s)`. This checks the implemented rules; it does not execute the software or prove schedulability. `npm run check` also runs the repository tests. Open the implementation matrix before interpreting a passed check as a stronger assurance claim.

A file begins directly with named packages. Use the 1.1 language core or pin SWAML 1.1 in the editor host; there is no per-file version header. The 1.1 parser rejects historical headers rather than guessing a version. Names are case sensitive. Keywords cannot be identifiers. Use `orderLink`, for example, rather than the reserved word `ordering`.

2. Write the contract first

Choose the communication model by the meaning of the exchange:

Need	Interface	Roles	Operation statement
A current sensor value	`senderReceiver`, `sampled` data	sender → receiver	`send p.member(value);`
Every queued reading	`senderReceiver`, `queued` data	sender → receiver	`send p.member(value);`
Request and correlated response	`clientServer`	client → server	`request p.member(...) as result within … otherwise …;`
One event to many subscribers	`publishSubscribe`	publisher → subscriber	`publish p.member(...);`
RPC plus server notifications	`service`	requires → provides	request and publish

Each connector uses one nominal interface definition. Equal-looking interfaces are not automatically interchangeable. Create an adapter unit when their meaning, payload, or version differs.

This is a complete small model; save it as `example.swaml`:

```swaml
package First {
    interface senderReceiver Reading version "1.0.0" {
        data value: Real sampled;
    }
    unit def Source {
        port output: Reading sender;
```

SWAML / CONTINUED

```
        operation sample() on start {
            send output.value(21.5);
            return;
        }
    }
    unit def Sink {
        port input: Reading receiver;
        var last: Real = 0.0;
        operation update(value: Real) on receive input.value {
            last = value;
            return;
        }
    }
    component def Pair {
        unit source: Source;
        unit sink: Sink;
        connect readings: source.output -> sink.input qos {
            delivery atMostOnce; ordering fifo; capacity 1; overflow dropOldest;
        };
    }
    component Demo: Pair;
    view overview GV of Demo;
    view wiring IV of Pair;
    view updateFlow FV of Sink.update;
}
```

`component def Pair` is a reusable type; `component Demo: Pair;` selects one root occurrence for execution. Its omitted profile means service. `Reading` defines the payload. Both ports use it with complementary roles. `sample` sends through its own port, and `update` declares that same port-member contract as its activation source. The connector allows the transfer. The variable `last` belongs to each `Sink` instance, not to every instance of its definition.

3. Choose component and unit boundaries

Use a **component** for a reusable architecture boundary: an embedded controller, a service backend, or a subsystem. Use a **unit** for a cohesive implementation boundary that owns operations and private data. A unit need not be a thread, file, class, or process; deployment is separate.

A component definition owns component/unit instances, ports and connectors. `def` makes this type declaration explicit, following the familiar SysML definition/usage distinction. A root is an occurrence: use `component Demo: Pair;` to instantiate a definition, or `component Demo { unit source: Source; … }` to assemble units/components directly. It does not own executable statements. A unit owns operations, pure functions and private variables. This keeps executable ownership unambiguous.

An operation can call a local helper with `invoke helper(...);`. The helper must be an untriggered `Void` operation in the same unit. Use `functionName(...)` for pure calculation results. These local calls do not need ports because they do not cross the agreed unit boundary.

Never write `otherUnit.operation()` or `otherUnit.privateValue`. The syntax and validator reject that access. To reach another unit, declare a port, choose an interface, connect it in their common parent, and use `request`, `send` or `publish`.

For a nested component, expose a boundary port and delegate it to one immediate child port. The service example demonstrates `Backend.api` delegating to `serviceUnit.api`. Delegation preserves the port's role and interface; it does not reverse them or convert data.

4. Describe operation control flow

Operations contain ordered statements: typed local declarations, assignments, assertions, conditionals, bounded loops, local calls, interface exchanges, returns and failures. Read Values and flow as a worked example.

Operations and interface signatures may omit `: Void`; a missing result annotation means exactly Void. Functions are expressions with typed parameters and an explicit non-Void result. They cannot read mutable unit data, issue requests, or emit events. Pass their inputs explicitly. Records, enums and arrays are values, so sending them does not expose a remote reference to local mutable data. Write `count = count + 1;` for whole-value assignment, `clamp(value)` for a pure local calculation, `[1, 2, 3]` for an array, and `Status.Ready` for an enum value. Use `Contracts::Status.Ready` when a local value named `Status` would otherwise hide that enum.

A `while` includes a maximum iteration count. Exceeding that count at runtime is a failure, not permission to silently stop the loop. This makes the bound useful to a later analysis or code generator.

Use `assert` for a model-level assumption that must hold at that point. A requirement is a named prose obligation. A case can link to requirements using `satisfies`; that trace link is not proof that the requirement is met.

5. Model a service call and its failure

Read Services and pub/sub. The service declares `submit` in its interface; the server unit implements it with `on call api.submit`. The client's `request` includes a timeout and an `otherwise` block.

A request binds its result only on success. The failure block has a local `failure: String` value and must return from or fail the current operation. This prevents subsequent statements from reading an uninitialized result. A timeout says the caller stopped waiting; it does not prove the server did nothing. Later responses to that timed-out request are ignored.

For application-specific errors, use an enum or result record in the operation's declared return type. Transport failures and application results are different concepts. Do not retry a request unless its idempotency and retry budget are established. The core rejects `atLeastOnce` RPC when the operation is not declared idempotent; proving that declaration is an implementation obligation.

6. Use publish/subscribe without a hidden event bus

An event is a member of a `publishSubscribe` or `service` interface. Only an operation can publish it, through one of its own outgoing ports. Subscribers receive it through their own typed ports. The normal handler uses `on receive input.eventName` with the event's parameter types in order.

Fan-out is multiple explicit connectors from a publisher port to subscriber ports. Each subscription has its own queue and QoS. A slow subscriber does not silently change the queue of another subscriber. Multiple publishers may connect to one subscriber; ordering is per connector, not global.

In a combined service, the connector is drawn from `requires` to `provides` for requests. Service events travel back from provider to consumer, and replies return on the originating request's correlation. Both remain members of the same declared interface.

Model a broker as a component with incoming and outgoing interfaces if broker behavior, failure, buffering or deployment matters. A connector alone represents a logical subscription, without selecting MQTT, DDS, Kafka or SOME/IP. Dynamic discovery and unsubscribe operations are reserved for future language versions.

7. Put states inside an operation

The stateful operation example owns `Idle`, `Running`, and `Stopped` inside `Device.lifecycle`. It starts once with its unit. No performer or external behavior allocation is required.

A stateful operation has no parameters, a Void result (usually omitted), and `on start { initial Idle; … }`. The initial reference is the first item in its body. There is no extra behavior wrapper. It declares an initial state, states with optional entry/exit blocks, and transitions. Triggers are an incoming port member, a positive delay after entering a state, or completion. A receiving transition makes event parameter names available in its guard and effect; sender-receiver data uses `payload`.

This complete model places the states directly where they execute:

SWAML

```
package TimerExample {
    unit def Device {
        var cycles: Int = 0;
        operation lifecycle() on start {
            initial Waiting;
            state Waiting {
                entry { cycles = cycles + 1; }
            }
            state Finished final {}
            transition complete: Waiting -> Finished on after 10 ms {}
        }
    }
    component Demo { unit device: Device; }
    view lifecycle FV of Demo.device.lifecycle;
}
```

The root contains its unit directly. The delay is local to that operation; it is not a free event or cross-unit channel. The interface-triggered lifecycle model adds commands and status ports.

On a transition: run the source exit block, the transition effect, then the destination entry block. A final state completes the stateful operation. A received event with no enabled transition in the current state is consumed and ignored. Add an explicit transition if ignoring it is unacceptable.

Use one stateful operation as the receiver for a given member, or one ordinary operation. Do not bind the same member to both: that would create ambiguous delivery.

8. Model realtime constraints

The controller example uses `on every 10 ms`, execution-time bounds and deadlines. A root with a realtime or mixed profile requires timing contracts for every used operation and maximum connector latency. Sampled realtime data also needs `maxAge`.

The validator checks positivity, `wcet <= deadline`, and `deadline <= period` for a periodic operation. These are necessary checks, not a scheduling proof. Document the actual RTOS policy, interrupt load, CPU allocation, priorities, clock assumptions, transport timing and measured/analyzed WCET in the implementation project. A production backend must establish them before claiming hard realtime execution.

`serial` is the default unit-local execution policy. `reentrant` requires independent activations and must not mutate shared unit state, including through helpers. An operation waiting for a service response consumes elapsed deadline time even when it consumes no CPU execution time.

9. Separate deployment from architecture

A root component can declare execution nodes with kind `mcu`, `process`, or `container`, and `deploy instance.path on nodeName;`. A typed root may have a deployment-only body, for example `component Product: Controller realtime { node ecu mcu; deploy sensor on ecu; }`. Its `sensor` must already exist in `Controller`; this body cannot add children or connections. An inline root may put composition and deployment together. A component deployment applies to descendants unless a more specific descendant deployment overrides it. Each explicit instance path may be deployed only once.

The component roots example shows a reusable assembly, a typed realtime deployment binding and a small inline root with direct unit children. No `system` wrapper is involved.

Nodes are allocation targets, not units and not communication shortcuts. Moving a unit to another process cannot make a previously illegal direct data access legal. Network addresses, credentials, task schedules and generated code are outside the 1.1 model.

The mixed gateway shows how one unit explicitly converts sampled values into event notifications. An adapter owns the conversion, timing and failure policy.

10. Organize a larger model

Packages are flat namespaces in 1.1. Write `import Contracts::*;` to use another package's declarations unqualified, or name a declaration `Contracts::Reading`. All relevant files must be loaded together. Two imports exposing the same short name make it ambiguous; use the qualified name.

The editor host should load the whole model workspace. The CLI deliberately loads the files you supply and does not search unrelated directories. Do not put secrets in model payloads, examples or deployment notes.

11. Navigate through views

Surface	Purpose
Browser	Navigate packages → definitions → instances → ports/operations → states
GV - General View	Inspect structure, declarations and their relationships
IV - Interconnect View	Inspect connected ports and component delegation
CV - Case View	Inspect cases and linked requirements
SV - Sequence View	Inspect the partial ordering of interface exchanges in a case
GRV - Grid View	Inspect sortable model, interface, timing and deployment tables
FV - Flow View	Inspect an operation's control flow or owned state transitions

These are specified editor behaviors. The current package supplies the model syntax and language server, not a diagram renderer. View declarations can already be parsed and checked. See graphical notation for the drawing and editing contract.

12. Resolve diagnostics

Diagnostic	Typical fix
`SW030`	Connect immediate child ports; expose/delegate a port across each component boundary
`SW031` / `SW032`	Use the same interface definition and complementary port roles
`SW034`	Connect the port in a closed root component, or mark an intentionally unbound port `optional`
`SW050` / `SW051`	Use a local port member and the correct communication verb/direction
`SW052` / `SW055`	Match handler parameters/results and give each incoming member one owner
`SW057`	Make every failure path return or fail
`SW062`	Give competing transitions distinct priorities

A parser error usually means a reserved identifier, missing delimiter, or malformed block. Resolve syntax errors first; semantic checks are skipped on syntactically incomplete documents to avoid misleading secondary errors.

Service fields, repeated consumers and checked traces

Service field models one temperature setting with explicit `get`, `setPoint` and `changed` interface members. The service stores the value privately. Its serial setter publishes through its provided port, which fans out to two required ports on separate panel instances. Each panel owns a distinct displayed value and its own receiving operation.

The getter is reentrant and reads the private-state snapshot taken when it is dispatched. It can safely overlap the serial setter because it does not mutate storage, directly or through a helper. Snapshot implementation is an execution-backend obligation; parsing this example does not simulate it.

A case reply must follow a unique unanswered matching request through its `after` ancestry. Two outstanding requests on the same route/member cannot be disambiguated by drawing one reply lower on the page. Order the pairs or use separate cases. Requests may remain unanswered in a partial trace. A route-valid case still needs an operation-level witness before it counts as verified behavior.

Use the owning package's name as a GV/GRV subject for an inventory. IV takes a component definition/root component; FV takes an operation, including an occurrence path such as `Cabin.settings.store.update`. CV/SV take a case or root component. A view cannot use another view as its semantic subject.

If a request returns Void, its required `as` name marks completion but cannot be read as a value. Reserve `failure` for the implicit error-path binding. Write string line breaks with `\n`; raw line breaks, tabs and unsupported escapes inside strings are rejected.

The operational contract defines execution edge cases, and the conformance obligations distinguish validator acceptance from runtime and analysis evidence.

Find a construct or run an example

Use the alphabetical keyword reference for exact words and context, the concern guide for architectural questions, or the complete Markdown book for a single user-facing document. The maintained examples have executable simulation configurations; follow the reference simulation guide to run them and inspect their traces.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Architecture analysis and SWAML design rationale

This is an original design comparison, not a declaration of compatibility with the referenced standards. Research checked on 2026-10-14. The detailed AUTOSAR comparisons use the accessible R21-11/R19-11 specification material as a stable conceptual baseline; they are not a feature-by-feature review of the latest release.

Comparison

Concern	AUTOSAR Classic	AUTOSAR Adaptive	UML 2.5.1 / SysML 2.0	SWAML decision
Architectural focus	Embedded software components, RTE and BSW	Service-oriented adaptive applications	General software / systems modeling	Software units and components with separate deployment
Contract style	Sender-receiver and client-server interfaces	Methods, events and fields	Interfaces, ports, connectors and behavioral constructs	Four explicit interface kinds; fields expand to operations/events
Behavior ownership	Runnable entities and activation events	Application behavior behind service APIs	UML behaviors/operations; SysML actions, states and usages	Operations own flow/states directly in their unit
Exchange boundary	Ports and runtime integration	Service communication	General-purpose mechanisms can be used in several modeling styles	Cross-unit communication must always use ports/interfaces
Timing	Strong embedded integration context	Distributed execution and communication concerns	Expressible constraints; target analyses vary	Period, WCET, deadline, latency and age; no implied scheduling proof
Deployment	ECU/BSW/RTE integration	Processes/machines and middleware	Deployment/allocation abstractions	Logical nodes and separate target binding
Graphics	Tool-specific architecture views	Tool-specific service views	Broad diagram/view families	Browser + GV/IV/CV/SV/GRV/FV

The table's SWAML column is our design judgment. The following paragraphs identify the source concepts and the choices drawn from them.

AUTOSAR Classic

Classic's application, RTE and basic-software layers separate application software from its integration infrastructure. Its virtual functional bus uses dedicated ports. The software-component template distinguishes sender-receiver data and operation-oriented client-server communication; runnable activation links interface activity to executable behavior. Classic Platform overview, Software Component Template, R21-11, R19-11 runnable/operation specification.

SWAML inference: retain strict port contracts and explicit activation, but make them platform-neutral. A unit is not automatically a runnable, and a component is not automatically an AUTOSAR atomic SWC. A

Classic backend could map a suitable unit operation to a runnable, `on every` to timing activation and `on call` to operation activation, subject to target restrictions. WCET and queue semantics must survive the mapping.

Do not infer: SWAML does not supply RTE generation, ECU extract completeness, BSW configuration or ARXML conformance. A valid SWAML model may need additional target constraints before export.

AUTOSAR Adaptive

Adaptive communication management distinguishes service methods, events and fields, with proxy/skeleton roles and subscription behavior. That separation is useful for a contract that contains both RPC and asynchronous notification. Communication Management, R21-11, Communication Management, R19-11.

SWAML inference: a `service` interface combines operation and event members; requires/provides ports capture consumer/provider roles. A field is explicitly represented by getter, setter and changed-event members, with storage private to the provider. Separate `publishSubscribe` contracts also support event-only architecture without forcing an RPC service.

Do not infer: a combined service connector is not automatically SOME/IP, DDS, ara::com, discovery configuration or an Adaptive manifest. Dynamic discovery and subscription lifecycle are deferred rather than approximated by undocumented behavior. Adapter units provide explicit mappings between Classic-style data and service events.

UML

UML supplies rich structural and behavioral modeling: components, interfaces, ports, operations, activities, state machines and interactions. Its breadth permits many modeling styles; a software-architecture discipline must choose which relationships constitute legal runtime exchange. OMG UML 2.5.1.

SWAML inference: preserve familiar components, operation signatures, state/flow and sequence concepts while making the communication boundary a language invariant. Replace free signals and arbitrary cross-object access with interface-member paths. Keep simple value types rather than importing the full UML class/object feature set.

“Unofficial UML 3.0” is a useful aspiration - modern textual software modeling on KerML - but not a version or compliance claim. SWAML is named independently and does not reuse UML certification marks.

SysML 2.0 and KerML 1.0

SysML 2.0 uses KerML as its foundation and provides definition/usage modeling, typed features and behavior constructs. KerML offers general modeling elements suitable for a separate domain language. SysML 2.0, KerML 1.0.

SWAML inference: keep type versus occurrence and a common semantic foundation, but scope the authoring language to software. Operations and states live where they are used, removing a separate performer allocation from the user model. The mapping contract states what kernel normalization must preserve; the reference Langium AST is not advertised as a conforming kernel implementation.

SWAML defines Browser navigation and six view families: General, Interconnect, Case, Sequence, Grid and Flow. Each view projects the same semantic model and retains links to its source declarations and occurrences.

SWAML 1.1 retains `component def` and `unit def` so reusable types remain visibly distinct from occurrences. A top-level component usage also supplies the closed execution root; no second `system` concept is needed. States remain directly inside their owning operation, distinguished by its initial-state declaration, rather than an extra authored `machine` wrapper. This reduces authoring vocabulary without changing interface boundaries or runtime state semantics.

Additional architecture lessons

AADL emphasizes architecture-level analysis of performance-critical properties and real-time embedded systems. SEI introduction to AADL, SEI control-systems modeling guide.

SWAML inference: timing and deployment need first-class declarations, but an operation deadline is not evidence of schedulability. Keep the logical architecture and platform execution assumptions separate. A backend can translate suitable subsets to an analysis model rather than claim the small reference validator performs scheduling analysis.

For service and event-driven software, our design additionally makes queue capacity, overflow, duplicate delivery, partial publication and timeout uncertainty explicit. These are original SWAML semantic choices. An event broker is a modeled intermediary when its behavior matters. Local shared-memory coupling, hidden topic names and transport-specific callbacks cannot erase the interface boundary.

Mapping examples and limits

Source concept	Candidate SWAML representation	Additional evidence needed for an exporter
Classic SW component composition	component definitions and instances	supported target categories and composition rules
Runnable	unit operation	activation, argument, task and scheduling mapping
S/R data element	sampled or queued data member	precise buffering and freshness correspondence
Client/server operation	interface operation + handler	error/result mapping, timeout and invocation policy
Adaptive method	service operation member	target method signature and middleware binding
Adaptive event	service event + publishing/receiving operation	subscription/delivery behavior
Service field	explicit read/write/change members	coherent state and notification protocol
UML activity	operation statement flow	subset restrictions and exception/concurrency mapping
UML state machine	stateful operation	flattening or rejection of unsupported hierarchy/regions
SysML part definition/usage	component/unit definition and instance	software-specific ownership interpretation
Physical/external endpoint	adapter unit with typed ports	how actual I/O enters the implementation

Decisions retained in 1.1

1. Unit is the smallest interface-protected software boundary. Pure local computation does not require a port.
2. Components compose; units execute. Do not make both own independent execution semantics.
3. Interface signatures are contracts; unit operations provide their implementation.
4. All message/data/event exchange names a local port and interface member.
5. Definitions are reusable, but behavior is owned directly where used. No performers.
6. Publish/subscribe is explicit and supports fan-out without assuming a particular broker.
7. A service can combine requests and events, but implicit fields are avoided.

8. State and flow share FV because both belong to operations.
9. A complete, bounded 1.1 language is preferable to undefined syntax for advanced features. Deferred constructs are listed in the implementation/roadmap documents.
10. Tool claims must distinguish parsing, validation, kernel mapping, execution and graphical editing.

Source handling

Links point to the primary standards/publisher sources. Some AUTOSAR PDF endpoints intermittently failed full downloads during research; their official indexed material and platform pages were available. No standard PDF or proprietary template content is redistributed in this repository. The comparison intentionally focuses on the concepts described above rather than claiming exhaustive standard coverage.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Language tooling and integration

Package contract

The package supplies a reusable language core, command-line tools and a language-server adapter. It is marked `private` to prevent accidental npm publication. Consumers can build it locally and use the documented APIs with an explicitly selected language version.

This edition parses headerless **SWAML 1.1** only. A host must pin its language-core dependency and identify 1.1 in project/workspace settings before source loading; these settings belong to the host, not to an undocumented model grammar. The repository CLI is itself pinned to 1.1. Unsupported requested editions must be rejected explicitly; a host must never silently upgrade models to the latest parser. Read the migration guide before consuming this incompatible source update.

The source of concrete syntax is swaml.langium. Langium generates AST types, grammar metadata, reflection and TextMate syntax. Generated files and `out/` are ignored; rebuild them after checkout.

The implementation follows Langium's grammar → generated AST → references → validation workflow. Langium grammar reference, reference-resolution guide.

Reproducible commands

SH

```sh
npm ci
npm run build
npm test
npm run validate:examples
npm run check
node out/src/cli/main.js examples/01-hello.swaml
node out/src/server/main.js --stdio
```

Node 22 and 24 are the CI targets; Node 22+ is required. Dependencies are pinned in `package-lock.json`. `npm run build` regenerates the grammar before compiling. Never hand-edit generated files.

The package exports `createSwamlServices` from `out/src/language/swaml-module.js`, and AST declarations from `out/src/language/generated/ast.js`. Example usage after a build:

TS

```ts
import { EmptyFileSystem } from 'langium';
import { parseHelper } from 'langium/test';
import { createSwamlServices } from './out/src/language/swaml-module.js';

const { Swaml } = createSwamlServices(EmptyFileSystem);
const parse = parseHelper(Swaml);
const document = await parse(source, { validation: true });
console.log(document.diagnostics);
```

For multiple files, create all documents first and build them together so that imported declarations are available. CLI arguments are the complete loaded workspace. Definitions in unloaded files cannot resolve.

Services and source ownership

File	Responsibility
swaml-module.ts	Dependency injection and registration
swaml-document-builder.ts	Conservative workspace revalidation for paths and negative lookups
swaml-scope.ts	Local-owner scopes, imported/qualified declarations, ambiguity
semantic.ts	Instance/port/member paths, type identities, time conversion
swaml-validator.ts	Model invariants and typed behavior validation
CLI	File loading, diagnostics, exit status
LSP server	Standard language-server transport and startup

Cross-references use Langium references for declarations, functions, local operations, states and nodes. Composite endpoint/value/view paths are strings with explicit SWAML resolution. This keeps encapsulation checks explicit, but **generic rename/find-references/navigation does not cover those strings**. A host must supply path-aware providers before advertising full refactoring.

The current scope provider scans loaded documents for clarity. Every changed/added/deleted document conservatively relinks and revalidates all loaded SWAML models, including string paths and previously absent declarations. This favors correct editor diagnostics over incremental performance. Large-model dependency indexing and cancellation-aware expensive checks remain optimization work. The validator skips semantic checks after parser/lexer errors; built-in diagnostics still report malformed input.

Language-server integration

The server implements the Language Server Protocol over a standard transport. Start it with `node out/src/server/main.js --stdio`; reserve stdout for protocol messages and send diagnostic logging to stderr. Clients supply document URIs, text versions, changes and the complete loaded workspace. They consume diagnostics with source ranges and stable diagnostic codes.

Language hosts must select the supported SWAML edition before loading a document, preserve source ranges, and rebuild affected documents after changes. A formatter or graphical host must apply semantic changes to source and validate the result. File associations, transport startup and presentation are host responsibilities.

Browser hosts

The language core has no Node filesystem dependency. Browser applications can create services with an in-memory filesystem and supply document text directly. The Node CLI and language-server adapter own their filesystem access. A worker-based browser host needs its own transport adapter; test document loading, dependency bundling and import resolution in that environment.

Graphical architecture

Implement a semantic projection layer between Langium and diagrams. It should produce stable node/edge IDs, instance-aware paths, source ranges, declaration references, roles, signatures, diagnostics and presentation kinds. View renderers consume that projection, never parse source text independently.

All edits go through a source-edit service. Layout-only edits update a presentation sidecar. Semantic edits update `.swaml`, then parse/link/validate. Reject silent creation of unmodeled exchanges. Graphical contract.

Kernel backend boundary

Build a separate normalized model from the linked/validated authoring AST, then lower through a versioned SWAML semantic library onto KerML. Keep kernel IDs and authoring IDs traceable. The mapping chapter defines obligations; the JSON inventory is design metadata, not serialized KerML.

Validation and release gates

Before a host consumes a new language-core version, run `npm ci && npm run check`, inspect the spec/grammar changes and update versioned fixtures. A grammar change, even if additive, can reserve an identifier previously used in source. Release notes must explain migration.

Tests cover valid fixtures, illegal interface access, scoping, type errors, QoS, state/timing rules and the LSP transport lifecycle. Broader conformance claims require the additional tests in implementation status.

Simulation API for a separate host

After linking and validating every referenced document, call `simulate(models, "Package::Root", options)` from the package's `./simulation` export. It has no Node filesystem dependency. The host supplies a finite horizon and may supply manual releases, transport fault schedule and resource limits. The profile contract defines the exact scheduler, observability and limits. The CLI adapter adds file loading and SHA-256 source digests.

Present `outcome` and `status` together: a run can quiesce after a failed activation, and a bounded horizon is not a proof of liveness. Hosts can connect trace source lines and occurrence paths to FV/IV/GRV selections. They must preserve the simulation contract and validate every semantic edit.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Implementation status - 1.1.0

The normative specification defines the complete bounded 1.1 language. The repository implements the Langium parser, CLI, language-server foundation and broad static checking. It also implements a bounded logical-time reference simulator; it does not implement a production runtime, KerML exporter or graphical editor. See the 1.1 migration, historical 1.0 evidence and historical 0.1 audit and conformance obligations.

Implemented and tested

Capability	Evidence
Headerless 1.1 grammar, portable string escapes, generated AST/TextMate syntax	generator, all examples, lexical regressions
Explicit definitions, inline/typed root components, direct unit children, deployment-only typed binding and default service profile	1.1 root/migration regressions
Direct operation states, optional Void results, assignment/function/array/enum syntax	1.1 parser, type, shadowing and simulation regressions
Imports, qualified names, ambiguity across declaration kinds and owner-scoped references	multi-document and namespace tests
Interface member/role/signature checks; no cross-unit operation/data bypass	contract and isolation tests
Composition, delegation, required boundary/leaf closure, route uniqueness and cardinality	topology tests
Emitting-operation provenance and unique handler ownership	SW050/SW055 tests
Value typing, records/enums/arrays, expressions, pure functions, Void completion restrictions	values/flow and audit tests
Local calls, acyclic call/containment graphs, transitive reentrant write analysis	call-cycle and helper-effect tests
Explicit QoS, sampled buffering and idempotency declaration checks	QoS tests
Positive safe bounds, WCET/deadline/period consistency and realtime completeness	timing and numeric tests

Capability	Evidence
Stateful signature, transition priorities, graph reachability, finality and unconditional-cycle checks	state tests
Deployment references, case direction/dependencies and unique causal reply correlation	allocation/case regressions
All six view subject categories, package and occurrence contexts	view regressions
Conservative workspace relink/revalidation after change/addition/deletion	editor update tests including newly ambiguous names
LSP initialize/diagnostics/shutdown and CLI exit behavior	integration tests
Reference execution	all statements/expressions, startup/periodic/manual activation, machines, queue policies, RPC, snapshots, fault injection and bounded trace evidence
Production-to-spec/normalized mapping inventory	automatic coverage for all 80 grammar rule/type names

`npm run check` rebuilds, runs tests, validates every example, checks documentation links and executable snippets, and detects missing/duplicate grammar mappings. Mapping coverage is a documentation consistency check, not a KerML validation result.

Limits of static acceptance

The reference core checks the static groups listed in chapter 8; it remains labeled **S, partial V** until an independent conformance suite audits every normative clause and boundary case. It is not a model checker. Acceptance does not establish useful emissions, guard feasibility, case realization, idempotency of implemented effects, termination without runtime failure, schedulability, absence of request cycles, or snapshot support on a particular target.

General arithmetic ranges and computed array bounds are runtime checks. Guarded completion cycles are protected by a specified runtime limit. Activation backlog is bounded by a required execution binding. These are specified execution/analysis obligations, not unimplemented syntax disguised as successful static proof.

String-valued paths now receive conservative dependency invalidation. Path-aware rename, navigation, reference search and persisted editor IDs still need dedicated IDE providers. The all-document rebuild is correct for the loaded workspace but may be expensive on large projects.

Specified, not implemented

Capability	Remaining implementation
KerML K backend	versioned SWAML library, normalized mapper, kernel validation, reverse traces and round-trip tests
Production E backend	physical CPU/scheduling, real middleware/serialization, platform clock/resource evidence and independent assessment; logical reference simulation is implemented
Timing/behavioral analysis	target scheduling model, case witnesses, idempotency and safety/liveness evidence

Capability	Remaining implementation
Browser and graphical G workbench	projections, layout sidecars, Properties, validated source edits and undo
Complete IDE refactoring	path-aware rename/find references/navigation and stable identity map
Platform exports	target bindings and serialization such as AUTOSAR ARXML

No blanket K, E or G certification is claimed. The reference simulation profile documents its executable scope and time abstraction. These implementation milestones have complete language-side contracts; they are not necessary to claim that the 1.1 specification has defined its syntax and meaning.

Intentionally outside 1.1

Inheritance/redefinition, generics, dynamic object creation, unbounded arrays, reference-valued payloads, hierarchical/orthogonal state regions, fork/join syntax, host-language code blocks, dynamic service discovery, runtime subscribe/unsubscribe, topic wildcards, exactly-once delivery, full requirements algebra and geometry require a versioned language proposal. They cannot be introduced as undocumented backend behavior.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

SWAML 1.1.0 release evidence

Version 1.1.0 retains explicit reusable definitions and simplifies model authoring. The parser and host select language 1.1; files start with packages. Concrete component usages replace system declarations, states remain directly owned by operations, and unnecessary expression markers are removed. The migration guide documents the intentionally incompatible syntax revision introduced in version 1.1. Runtime and interface isolation obligations are preserved.

Tracked iterations

Issue	Acceptance and evidence
17: Compact language core	Grammar, explicit definition/usage identities, root closure, state ownership, expression resolution and static regressions.
18: Reference execution	Typed and inline root expansion, private stores, deployment bindings, direct state behavior, pinned CLI version, behavioral regressions and reproducible examples.
19: Complete language specification	Normative clauses, formal semantics, migration, user guide, grammar mapping and alphabetical keyword/concern navigation.
20: Publication	Full Markdown/PDF, reviewed vector notation examples, exact source attachments and manifests, repository guidance, CI and release assets.

Each issue is resolved by a focused commit with its verification evidence. CI runs on Node 22 and 24 and checks the committed PDF, an independent PDF rebuild and figure reproducibility. Earlier review fixes for required boundaries and terminated-state input deadlines are included.

Verification and scope

The release gate passes **212 tests**, validates **nine example models**, checks **11 simulation runs**, covers **80 grammar rule/type mappings** and **122 reserved keywords**, and verifies **eight graphical subjects** across all seven view families and both FV forms. Cross-review regressions include imported-root diagnostic locations, enum-only workspace dependencies and lexical enum/value shadowing.

The publication gate runs `npm run check`, covering the full test suite, every model validation and expected simulation, the complete grammar mapping, keyword coverage and graphical subjects. PDF verification checks the source bytes and hashes, bookmarks, links and page bounds. Every rendered page is visually inspected before release.

The reference core provides parsing, broad static validation and bounded logical-time simulation. It remains S and partial V pending an independent conformance audit. It does not claim a kernel-validated exporter, physical realtime schedulability, universal safety/liveness, automatically realized cases, production middleware or a graphical editor.

The v0.1.0 release and v1.0.0 tagged edition remain immutable. Current artifacts are named 1.1.0 and embed their exact maintained sources; current syntax is not retroactively applied to historical editions.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

ADR 0001 - Ownership and communication boundaries

Status: accepted for SWAML 0.1. Date: 2026-10-14.

The repository owner confirmed that units and components communicate only through interfaces, while local calls and variables can remain internal to a unit. The authoring hierarchy is component → unit → operation, with states directly inside stateful operations. Components may compose other components. Pure functions support local value computation.

An operation is the executable primitive. A contract operation signature and its implementation are separate because the interface must remain reusable across implementations. The handler declaration binds them explicitly. There is no performer or independently allocated action object in the user language.

Events belong to interfaces; publishing statements identify a unit-owned port and member. Each outgoing event/data member has a syntactic emitting operation. Each incoming member has exactly one handler owner. This supplies traceable provenance without pretending to prove runtime reachability.

Consequences: more ports are required when decomposing a component into several units. Shared-memory shortcuts cannot be introduced as “internal implementation” across a modeled unit boundary. Keep tightly coupled helpers inside one unit, or model the shared-state service through a contract.

The view families are General, Interconnect, Case, Sequence, Grid and Flow. Browser provides ownership navigation. Flow View covers operation statement flow and owned-state behavior.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Decision 0002: Complete semantics before claiming complete tooling

Status: accepted for the 0.1.1 draft revision, 2026-10-14.

The specification audit found implementation gaps and ambiguous operational details. The language remains a bounded 0.1 design with direct ownership and interface-only boundaries. This revision keeps the syntax/header and clarifies interpretation while tightening previously incomplete validation.

- A reentrant root and all local helpers use one dispatch-time private-state snapshot and cannot write unit state. Static analysis follows helper calls.
- Initialization installs all handlers/routes before startup dispatch. Lifecycle backlog has an explicit bounded target contract; overflow is observable and stops execution.
- All fanout offers are attempted; a rejected offer cannot undo successful ones. Request timeout wins a response observed at the exact expiry instant.
- State timers start at entry completion and belong to a state-entry generation. Completion is processed before the next external stimulus, with a defined microstep bound.
- Replies infer correlation from one unique unanswered request ancestor. The current grammar has no explicit correlation selector; ambiguous partial traces must be split or ordered rather than guessed.
- Required port status survives every delegation level. Normalized routes reject duplicate leaf pairs.
- Every grammar production has a normative home and a mapping-design entry. These entries do not stand in for an implemented KerML library/exporter.
- Static validity, runtime failures, analysis evidence and IDE work have separate conformance obligations. The tool continues to claim S and partial V honestly.

The stricter lexer rejects raw controls/unknown escapes; Void completions cannot be read; imported names are ambiguous even across target kinds. Models accepted accidentally by 0.1.0 may now receive errors. No valid example from the initial draft requires semantic migration. Keeping the 0.1 header is a draft clarification policy, not permission for silent breaking changes after stabilization.

See issues 1-5 and evidence, the operational contract and normalized model.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Decision 0003: stabilize the language with reproducible execution evidence

Status: accepted for SWAML 1.0.0, 2026-10-14.

The repository is the public language authority. Version 1.0 retains the bounded 0.1 construct set and promotes its clarified rules into explicit typing and operational judgments. The header changes to 1.0 so consumers cannot silently treat this edition as an old draft. Specification and package versions are aligned at 1.0.0; interface versions remain independent.

A small deterministic interpreter makes the language testable through observed behavior. It interprets all supported values, statements and state machines; it resolves the same interface routes checked statically. The execution binding deliberately uses zero local CPU cost and explicit transport delay, resource limits, timeout ties and fault schedules. These assumptions produce useful reproducible traces without claiming hardware scheduling proof or covering every possible schedule.

The full user-facing Markdown/PDF references provide two access paths: an alphabetical concern guide and all grammar keywords in alphabetical order. A machine-checked inventory prevents a new keyword from entering the grammar without a user-facing explanation. Maintained chapters, exact grammar and complete examples are assembled into the edition; generated files are checked for drift.

Browser and diagrams retain the established view vocabulary. Eight rendered examples demonstrate definition/occurrence identity, port-bound exchange, case partial order, grid properties and both FV forms. Their semantic subjects are resolved against real models; static illustrations do not imply that an editor has been implemented.

KerML mapping remains an explicit normative design contract requiring a separately implemented library/exporter and kernel validation before K conformance. The reference execution binding is independently testable without pretending that its Langium AST is a KerML serialization. Future production or editor backends must preserve the defined meaning rather than importing undocumented behavior from their frameworks.

Publication preserves the historical v0.1.0 tag/assets. Each 1.0 work item has a focused issue, resolution commit and acceptance evidence. A release-ready tag and draft package can be reviewed before a separate public release action. Once published, assets must not be silently replaced to change language meaning.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

ADR 0004 - Compact component language

Status: accepted for SWAML 1.1. Date: 2026-10-14.

The owner requested a smaller authoring language, retaining `def` to make reusable types clear in the SysML style, removing redundant `system`/`machine` layers, and preserving the original architecture and simulation requirements. The selected release is 1.1.0. This is an intentional incompatible syntax migration, despite using a minor version; no backward-compatible parser claim is made.

Decision

A component usage can be the selected execution root. `component Demo { … }` supplies inline composition; `component Demo: Assembly;` instantiates a reusable component definition. Roots may contain units directly. Omitted profile means service. A typed root may add nodes/deployment in a binding body, but cannot extend topology. Definition identity, root occurrence identity and child occurrence paths remain distinct.

Operations own states directly. A state body starts with `initial`, followed by states and transitions, so parsing can distinguish it from a statement block. `StateBody` is syntax grouping and normalization contributes to the owning operation. No independent machine, action or performer allocation is introduced.

The language header is removed. The CLI/parser is pinned to 1.1; external hosts must pin an explicit supported edition before loading source. A file does not silently choose the latest parser, and no unimplemented configuration format is presented as language syntax.

No-result operation/signature annotations may be omitted. Assignment, local pure function calls, arrays and enum values use familiar punctuation without `set`, `compute`, `array` or `enumValue`. Enum resolution is lexical-first; qualification selects the nominal package enum explicitly. Functions retain explicit non-Void results, and reusable component/unit declarations retain `def`.

Consequences

The reserved vocabulary decreases from 129 to 122 words. This count includes primitive types, time units, role names, view kinds and policy values; it is not the number of independent modeling concepts. Further removal needs evidence of a redundant meaning, not only a lower token count. Explicit QoS, timeouts, interface identity, port ownership and boundary routing continue to express necessary architectural choices.

The specification, grammar, validators, normalized mapping, simulator, examples, keyword index and graphical illustrations must describe the same 1.1 edition. Old release assets remain immutable. The migration guide documents changes to source and host API. The specification defines every supported construct and its simulation contract; graphical tooling and KerML export have their own implementation and conformance obligations.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

PDF specification and release publication

The complete PDF contains the normative chapters, notation, grammar, mapping inventory, nine models and project guidance. It includes clickable contents, bookmarks, links, vector figures, selectable numbered code, repeated table headings and embedded sources.

Version identifiers

This edition is specification/package **1.1.0**, with a pinned **1.1** parser and headerless source. The full Markdown reference and PDF are generated from maintained chapters. The PDF additionally includes project guidance, a complete mapping inventory and embedded source/figure files. Keyword navigation covers every current reserved word; the concern guide is alphabetical.

The v1.0.0 tagged Markdown/PDF edition is preserved unchanged; its source header and keyword inventory belong to the historical grammar. Current publication checks target 1.1 sources.

Historical v0.1.0 packages draft/core revision 0.1.1 with header 0.1; its assets remain unchanged. Version 1.1 provides logical-time simulation; physical timing proof, KerML export and graphical editing remain separate implementations.

Rebuild the book

Use Python 3.10+ and DejaVu fonts (`fonts-dejavu-core` on Debian/Ubuntu). Elsewhere, supply DejaVu Sans regular/bold/oblique/bold-oblique and Sans Mono TTF files with `--font-dir`.

SH

```sh
python3 -m venv .venv-pdf
. .venv-pdf/bin/activate
python3 -m pip install -r scripts/pdf-requirements.txt
python3 scripts/build-spec-pdf.py --font-dir /usr/share/fonts/truetype/dejavu
python3 scripts/check-spec-pdf.py
```

First run `python3 scripts/build-graphical-examples.py` and `npm run docs:build` after changing source chapters or figures. The explicit source list is built directly; unsupported Markdown fails the build. Mermaid illustrations become vector figures. The eight examples share SVG/vector-scene sources across Markdown/PDF. Keyword links jump to individual entries. Every grammar line is preserved; wrapped continuations have no new line number. Attachments preserve exact source bytes.

Outputs are the PDF, a JSON source/hash manifest and a SHA-256 checksum under `output/pdf/`. The checksum identifies the generated artifact; PDF metadata may vary between builds. The manifest allows readers to verify exactly which source content an edition contains.

Quality and release gate

Run `npm run check`, build the PDF, and run the PDF checker. Render all pages with Poppler and inspect them for clipped text, overlapping elements, unreadable tables, broken figures, orphan headings and excessive blank pages. Check dense tables and code at full resolution as well as page contact sheets. Automated extraction cannot replace that visual review.

Commit the reviewed source, builder and final artifacts together. The release must target the reviewed commit, with the PDF, source manifest and checksum attached. Use `--notes-file` for release notes and verify the published tag and assets after upload. Preserve release assets once published; issue a new documented edition for corrections.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Contributing to SWAML

SWAML is developed as a language: specification, grammar, semantics, diagnostics and examples must evolve together. The 1.1 syntax keeps explicit `def` declarations and distinguishes them from concrete component usages. Source files are headerless; a host must pin the grammar version. Read the 1.1 migration when converting old models.

Start with the language specification, ownership decision and implementation status.

Development setup

Use Node.js 22+ and npm 10.2.3+:

```sh
npm ci
npm run check
```

This generates the AST/TextMate files, compiles TypeScript, runs the test suite, validates the examples and checks documentation. Generated files, dependencies and compiled output are not committed. The dependency lockfile is committed.

Propose a language change

Describe the actual modeling problem and a small before/after model. Identify the affected grammar productions, normative clauses, KerML representation, validation rules, view projections and migration behavior. Distinguish a new language feature from a new target-platform binding.

A proposal that adds cross-unit communication must identify the port/interface/member route. A proposal for an event must identify its declaration, emitting operation, receiving owner, delivery policy and failure behavior. A proposal for state or control flow must define ordering, concurrency and termination. Avoid accepting syntax before its semantics exists.

Breaking changes require a new language version and migration notes. Additive keywords can still break existing identifiers; document that impact. Record substantial decisions under `docs/decisions/`.

Implement and test

1. Update the normative prose and grammar together.
2. Update the grammar mapping inventory and classify each obligation as static, runtime, analysis, kernel or graphical. Every rule/type needs a normative home. Regenerate with `npm run langium:generate`; update scoping and validators as needed.
3. Add a valid model and a negative regression that would have caught the original issue. Cross-file changes need imported/qualified and ambiguous-name cases.
4. Update the keyword inventory, user guide, affected examples, simulation configurations, graphical projections, implementation matrix and changelog. Regenerate the complete reference with `npm run docs:build`.
5. Run `npm run check` and report the result in the pull request.

Tests must verify observable language behavior, not copy the implementation's internal algorithm. Preserve malformed-input recovery: editor diagnostics must not crash the language server. New communication features need tests that attempts to bypass interfaces are rejected.

Pull requests and review

Use a focused branch and describe the user-visible modeling behavior. Include a concise syntax/semantic example, the affected specification clauses, validation evidence and any remaining conformance gap. Do not describe a proposed feature as implemented merely because the grammar accepts it.

Reviewers should check that definitions and instances remain distinct, private data cannot escape through paths, all exchanges remain typed, and diagrams do not invent semantics. Compatibility or migration decisions should be explicit before merging.

Documentation and standards

Link to primary sources for external standards. Write original explanations; do not copy standard PDFs or proprietary specification text into the repository. Preserve the existing MIT license for original code and documentation and retain any applicable third-party notices.

Executable `swaml` code fences must be complete, valid models: the documentation checker parses and validates them. Use inline snippets or `text` fences for deliberate fragments, and put intentionally invalid models under `examples/invalid/` with their expected diagnostic.

Resolve specification issues

Track a concrete contradiction, missing semantic choice or validator defect as a GitHub issue with acceptance criteria. Resolve it with matching normative text, implementation where in scope, examples and regression evidence. Close the issue only when its stated criteria are met; do not close runtime/editor implementation work because a specification paragraph exists. Link the resolution commit or merged pull request and record any remaining conformance limitation.

Report a defect

Include the language/core version, a minimal model, exact CLI command or editor action, observed diagnostics and expected rule. For graphical work include the view ID and semantic subject path. For timing/delivery concerns state the target assumptions; a compiler acceptance result alone is not execution evidence.

Do not include credentials, private service URLs or production payloads in examples. Replace them with minimal synthetic values.

Publication changes

The source chapters and `spec/keywords.json` are maintained inputs. The complete Markdown and keyword reference are generated. Run `npm run docs:build` after editing those inputs. Figure changes require `python3 scripts/build-graphical-examples.py` and semantic subject checks; commit SVG and scene pairs together. Do not silently edit one rendition.

For an edition, follow the PDF workflow: build, verify exact source attachments and hashes, render all pages, and visually inspect figures, code, tables and navigation. Attach the full Markdown, PDF, manifest and checksum to the prepared release. A public release is a separate explicit publication action; a release-ready commit/tag or draft preserves reviewability.

Simulator changes must retain the profile contract or introduce a new named profile/version. Include a trace-level regression and record configuration/source identity. Review timeout ties, partial fanout, stale timers, queue limits and snapshot behavior whenever communication or dispatch changes. Never equate a bounded successful run with physical schedulability or a proof of all possible executions.

INFORMATIVE / GUIDANCE AND PROJECT STATUS

Instructions for agents working on SWAML

Read first

- `docs/LANGUAGE_SPECIFICATION.md` and the normative chapter affected by the task.
- `docs/IMPLEMENTATION_STATUS.md` to distinguish specified and implemented behavior.
- `docs/spec/06-execution-contract.md`, `07-normalized-model.md`, `08-conformance-obligations.md` and `09-formal-semantics.md` for execution, mapping and evidence boundaries.
- `docs/decisions/0001-language-boundaries.md` for the owner's accepted architecture choices.
- `CONTRIBUTING.md` for contribution and validation expectations.

Preserve the language invariants

- Components compose units/components. Units own private data, functions and operations. Operations own statement flow or states directly.
- No performers, separate action allocations, global mutable state or free events.
- Every exchange between distinct units/components uses an owned typed port, interface member and explicit connector/delegation route.
- Local calls/data may stay inside a unit. Never resolve a local operation or function from a different unit as a convenience fallback.
- Outgoing events/data must be tied to emitting operations; incoming members must have one handler owner. Service responses remain correlated through their interface route.
- Keep Browser, GV, IV, CV, SV, GRV and FV. FV covers statement flow and operation-owned states.
- Distinguish definition identity from instance paths. Never make same-type instances share private state.

SWAML 1.1 authoring contract

Keep `component def` and `unit def` for reusable types. A top-level component usage is an execution root; it may contain inline topology or instantiate a component definition with optional deployment bindings. Typed roots cannot extend topology. Every concrete root is checked for closure, independently of reusable library definitions. Omitted root profile means service; omitted operation/signature result means Void. Functions still require an explicit non-Void result.

Files are headerless and interpreted by the explicitly selected 1.1 parser/host. Do not restore `swaml`, `system`, `machine`, `set`, `compute`, `array` or `enumValue` as required markers. States are directly authored inside an operation's body; the StateBody AST grouping has no separate executable identity. Preserve the documented ambiguity rules for enum values and local records. Historical tagged syntax requires migration, not silent fallback.

Source and architecture

The normative concrete grammar is `src/language/swaml.langium`. Prose specifies validity and semantics. Update both when meaning changes. Generated files in `src/language/generated/`, TextMate output and `out/` must not be hand-edited or committed.

Keep the language core free of Node filesystem dependencies; CLI/server adapters own platform I/O. Composite paths currently use explicit semantic resolution and need path-aware IDE services. Do not claim generic Langium rename covers them.

KerML mapping is a proposed contract until a real exporter and kernel validation exist. Do not label Langium AST generation as KerML conformance. Likewise do not call an accepted model schedulable, race-free, executable or graphically editable without the corresponding implementation/evidence.

Required checks

For grammar, semantic or source changes run `npm run check`. Use `npm ci` after dependency changes or on a fresh checkout. Add meaningful positive and negative tests for changed language rules, including malformed input when parser recovery is affected. Report failures honestly and fix regressions within the task scope.

For documentation changes run `node scripts/check-docs.mjs` after a build. Complete `swaml` fences are executable fixtures; deliberate fragments should use inline text or a `text` fence.

Working practices

Prefer small, reviewable changes with stable diagnostic codes. Keep every grammar rule/type covered in `spec/kerml-mapping.json`; the documentation checker enforces coverage. Add a decision record for substantial language changes and update the changelog and implementation matrix. Do not silently widen access scopes to make a fixture pass.

Preserve the existing license, unrelated user edits and repository history. Never add credentials or downloaded proprietary standards. Use primary-source links and original analysis. Follow the user's authorization for GitHub changes; local edits alone do not imply permission to publish a release or alter another repository.

Reference execution and publication

Keep the reference simulator portable: file access and source hashing belong to CLI/host adapters. Preserve the documented scheduler, timeout ties, queue identity, copied values, snapshot scope, direct operation/state ownership and explicit resource failures. Test observable traces and final stores, including adverse cases. A simulator run is not production scheduling proof or universal verification.

Maintain every current keyword entry in `spec/keywords.json` as the grammar evolves. Regenerate the consolidated Markdown with `npm run docs:build`; do not hand-edit generated `KEYWORD_REFERENCE.md` or `SWAML-1.1.0-Reference.md`. Regenerate vector examples with `python3 scripts/build-graphical-examples.py`. Their scene/SVG sources and model-subject manifest must stay synchronized.

Before publishing an edition, run the complete language gate, figure reproducibility check and PDF build/check described in `docs/PDF_PUBLICATION.md`, then render and inspect every page. Commit synchronized Markdown, PDF, manifest and checksum. Preserve published historical assets. Keep the public specification independent of specific authoring tools.

D. External references

Primary-source and project links cited in the included text. The comparisons identify their research baseline; this list does not imply conformance to any external standard.

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13. reference-resolution guide
https://langium.org/docs/learn/workflow/resolve_cross_references/

14. 17: Compact language core
https://github.com/voidaliot/swaml/issues/17

15. 18: Reference execution
https://github.com/voidaliot/swaml/issues/18

16. 19: Complete language specification
https://github.com/voidaliot/swaml/issues/19

17. 20: Publication
https://github.com/voidaliot/swaml/issues/20

E. License

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